

MAS312 Modern Algebra II

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Part I

Ring Theory

1 Rings

28-08-23
Recall.

Definition. A **ring** R is a set with $a + b$ and $ab \in R$ for all $a, b \in R$ satisfying

- (1) $(R, +)$ is an abelian group.
- (2) $(ab)c = a(bc)$ for all $a, b, c \in R$.
- (3) $(a + b)c = ac + bc$, $a(b + c) = ab + ac$ for all $a, b, c \in R$.
- (4) $1 \in R$ with $a1 = 1a = a$ for all $a \in R$ exists.

Definition.

1. **Commutative ring:** $ab = ba$ for all $a, b \in R$.
2. **Division ring:** $1 \neq 0$ ($\leftrightarrow$ nonzero ring), $R^\times = R \setminus \{0\}$.
 $R^\times = \{a \in R \mid a \text{ has an inverse}\}$, called group of units.
3. **Field:** commutative division ring.
4. An element a of a ring R is called a **left zero divisor** (resp. **right zero divisor**) if there exists a nonzero element $b \in R$ such that $ab = 0$ (resp. $ba = 0$).
 If R is a commutative ring, left zero divisors $\equiv$ right zero divisors.
5. **(Integral) domain:** commutative and 0 is the only zero divisor.

$$\left[\begin{array}{c} \text{Commutative} \\ \text{rings} \end{array} \right] \supset \left[\begin{array}{c} \text{Integral} \\ \text{domains} \end{array} \right] \supset \left[\begin{array}{c} \text{Unique} \\ \text{factorization} \\ \text{domains} \end{array} \right] \supset \left[\begin{array}{c} \text{Principal ideal} \\ \text{domains} \end{array} \right] \supset \left[\begin{array}{c} \text{Euclidean} \\ \text{domains} \end{array} \right] \supset [\text{Fields}]$$

$$[\text{Integral domains}] \supset [\text{Integrally closed domains}] \supset [\text{Unique factorization domains}]$$

Definition. A commutative ring R is called a **principal ideal ring (P.I.R.)** if every ideal of R is principal (i.e., every ideal of R is generated by a single element), or equivalently, if I is an ideal of R , then $I = Ra (= \{ra \mid r \in R\})$ for some $a \in R$. We use the notations $Ra = \langle a \rangle = (a)$.

An integral principal ideal ring (= P.I.R. which is an integral domain) is called a **principal ideal domain (P.I.D.)**.

Examples.

1. Fields, $\mathbb{Z}$: P.I.D.s
For a field F , $\{0\} = (0)$ and $F = (1)$ are all ideals of F .
Every ideal of $\mathbb{Z}$ is of the form $n\mathbb{Z} = (n)$ for some $n \in \mathbb{Z}$.
2. $\mathbb{Z}[x]$ is not a P.I.D. as $(2, x) = 2\mathbb{Z}[x] + x\mathbb{Z}[x]$ is not a principal.
(If $(2, x) = (f)$ for some $f \in \mathbb{Z}[x]$, then $\deg f = 0$, but $x \notin (f)$.)
3. $\mathbb{R} \times \mathbb{Z}$ is a principal ideal ring but not principal ideal domain.
Ideals of $\mathbb{R}$ are $\{0\}$ and $\mathbb{R}$, and ideals of $\mathbb{Z}$ are of the form $n\mathbb{Z}$. Thus, an ideal of $\mathbb{R} \times \mathbb{Z}$ is of the form $\{0\} \times n\mathbb{Z} = \langle (0, n) \rangle$ or $\mathbb{R} \times n\mathbb{Z} = \langle (1, n) \rangle$.
 $(0, 1) \cdot (1, 0) = (0, 0)$ are zero divisors.

Proposition 1.1. *Every nonzero prime ideal in a P.I.D. is maximal.*

Proof. Let $P = (a)$ be a prime ideal for some $a \in R$, where R is a P.I.D. By Zorn's lemma, $P \subset I = (b)$ for some maximal ideal I of R and some $b \in R$. Then, $a = br$ for some $r \in R$. As P is prime, $b \in P$ or $r \in P$. If $r \in P$, then $r = as = brs$ for some $s \in R$, $1 = bs$. Hence, $I = R$, a contradiction. Therefore, we have $b \in P$, so $P = I$. ■

Definition (Euclidean Domain). A commutative ring R is called a **Euclidean ring** if there exists a map $\varphi : R \setminus \{0\} \rightarrow \mathbb{N} \cup \{0\}$ such that for any $a, b \neq 0 \in R$, there exists $q, r \in R$ such that $a = bq + r$ with either $r = 0$ or $\varphi(r) < \varphi(b)$.

A Euclidean ring which is an integral domain is called a **Euclidean domain (E.D.)**.

Examples.

1. Every field is a Euclidean domain with any φ because $a = b(b^{-1}a) + 0$ if $b \neq 0$.
2. $R = \mathbb{Z}$, $\varphi(n) = |n|$ is a E.D.
3. $R = F[x]$ with a field F , $\varphi(f) = \deg f$ is a Euclidean domain.
4. $R = \mathbb{Z}[x]$ is not a E.D. as R is not a P.I.D.
(E.D. $\Rightarrow$ P.I.D.)
5. $R = \mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ with $\varphi(x) = |x|^2$ (i.e., if $x = a + bi$, then $\varphi(x) = (a + bi)(a - bi) = a^2 + b^2$) is a E.D.

Let $x, y \neq 0 \in R$. Then

$$\frac{x}{y} \in \mathbb{Q}[i] = \{u + iv \mid u, v \in \mathbb{Q}\}.$$

Write $x/y = u + iv$. If $u, v \in \mathbb{Z}$, then $x = y(u + iv) + 0$. Otherwise, find a and b in $\mathbb{Z}$ such that $|u - a| \leq 1/2$ and $|v - b| \leq 1/2$. Set $q = a + bi$. Then $x/y - q = (u - a) + i(v - b)$ and

$$\left| \frac{x}{y} - q \right|^2 = (u - a)^2 + (v - b)^2 \leq \frac{1}{4} + \frac{1}{4} < 1$$

Then $|x - qy|^2 < |y|^2$. Set $r = x - qy$. Then $\varphi(r) < \varphi(y)$.

Theorem 1.2. Every Euclidean domain is a principal ideal domain.

Proof. Let $I \neq 0$ be an ideal of a E.D. R with φ . Consider $S = \{\varphi(a) \mid a \neq 0 \in I\}$. Choose $a \in I$ such that $\varphi(a)$ is the smallest integer in S . Then $I = Ra$.

(Let $b \in I$. Then, $b = qa + r$ for some $q, r \in R$. If $r \neq 0$, then $\varphi(r) < \varphi(a)$ and $r = b - qa \in I$, a contradiction. Hence, $r = 0$, $b = qa \in Ra$.) ■

Remark. The converse statement may not hold in general:

$$\mathbb{Z}[\alpha] = \{a + b\alpha \mid a, b \in \mathbb{Z}\}, \quad \alpha = \frac{1 + \sqrt{-19}}{2}$$

is a P.I.D. but not a E.D.

30-08-23

Definition. Let R be a commutative ring and let $a, a', b \in R$ with $b \neq 0$.

1. We say that a is **divisible** by b if $a = bc$ for some $c \in R$.
(We use the notation $b \mid a$.)
2. Two elements a and a' are called **associate** if $a \mid a'$ and $a' \mid a$.
(We use the notation $a \sim a'$.)

Remarks.

1. $b \mid a \iff aR \subset bR$ and $a \sim b \iff aR = bR$.
2. Let $u \in R^\times$. Then,
 - $u \mid a$ for all $a \in R$ (as $a = u \cdot (u^{-1}a)$).
 - $u \sim 1$ (as $uR = R = 1 \cdot R$).
 - $a \sim au$ (as $auR = aR$).
3. If R is a domain, then $a \sim b \iff a = bu$ for some $u \in R^\times$:
 $\leftarrow aR = buR = bR$
 $\rightarrow a \in aR = bR$. Then $a = bc$ for some $c \in R$. As $b \in aR = bR$, $b = ad$ for some $d \in R$.
 Hence, $a = bc = ad$, thus $1 = dc$, i.e., $c \in R^\times$.

Definition. Let R be a domain.

1. An element $c \in R$ is called **irreducible** if
 - (1) $c \neq 0$, $c \notin R^\times$
 - (2) $c = ab$ for some $a, b \in R$ implies $a \in R^\times$ or $b \in R^\times$.
2. An element $p \in R$ is **prime** if
 - (1) $p \neq 0$, $p \notin R^\times$
 - (2) $p \mid ab$ for some $a, b \in R$ implies $p \mid a$ or $p \mid b$.

Examples.

- 1.
- $R = \mathbb{Z}, R^\times = \{\pm 1\}$

Let $a \in R$. Then, a is irreducible $\iff a$ is prime $\iff a$ is a (possibly negative) prime integer.

- 2.
- $R = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\}$

2 is not prime but irreducible: $2 \mid (1 + \sqrt{-5})(1 - \sqrt{-5}) = 6$ but $2 \nmid a$ and $2 \nmid b$ as $\frac{1 \pm \sqrt{-5}}{2} \notin R$.

We show 2 is irreducible:

$$\begin{aligned} (a + b\sqrt{-5})(c + d\sqrt{-5}) &= 1 \\ \implies |a + b\sqrt{-5}|^2 |c + d\sqrt{-5}|^2 &= 1 \\ &= (a^2 + 5b^2)(c^2 + 5d^2) = 1 \\ \implies a^2 + 5b^2 = 1 = c^2 + 5d^2 &\implies a = \pm 1, b = 0 \\ \implies R^\times &= \{\pm 1\} \end{aligned}$$

Assume that $2 = xy$, $x, y \in R$. Then $4 = |x|^2 |y|^2$. As $|x|^2 \in \mathbb{Z}^+ \cup \{0\}$, $|x|^2 = 1$, $|y|^2 = 4$ or $|x|^2 = 2 = |y|^2$.

If $|x|^2 = 1$, $|y|^2 = 4$, then $|x|^2 = a^2 + 5b^2 = 1 \implies a = \pm 1 \in x \in R^\times$ ($x = a + b\sqrt{-5}$).

If $|x|^2 = a^2 + 5b^2 = 2$, this cannot happen since $a, b \in \mathbb{Z}$, i.e., $a^2 + 5b^2 \neq 2$.

Note that R is not a P.I.D. as $(2, 1 + \sqrt{-5})$ is not principal.

Lemma 1.3. *Let R be an integral domain.*

- (1) *Every prime element of R is irreducible.*
- (2) *Let $p \in R$. Then, p is prime if and only if Rp is a nonzero prime ideal.*
- (3) *Let $c \in R$. Then c is irreducible if and only if Rc is maximal in $\{Ra \mid a \in R, a \neq 0, a \notin R^\times\}$.*
- (4) *If R is a P.I.D., then an element a of R is irreducible if and only if a is a prime.*

Proof. (1) Let p be a prime element. If $p = xy$ for some $x, y \in R$, then $p \mid x$ or $p \mid y$, thus $x = pa$ or $y = pb$ for some $a, b \in R$. Hence, $p = pay$ or $p = xpb$. As R is a domain, $1 = ay$ or $1 = xb$, i.e., $y \in R^\times$ or $x \in R^\times$. Hence, p is irreducible.

(2) We may assume that $p \neq 0$, $p \notin R^\times$.

$$\begin{aligned} p \text{ is a prime} &\stackrel{\text{def}}{\iff} p \mid ab \text{ implies } p \mid a \text{ or } p \mid b \\ &\iff ab \in Rp \text{ implies } a \in Rp \text{ or } b \in Rp \\ &\stackrel{\text{def}}{\iff} Rp \text{ is a prime ideal.} \end{aligned}$$

(3) ($\implies$) Let $Rc \subset Ra$ for some $a \in R$. Then, $c = ax$ for some $x \in R$. As c is irreducible, $a \in R^\times$ or $x \in R^\times$, thus $Ra = R$ or $Rc = Rax = Rxa = Ra$.

($\impliedby$) Assume that $c = xy$ for some $x, y \in R$. Then, $Rc = Rxy \subset Rx$. As Rc is maximal, $Rx = R$ or $Rc = Rx$. If $Rx = R$, then $x \in R^\times$. If $Rc = Rx$, then $x = rc$ for some $r \in R$. Hence, $c = xy = rcy$, i.e., $y \in R^\times$.

(4) By (1), it suffices to show that every irreducible element is prime. Assume that p is irreducible. Then, by (3), Rp is maximal (as R is a P.I.D.). Hence, Rp is a prime ideal. By (2), p is prime. ■

Lemma 1.4. Let R be a commutative ring. Then, the following statements are equivalent:

- (1) Every ideal I of R is finitely generated, i.e., $I = (a, \dots, a_n) = Ra_1 + \dots + Ra_n$ for some positive integer n and some $a_i \in R$.
- (2) Every nonempty set $\mathcal{S}$ of ideals of R has a maximal element in $\mathcal{S}$.
- (3) R satisfies the ascending chain condition (A.C.C.): for any chain of ideals $I_1 \subset I_2 \subset \dots \subset \dots$, there exists some $N \in \mathbb{N}$ such that $I_N = I_n$ for any $n \geq N$.

Definition. A commutative ring R is called **Noetherian** if it satisfies (1) $\sim$ (2) $\sim$ (3) in Lemma 1.4.

Proof. (1) $\Rightarrow$ (2). Use Zorn's Lemma. Define $I_1 \leq I_2$ if $I_1 \subset I_2$. Let $\{I_\alpha\}_{\alpha \in \Lambda}$ be a chain of ideals in $\mathcal{S}$. Then, $I := \bigcup_{\alpha \in \Lambda} I_\alpha$ is an ideal and I is an upper bound of $\{I_\alpha\}_{\alpha \in \Lambda}$ ($\forall \alpha \in \Lambda: I_\alpha \leq I$).

We show $I \in \mathcal{S}$. By the assumption of (1), $I = (a_1, \dots, a_n)$ for some $a_i \in R$. Let $a_i \in I_{\alpha_i}$, $\alpha_i \in \Lambda$ for all $i = 1, \dots, n$. Choose j such that $I_{\alpha_i} \leq I_{\alpha_j}$ (i.e., $I_{\alpha_i} \subset I_{\alpha_j}$) for all i . Then, $a_i \in I_{\alpha_j}$, thus $I \subset I_{\alpha_j}$, so $I = I_{\alpha_j}$.

(2) $\Rightarrow$ (3). Let $I_1 \subset I_2 \subset \dots$. Let $\mathcal{S} = \{I_1, I_2, \dots\}$. By the assumption of (2), $\mathcal{S}$ has a maximal element I_N in $\mathcal{S}$. Then $I_N = I_n$ for all $n \geq N$.

(3) $\Rightarrow$ (1). Let I be an ideal of R . Choose $a_1 \in I$. Set $I_1 := Ra_1 \subset I$. If $I_1 \neq I$, then choose $a_2 \in I \setminus I_1$. Set $I_2 := I_1 + Ra_2$. Then $I_1 \subsetneq I_2$. If $I_2 \neq I$, then choose $a_3 \in I \setminus I_2$. Set $I_3 = I_2 + Ra_3$. Then, $I_1 \subsetneq I_2 \subsetneq I_3$. By the assumption of (3), there exists some $N \in \mathbb{N}$ such that $I_N = I$. Hence

$$I = Ra_1 + \dots + Ra_N = (a_1, \dots, a_N). \quad \blacksquare$$

04-09-23

Examples.

1. Every P.I.D. is Noetherian; e.g., F (field), $\mathbb{Z}$, $\mathbb{Z}[i]$.
2. $\mathbb{Z}[x]$ and $F[x]$ (F is a field) are Noetherian (as E.D. $\Rightarrow$ P.I.D. $\Rightarrow$ Noetherian).

Theorem 1.5 (Hilbert basis theorem). If R is Noetherian, then so is $R[x]$.

Proof. Let I be an ideal of $R[x]$. Write $l(f)$ for the leading coefficient of $f \in R[x]$, i.e., $f = a_n x^n + \dots + a_0$ with $a_n \neq 0$ then $l(f) = a_n$.

For each $n \geq 1$, the set

$$I(n) := \{a \in R \mid a = l(f) \text{ for some } f \in I, n = \deg(f)\} \cup \{0\}$$

is an ideal of R . Moreover, $I(n-1) \subset I(n)$ ($a \in I(n-1) \implies a = l(f) \implies a = l(xf)$).

As R is Noetherian, there exists an integer m such that $I(m) = I(m+1) = \dots$. Write $I(i) = \langle a_{i1}, a_{i2}, \dots, a_{ik_i} \rangle$, where $a_{ij} = l(f_{ij})$, $1 \leq j \leq k_i$, $1 \leq i \leq m$. Let $J = \langle f_{ij} \mid 1 \leq i \leq m, 1 \leq j \leq k_i \rangle$ and let

$$J(n) = \{a \in R \mid a = l(f) \text{ for some } f \in J \text{ and } \deg(f) = n\} \cup \{0\}.$$

Then $J \subset I$ and $J(n) = I(n)$ for all $n \geq 1$.

We show $I = J$. Assume that there exists $g \in I \setminus J$ such that $\deg(g) = i$ is the smallest. Then $I(i) = J(i)$, there exists $h \in J$ such that $\deg(h) = i$ and $l(g) = l(h)$. As $\deg(g - h) < \deg(g) = i$, we have $g - h \in J$. Hence, $(g - h) + h \in J$, a contradiction ($g \notin J$). Thus $I = J$. ■

Definition. A commutative ring R is called **Artinian** if it satisfies

- (1) Every nonempty set $\mathcal{S}$ of ideals of R has a minimal element in $\mathcal{S}$
- or equivalently,
- (2) R satisfies the descending chain condition (D.C.C.):

$$I_1 \supset I_2 \supset \cdots \implies \exists m \in \mathbb{N} : I_m = I_{m+1} = \cdots$$

Proof. (1) $\implies$ (2). $\{I_1, I_2, \dots\}$ has a minimal element. Hence, there exists $m \in \mathbb{N}$ such that $I_m = I_{m+1} = \cdots$.

(2) $\implies$ (1). Prove by the contrapositive statement. Let $\mathcal{S}$ be a set which has no minimal element. Then, there exists a chain $I_1 \supsetneq I_2 \supsetneq I_3 \supsetneq \cdots$. ■

Examples.

1. Every field is both Noetherian and Artinian as $\{0\}$ and F are all ideals of a field F .
2. $R = \mathbb{Z}$ is Noetherian (as every ideals of R is of the form $n\mathbb{Z}$) but not Artinian as

$$2\mathbb{Z} \supsetneq 4\mathbb{Z} \supsetneq 8\mathbb{Z} \supsetneq \cdots \supsetneq 2^n\mathbb{Z} \supsetneq \cdots$$

3. $R = \mathbb{Z}/n\mathbb{Z}$ is both Noetherian and Artinian as every ideal of R is of the form $m\mathbb{Z}/n\mathbb{Z}$, $m \in \{0, \dots, n-1\}$. This is a P.I.R. but not a domain (R is a domain $\iff n$ is a prime).
4. $R = F[x_1, \dots, x_n]$, where F is a field. By Hilbert's basis theorem, R is Noetherian. (use induction on $R = F[x_1, \dots, x_{n-1}][x_n]$). R is not Artinian as

$$(x_1) \supsetneq (x_1^2) \supsetneq \cdots \supsetneq (x_1^n) \supsetneq \cdots$$

5. $R = F[x_1, x_2, \dots]$ is neither Noetherian nor Artinian.

$$(x_1) \subsetneq (x_1, x_2) \subsetneq (x_1, x_2, x_3) \subsetneq \cdots$$

$$(x_1) \supsetneq (x_1^2) \supsetneq (x_1^3) \supsetneq \cdots$$

Indeed, every Artinian ring is Noetherian. More precisely,

Artinian ring = Noetherian ring such that every prime ideal is maximal.

Lemma 1.6. *An Artinian integral domain is a field.*

Proof. Let $0 \neq a \in R$, where R is an Artinian domain. Consider $(a) \supset (a^2) \supset (a^3) \supset \cdots$. As R is Artinian, there exists some $n \in \mathbb{N}$ such that $(a^n) = (a^{n+1}) = \cdots$. Hence, $a^n = ba^{n+1}$ for some $b \in R$. As R is a domain, $1 = ba$, i.e., $a \in R^\times$. ■

Definition. An integral domain R is called a **unique factorization domain (U.F.D.)** if

1. Every nonzero element x of R can be written as $x = uc_1 \cdots c_n$ for some $n \in \mathbb{N}$, where $u \in R^\times$ and $c_1, \dots, c_n$ are irreducible elements (i.e., every element of R has a factorization).
2. If $x = uc_1 \cdots c_n = vd_1 \cdots d_m$, where $v \in R^\times$ and $d_1, \dots, d_m$ are irreducible elements, then $n = m$ and there exists some $\sigma \in S_n$ such that $d_i = w_i c_{\sigma(i)}$ for all $1 \leq i \leq n$ and $w_i \in R^\times$ (i.e., $d_i \sim c_i$).

Theorem 1.7.

- (1) If R is a U.F.D., then every irreducible element is prime (i.e., if R is a U.F.D., then irreducible $\equiv$ prime).
- (2) Every P.I.D. is a U.F.D.

Proof. (1) Let p be an irreducible element and let $p \mid xy$. Then, $xy = pz$ for some $z \in R$. Write $x = uc_1 \cdots c_n$, $y = vd_1 \cdots d_m$ and $z = we_1 \cdots e_l$, where $u, v, w \in R^\times$ and $c_1, \dots, c_n, d_1, \dots, d_m, e_1, \dots, e_l$ are irreducible elements. Then,

$$(uv) \underbrace{c_1 \cdots c_n \cdot d_1 \cdots d_m}_{\text{irreducible elements}} = w \underbrace{e_1 \cdots e_l \cdot p}_{\text{irreducible elements}}.$$

Hence, $p \sim c_i$ or $p \sim d_j$ for some $1 \leq i \leq n$, $1 \leq j \leq m \Rightarrow p \mid x$ or $p \mid y$.

(2) Let $\mathcal{S} = \{Ra \mid a \neq 0, a \notin R^\times, a \text{ is not a product of irreducible elements}\}$. We show that $\mathcal{S} = \emptyset$. Assume that $\mathcal{S} \neq \emptyset$. As R is a P.I.D., it is Noetherian, thus there exists a maximal element $Rb \in \mathcal{S}$ for some $b \in R$. As b is not irreducible, $b = xy$ for some $x, y \notin R^\times$. Then, $Rb \subsetneq Rx$ and $Rb \subsetneq Ry$. (If $Rb = Rx$, then $x = bz$ for some $z \in R$, thus $b = bzy$, i.e., $1 = zy$, $y \in R^\times$.) By the maximality, $Rx, Ry \notin \mathcal{S}$. Hence, $b = xy = a$ product of irreducible elements, a contradiction, i.e., every nonzero element of R has a factorization.

Assume that $uc_1 \cdots c_n = vd_1 \cdots d_m$ where $u, v \in R^\times$ and $c_1, \dots, c_n, d_1, \dots, d_m$ are irreducible elements. Use the induction on n . As R is a P.I.D., by the theorem (if R is a P.I.D., every irreducible element of R is prime), $c_1 \mid d_i$ for some i . We may assume that $c_1 \mid d_1$ ($i = 1$). As d_1 is irreducible, $c_1 \sim d_1$, i.e., $d_1 = wc_1$ with $w \in R^\times$. Then,

$$uc_1 \cdots c_n = vwc_1 d_2 \cdots d_m \implies u \underbrace{c_2 \cdots c_n}_{n-1 \text{ elements}} = vw \underbrace{d_2 \cdots d_m}_{m-1 \text{ elements}}.$$

By the induction hypothesis, $n - 1 = m - 1$ and $c_i \sim d_i$ up to permutation. ■

06-09-23

Remarks. 1. The proof of the theorem shows that R is a U.F.D. $\Leftrightarrow R$ has a factorization and irreducible $\equiv$ prime.

2. The proof of the theorem shows that if R satisfies the ascending chain condition on principal ideals, then R has a factorization. In particular, every Noetherian ring has a factorization.

3. Combining Remarks 1 and 2,

R is a U.F.D. $\iff R$ satisfies the A.C.C. on principal ideals

and irreducible $\equiv$ prime.

4. A Noetherian domain may not be a U.F.D. For instance, $\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\}$. As $\mathbb{Z}[\sqrt{-5}] \simeq \mathbb{Z}[x]/(x^2 + 5)$ and every quotient of a Noetherian ring $\mathbb{Z}[x]$ is Noetherian, $\mathbb{Z}[\sqrt{-5}]$ is Noetherian. In $\mathbb{Z}[\sqrt{-5}]$, $6 = (1 + \sqrt{-5})(1 - \sqrt{-5}) = 2 \cdot 3$ or $21 = (1 + 2\sqrt{-5})(1 - 2\sqrt{-5}) = 3 \cdot 7$.
5. A U.F.D. need not be Noetherian. For instance, consider a polynomial ring $R = F[x_1, x_2, \dots]$ with infinite variables. As $(x_1) \subsetneq (x_1, x_2) \subsetneq \dots$, R is not Noetherian. But R is a U.F.D. Indeed, if S is a U.F.D., then so is $R = S[x_1, x_2, \dots]$. (Exercise.)

Sketch of proof. We will show that if S is a U.F.D. then $S[x_1, \dots, x_n]$ is a U.F.D. (a theorem which will be proven in next week). By Remark 3, it suffices to check that R satisfies the A.C.C. on principal ideals and every irreducible element is prime. Let $f \in R$. Then, $f \in S[x_{i_1}, \dots, x_{i_n}]$ for some $x_{i_1}, \dots, x_{i_n} \in \{x_1, x_2, \dots\}$. We may assume that $f \in S[x_1, \dots, x_n]$. Consider $Rf \subset Rf_1 \subset Rf_2 \subset \dots$. Then, f_{i+1} is a divisor of f_i (set $f := f_0$). As $S[x_1, \dots, x_n]$ is a U.F.D., it stabilizes. Let $f \in R$ be irreducible. Note that f is prime $\iff R/(f)$ is a domain. Then, $f \in S[x_1, \dots, x_n]$ is irreducible (as $f = gh$, g or $h \in (S[x_1, \dots, x_n])^\times = S^\times = R^\times$)

$$\implies R/(f) = S[x_1, \dots, x_n][x_{n+1}, \dots]/(f) \simeq \underbrace{(S[x_1, \dots, x_n]/(f))}_{\text{domain}}[x_{n+1}, \dots]$$

is a domain. ■

Definition. A polynomial $f \in R[x]$ is **irreducible over R** if it is irreducible in $R[x]$, i.e.,

$$f = gh, \quad g, h \in R[x] \implies g \in (R[x])^\times \text{ or } h \in (R[x])^\times.$$

Examples.

1. x^2 is irreducible over $\mathbb{R}$ but it is not irreducible over $\mathbb{C}$ ($x^2 + 2 = (x + i\sqrt{2})(x - i\sqrt{2})$). $\mathbb{R} \subset \mathbb{C}$ but irreducible $\nRightarrow$ irreducible.
 $3x + 3$ is irreducible over $\mathbb{Q}$ but is not irreducible over $\mathbb{Z}$. $\mathbb{Z} \subset \mathbb{Q}$ but irreducible $\nLeftarrow$ irreducible.
2. If R is a field, then $R[x]$ is a E.D. $\Rightarrow$ P.I.D. $\Rightarrow$ U.F.D.

Lemma 1.8. Let R be a domain and let $c \in R$. Then,

- (1) $(R[x])^\times = R^\times$.
- (2) c is irreducible in R if and only if c is irreducible over R .

Proof. (1) Obviously, $R^\times \subset (R[x])^\times$. Let $f, g \in R[x]$ with $fg = 1$. Then, we have

$$\deg(fg) = \deg(f) + \deg(g) = \deg(1) = 0.$$

Thus, $\deg(f) = \deg(g) = 0$, i.e., $f, g \in R$. Hence, $R^\times \supset (R[x])^\times$.

(2) $(\Rightarrow)$ Let $c = fg$ with $f, g \in R[x]$. Then $f, g \in R$. As c is irreducible, we have that $f \in R^\times = (R[x])^\times$ or $g \in R^\times = (R[x])^\times$.

$(\Leftarrow)$ Let $c = fg$ with $f, g \in R$. As c is irreducible in $R[x]$, $f \in (R[x])^\times = R^\times$ or $g \in (R[x])^\times = R^\times$. ■

Definition. Let R be a U.F.D. and let $0 \neq f \in R[x]$. The **content of f** is the principal ideal generated by all coefficients of f and is denoted by $C(f)$, i.e.,

$$C(f) = \gcd(a_0, \dots, a_n)R, \quad \text{where } f = a_0 + a_1x + \dots + a_nx^n, a_i \neq 0 \in R \text{ for all } i.$$

f is called **primitive** if $C(f) = R$ (i.e., the coefficients of f are relatively prime).

Let a and b be nonzero elements of R . An element $d \in R$ with $d \mid a$ and $d \mid b$ such that any $e \in R$ with $e \mid a$ and $e \mid b$ divides d is a gcd of a and b .

Properties. 1. $C(af) = aC(f)$ for all $0 \neq a \in R$.

2. $C(a) = aR$ for all $0 \neq a \in R$.

3. If f is monic (i.e., $f = 1 \cdot x^n + a_{n-1}x^{n-1} + \dots + a_0$), then $C(f) = 1 \cdot R$ (i.e., f is primitive).

4. $0 \neq a \in R$ is primitive if and only if $a \in R^\times$.

5. For any $0 \neq f \in R[x]$, there exists a primitive $g \in R[x]$ such that $f = a \cdot g$ with $C(f) = aR$.

Lemma 1.9 (Gauss). Let R be a U.F.D. If f and g are primitive, then so is fg .

Proof. Let c be an irreducible element. As R is a U.F.D., c is prime, thus cR is a prime ideal and $\bar{R} := R/cR$ is an integral domain. Hence, $\bar{R}[x]$ is a domain.

The homomorphism $R \rightarrow \bar{R}$ with $a \mapsto \bar{a} = a + cR$, induces a homomorphism $R[x] \rightarrow \bar{R}[x]$ with

$$f = a_0 + a_1x + \dots + a_nx^n \mapsto \bar{f} = \bar{a}_0 + \bar{a}_1x + \dots + \bar{a}_nx^n.$$

Assume that $\bar{f}\bar{g} = \bar{f}\bar{g} = 0$. Then, as $\bar{R}[x]$ is a domain, $\bar{f} = 0$ or $\bar{g} = 0$. If $\bar{f} = 0$, then all coefficients of f are divisible by c . But f is primitive, this cannot happen. Similarly, $\bar{g} \neq 0$. Therefore, $\bar{f}\bar{g} \neq 0$, which implies that at least one of the coefficients of fg is not divisible by c . Hence, fg is primitive. ■

Corollary 1.10. $C(fg) = C(f)C(g)$ for any nonzero polynomials $f, g \in R[x]$.

Proof. Write $f = af_1$ and $g = bg_1$ where f_1 and g_1 are primitive (by property 5), so that $C(f) = aR$ and $C(g) = bR$. Then, $fg = abf_1g_1$. Hence,

$$\begin{aligned} C(fg) &= C(abf_1g_1) \\ &= abC(f_1g_1) && \text{(by property 1)} \\ &= abR && \text{(by Gauss lemma)} \\ &= aR \cdot bR = C(f)C(g). \end{aligned}$$

■

Lemma 1.11. *R : U.F.D., K : quotient fields of R*

- (1) Every nonzero $f \in K[x]$ can be written as $f = ag$, where $a \in K$ and $g \in R[x]$ is primitive.
- (2) Let f be a nonconstant polynomial in $R[x]$. Then, f is irreducible over R if and only if f is irreducible over K .
- (3) Let f, g be two primitive polynomials in $R[x]$. If $f \sim g$ over K , then $f \sim g$ over R .

Proof. (1) Write

$$f = \sum_{i=0}^n \frac{b_i}{a_i} x^i, \quad a_i, b_i \in R.$$

Let $a = a_0 \cdots a_n$. Then, $af \in R[x]$ and $af = dg$ for some primitive $g \in R[x]$ (by property 5). Hence, $f = (d/a)g$ with $d/a \in K$ and $g \in R[x]$.

(2) ($\Rightarrow$) Write $f = ag$, where $a \in R$, g is primitive. As f is irreducible in $R[x]$ and $(R[x])^\times = R^\times$, we have $a \in R^\times$. Hence, $C(f) = C(ag) = aC(g) = R$, i.e., f is primitive. Assume that $f = gh$, where $g, h \in K[x]$ are nonconstant polynomials. By (1), write $g = \frac{a}{b}g_1$, $h = \frac{c}{d}h_1$, where $a, b, c, d \in R$ and g_1, h_1 are primitive. Then, $f = \frac{ac}{bd}g_1h_1 \Rightarrow bdf = acg - 1h_1 \Rightarrow bdC(f) = acC(g_1h_1) = acR \Rightarrow bds = ac$ for some $s \in R \Rightarrow f = (sg_1)h_1$ (nonconstant polynomials in $R[x]$), a contradiction.

($\Leftarrow$) Write $f = gh$, where $g, h \in R[x]$. As f is irreducible in $K[x]$, $g \in K^\times$ or $h \in K^\times$. Assume that $g \in K^\times$. Then, $g \in R[x] \cap K^\times = R$. Hence, $g \in R^\times$. Hence, $g \in R^\times$. As f is primitive, $R = C(f) = C(gh) = gC(h)$. Thus $g \in R^\times$.

(3) Let $f = ag$ for some $a \in K^\times$. Let $a = b/c$ where $b, c \in R$. Then, $cf = bg$, so

$$cR = \underbrace{c C(f)}_{=R} = \underbrace{b C(g)}_{=R} = bR.$$

Hence, $b = cw$ for some $w \in R$, $a = w \in R^\times$. Similarly, $c = bu$ for some $u \in R$. Hence, $b = buw$, i.e., $w \in R^\times$. ■

Theorem 1.12. *If R is a U.F.D., then so is $R[x]$.*

Proof. Existence of factorization. Let $f \in R[x]$. We prove by induction on $n := \deg(f)$. If $n = 0$, then as R has a factorization, we are done. Now let $n \geq 1$. If f is irreducible in $R[x]$, then f has a factorization in $R[x]$. Otherwise, $f = gh$, where g and h are nonconstant polynomials in $R[x]$. (We may assume that f is primitive as every element in R has a factorization, i.e., $f = af_1$, $a \in R$, $f_1 \in R[x]$ primitive.) As $1 \leq \deg(g), \deg(h) \leq n - 1$, by induction hypothesis, g and h have factorizations. So f has a factorization.

Uniqueness. We also use induction on $\deg(f)$. Write $f = ug_1 \cdots g_m = vh_1 \cdots h_n$, where $u, v \in R^\times$, $g_1, \dots, g_m, h_1, \dots, h_n$ are irreducible polynomials in $R[x]$. We show $m = n$ and $g_i \sim h_i$ for each $1 \leq i \leq n$. By Lemma 1.11(2), g_m is irreducible over K . As $K[x]$ is a U.F.D., g_m is prime. Hence, $g_m \mid h_j$ for some j . We may assume that $j = n$. By Lemma 1.11(2), h_n is irreducible over K .

So $g_m \sim h_n$ over K . By Lemma 1.11(3), $g_m \sim h_n$ over R , i.e., $g_m = wh_n$ for some $w \in R^\times$. Hence,

$$\begin{aligned} ug_1 \cdots g_{m-1}g_m &= ug_1 \cdots g_{m-1}(wh_n) = vh_1 \cdots h_{n-1}h_n \\ \implies uwg_1 \cdots g_{m-1} &= vh_1 \cdots h_{n-1}. \end{aligned}$$

By induction, $m - 1 = n - 1$ and $g_i \sim h_i$ for each $1 \leq i \leq n - 1$. ■

Corollary 1.13. *If R is a U.F.D., then so is $R[x_1, \dots, x_n]$.*

Proposition 1.14 (Eisenstein's Criterion). *Let R be a U.F.D. and let $f = a_0 + a_1x + \cdots + a_nx^n \in R[x]$ with $n \geq 1$. If there exists an irreducible (= prime) element c of R such that*

(i) $c \mid a_i$ for all $0 \leq i \leq n - 1$

(ii) $c \nmid a_n$ and $c^2 \nmid a_0$

then f is irreducible over K , where K is the quotient ring of R .

Proof. Write $f = ag$, where g is primitive and $C(f) = aR$. As a is a unit in K , it suffices to check g is irreducible over K . As $c \nmid a_n$, g satisfies conditions (i) and (ii). (If $g = b_nx^n + \cdots + b_0$ then

$$c \nmid b_i \quad \text{for all } 0 \leq i \leq n - 1, \quad c \nmid b_n, \quad c \nmid b_0.)$$

Hence, we may assume that f is primitive. By Lemma 1.11(2), it is irreducible over R . Let $f = gh$, where

$$g = \sum_{i=0}^m b_i x^i \quad \text{and} \quad h = \sum_{i=0}^k d_i x^i, \quad m \geq 1, \quad k \geq 1.$$

As $c \mid a_0 = b_0d_0$, $c \mid b_0$ or $c \mid d_0$. Say $c \mid b_0$. As $c^2 \nmid a_0$, $c \nmid d_0$. Let l be the smallest integer such that $c \mid b_l$ for all $i < l$ and $c \nmid b_l$. (Since f is primitive, such l exists.) Consider

$$a_l = \sum_{i=0}^l b_i d_{l-i} = \underbrace{b_0 d_l + b_1 d_{l-1} + \cdots + b_{l-1} d_1}_{\text{divisible by } c} + b_l d_0.$$

As $c \mid a_l$, $c \mid b_l d_0$, which contradicts to $c \nmid b_l$, $c \nmid d_0$. Hence, f is irreducible over R . ■

Examples.

1. $f = 2x^4 - 6x^3 + 18x + 15 \in \mathbb{Z}[x]$. By the proposition with $c = 3$, f is irreducible over $\mathbb{Q}$ and $\mathbb{Z}$.
2. $f = y^3 - x^2y^2 + x^3y + x \in \mathbb{Z}[x, y] = \mathbb{Z}[x][y]$. By the proposition with $c = x$ (x is an irreducible element of $R = \mathbb{Z}[x]$), f is irreducible over $\mathbb{Q}(x)$ (in $\mathbb{Q}(x)[y]$). As f is primitive, f is also irreducible in $\mathbb{Z}[x, y]$.
3. $f = \frac{x^p-1}{x-1} \in \mathbb{Z}[x]$, p a prime. Let $y = x - 1$. Then,

$$f = \binom{p}{1} + \binom{p}{2}y + \cdots + \binom{p}{p}y^{p-1}.$$

By the proposition with $c = p$, f is irreducible over $\mathbb{Z}$ and $\mathbb{Q}$.

Part II

Field Theory

2 Field Extensions

Definition. Let F be a field. Let E be a field containing F as a subfield. In this case, we say that E is a **field extension of F** and F is called the base field. We write E/F for the field extension of E over F .

Observe that E is a vector space over F by restriction of scalars:

$$a + b \in E \text{ where } a, b \in E, \quad k \cdot a \in E \text{ where } k \in F \subset E.$$

Definition. Let E/F be a field extension. The dimension of E over F (as a vector space) is called the **degree of E/F** and is denoted by $[E : F]$ (i.e., $\dim_F E =: [E : F]$). We call E/F a **finite (field) extension** if $[E : F] < \infty$, and an **infinite extension** if $[E : F] = \infty$.

Examples.

1. Let F be a field. Consider the field extension F/F . Then we have $[F : F] = 1$ and $\{1\}$ is a basis of F/F .
2. Consider $\mathbb{C}/\mathbb{R}$. Then we have $[\mathbb{C} : \mathbb{R}] = 2$ and $\{1, i\}$ a basis of $\mathbb{C}/\mathbb{R}$.
3. $\mathbb{R}/\mathbb{Q}$ is an infinite extension as $\mathbb{Q}$ is countable but $\mathbb{R}$ is uncountable.
4. F a field,

$$F(x) = \left\{ \frac{f(x)}{g(x)} \mid f, g \in F[x], g \neq 0 \right\}$$

$$= \text{quotient field of } F[x].$$

As $(F[x])^\times = F^\times$, we have that $F[x] \subsetneq F(x)$ (because $1/x \in F(x)$, $1/x \notin F[x]$). As $\{1, x, x^2, \dots\}$ is a basis of $F[x]$ over F , $\dim_F F[x] = \infty$. Hence, $[F(x) : F] = \infty$.

Proposition 2.1. Let $E/L/F$ be field extensions. Then

$$[E : F] = [E : L][L : F].$$

Proof. Let $[L : F] = m$ and $[E : L] = n$. Let $\{x_1, \dots, x_m\}$ be a basis of L over F and let $\{y_1, \dots, y_n\}$ be a basis of E over L .

We show that $\{x_i y_j \mid 1 \leq i \leq m, 1 \leq j \leq n\}$ is a basis of E/F . Let $z \in E$. Then

$$z = \sum_{j=1}^n u_j y_j \quad \text{and} \quad u_j = \sum_{i=1}^m v_{ij} x_i$$

for some $u_j \in L$ and $v_{ij} \in F$. Hence,

$$z = \sum_{i=1}^m \sum_{j=1}^n v_{ij} x_i y_j,$$

thus $\{x_i y_j\}$ spans E .

Assume

$$0 = \sum_{i=1}^m \sum_{j=1}^n v_{ij} x_i y_j = \sum_{j=1}^n \left(\sum_{i=1}^m v_{ij} x_i \right) y_j.$$

As $\{y_j\}$ is linearly independent,

$$\sum_{i=1}^m v_{ij} x_i = 0 \quad \text{for all } j.$$

As $\{x_i\}$ is linearly independent, $v_{ij} = 0$ for all i, j . Therefore, $\{x_i y_j\}$ is a basis of E over F . Hence, $[E : F] = [E : L][L : F]$. In particular, if both $[E : L]$ and $[L : F]$ are finite, then so is $[E : F]$. If one of $[E : L]$ and $[L : F]$ is infinite, then so is $[E : F]$. ■

Corollary 2.2.

(1) Let $F_1/F_2/\dots/F_n$ be field extensions. Then,

$$[F_1 : F_n] = [F_1 : F_2][F_2 : F_3] \cdots [F_{n-1} : F_n].$$

(2) Let $E/L/F$ be finite extensions. Then

$$[E : L] \mid [E : F] \quad \text{and} \quad [L : F] \mid [E : F].$$

Definition.

1. Let F be a field and let S be a subset of F . The intersection of all subfields of F that contain S is called the **subfield generated by S** , i.e.,

$$\text{the subfield generated by } S = \bigcap_{\substack{S \subset L \subset F \\ L \text{ a field}}} L.$$

2. Let E/F be a field extension and let S be a subset of E . The subfield (of E) generated by $F \cup S$ is denoted by $F(S)$. When $S = \{\alpha_1, \dots, \alpha_n\}$, we write $F(\alpha_1, \dots, \alpha_n)$ for $F(S)$. $F \subset F(S) \subset E$ and $F(S)$ is the smallest field between E and F containing S .

Proposition 2.3. Let E/F be a field extension and let $S = \{\alpha_1, \dots, \alpha_n\}$ be a subset of E . Then,

$$F(\alpha_1, \dots, \alpha_n) = \left\{ \frac{f(\alpha_1, \dots, \alpha_n)}{g(\alpha_1, \dots, \alpha_n)} \mid f, g(\alpha_1, \dots, \alpha_n) \neq 0; f, g \in F[x_1, \dots, x_n] \right\}.$$

Proof. Let L be the right-hand side. As $F, \{\alpha_1, \dots, \alpha_n\} = S \subset F(\alpha_1, \dots, \alpha_n)$,

$$\frac{f(\alpha_1, \dots, \alpha_n)}{g(\alpha_1, \dots, \alpha_n)} \in F(\alpha_1, \dots, \alpha_n).$$

hence, L is a subfield of $F(\alpha_1, \dots, \alpha_n)$.

For any $a \in F$, $a = a/1 \in L$, thus $F \subset L$. As $\alpha_i = \alpha_i/1$, $\alpha_i \in L$, thus $S \subset L$. Since $F(\alpha_1, \dots, \alpha_n)$ is the smallest subfield, $F(\alpha_1, \dots, \alpha_n) \subset L$. ■

Example.

Consider $\mathbb{C}/\mathbb{R}$. Then,

$$\mathbb{C} = \mathbb{R}(i) = \left\{ \frac{a+bi}{c+di} \mid a, b, c, d \in \mathbb{R}, c^2 + d^2 \neq 0 \right\} = \{u + vi \mid u, v \in \mathbb{R}\}.$$

Definition. Let E/F be a field extension and let $S \subset E$ be a subset. The intersection of all subrings of E that contain S is the **subring generated by S** . The subring generated by $F \cup S$ is denoted by $F[S]$. We write $F[\alpha_1, \dots, \alpha_n]$ for $F[\{\alpha_1, \dots, \alpha_n\}]$.

Proposition 2.4. Let E/F be a field extension and let $S = \{\alpha_1, \dots, \alpha_n\} \subset E$. Then,

$$F[S] = \{f(\alpha_1, \dots, \alpha_n) \mid f \in F[x_1, \dots, x_n]\}.$$

The proof is similar to Proposition 2.3.

Definition. Let E/F be a field extension. An element $\alpha \in E$ is called **algebraic over F** if there exists $0 \neq f \in F[x]$ such that $f(\alpha) = 0$. If an element $\alpha \in E$ is not algebraic, α is called **transcendental over F** .

Examples.

1. $i \in \mathbb{C}$ is algebraic over $\mathbb{R}$ as i is a root of $x^2 + 1 \in \mathbb{R}[x]$.
2. $x \in F(x) = (\text{quotient field of } F[x])$ is transcendental over F .
(If x is algebraic over F , then there exists $0 \neq a_0 + \dots + a_n x^n$ with $a_0 + \dots + a_n x^n = 0$, then $a_i = 0$ for all i .)
3. $e = \lim_{n \rightarrow \infty} (1 + 1/n)^n$ (Euler's number) and π are transcendental over $\mathbb{Q}$.

Indeed, Lindemann-Weierstrass theorem says that

if $\alpha_1, \dots, \alpha_n \in \mathbb{C}$ are algebraic over $\mathbb{Q}$ which are linearly independent over $\mathbb{Q}$, then $e^{\alpha_1}, \dots, e^{\alpha_n}$ are algebraically independent over $\mathbb{Q}$.

$\beta_1, \dots, \beta_n$ are **algebraically independent over F** if there are no polynomials $f \in F[x_1, \dots, x_n]$ with $f(\beta_1, \dots, \beta_n) = 0$.

α is transcendental over F if and only if α is algebraically independent over F .

Let $0 \neq \alpha \in \mathbb{C}$ be algebraic over $\mathbb{Q}$. By Lindemann-Weierstrass theorem, e^α is transcendental over $\mathbb{Q}$.

If π is algebraic over $\mathbb{Q}$, then πi is algebraic over $\mathbb{Q}$ (as i is algebraic over $\mathbb{Q}$ and the product of two algebraic elements is algebraic¹). Thus, by Lindemann-Weierstrass theorem, $e^{\pi i} = -1$ is transcendental over $\mathbb{Q}$, a contradiction. Hence, π is transcendental over $\mathbb{Q}$.

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Theorem 2.5. Let E/F be a field extension and let $\alpha \in E$. Assume that α is algebraic over F . Then,

- (1) There uniquely exists the monic irreducible polynomial $m_\alpha \in F[x]$ with $m_\alpha(\alpha) = 0$. If $\varphi: F[x] \rightarrow E$ with $f \mapsto f(\alpha)$, then $\ker \varphi = (m_\alpha)$.
- (2) If $f(\alpha) = 0$ for some $f \in F[x]$, then $m_\alpha \mid f$.
- (3) $\{1, \alpha, \alpha^2, \dots, \alpha^{\deg(\alpha)-1}\}$ is a basis of $F[\alpha]$ over F and $F[\alpha] = F(\alpha)$.

Here, $m_\alpha = m_{\alpha,F}$ is called the **minimal polynomial** of α . We define $\deg(\alpha) = \deg(m_\alpha)$.

Proof. Let $\varphi: F[x] \rightarrow E$ be a ring homomorphism defined as $f \mapsto f(\alpha)$.

(1) Observe that $\text{Im } \varphi = F[\alpha] = \{f(\alpha) \mid f \in F[x]\}$. Moreover, $\ker \varphi$ is a nonzero ideal (as α is an algebraic element). As $F[x]$ is a P.I.D., $\ker \varphi = (m_\alpha)$, where m_α is a unique monic polynomial in $F[x]$ with $m_\alpha(\alpha) = 0$. Since $F[x]/(m_\alpha) \simeq F[\alpha]$ and $F[\alpha]$ is a domain, (m_α) is a prime ideal.

As $F[x]$ is a P.I.D., (m_α) is maximal. Hence, $F[\alpha]$ is a field. Since $F(\alpha)$ is the smallest field containing α , we get $F(\alpha) = F[\alpha]$. As $F[x]$ is a P.I.D. and m_α is prime, m_α is irreducible.

(2) Observe that $f \in \ker \varphi = (m_\alpha)$. Hence, $m_\alpha \mid f$.

(3) Let $n = \deg(\alpha) = \deg(m_\alpha)$. Assume

$$a_0 + a_1\alpha + a_2\alpha^2 + \dots + a_{n-1}\alpha^{n-1} = 0$$

for some $a_0, \dots, a_{n-1} \in F$. Let

$$g = a_0 + a_1x + a_2x^2 + \dots + a_{n-1}x^{n-1}.$$

Then, $g \in \ker \varphi$. Thus, $m_\alpha \mid g$. As $\deg g < \deg m_\alpha = n$, we get $g = 0$, i.e., $a_i = 0$ for all $0 \leq i \leq n-1$, which implies that $\{1, \alpha, \dots, \alpha^{n-1}\}$ is linearly independent over F .

Let $\beta \in F[\alpha]$ i.e., $\beta = h(\alpha)$ for some $h \in F[x]$. We divide h by m_α . Then, $h = m_\alpha q + r$ with $\deg r < \deg m_\alpha = n$ for some $q, r \in F[x]$. Let $r = b_0 + b_1x + \dots + b_{n-1}x^{n-1}$ for some $b_i \in F$. Then,

$$\beta = h(\alpha) = \underbrace{m_\alpha(\alpha) \cdot q(\alpha)}_{=0} + r(\alpha) = r(\alpha).$$

Hence $\{1, \alpha, \dots, \alpha^{n-1}\}$ spans $F[\alpha]$. ■

Examples.

1. $\sqrt{5} \in \mathbb{R}$ is algebraic over $\mathbb{Q}$ as it is a root of $x^2 - 5 \in \mathbb{Q}[x]$ (that is, $m_{\sqrt{5}} = x^2 - 5$) and

¹Refer to Corollary 2.6(4).

$\mathbb{Q}[\sqrt{5}] = \mathbb{Q}(\sqrt{5}) = \{a + b\sqrt{5} \mid a, b \in \mathbb{Q}\}$. $\{1, \sqrt{5}\}$ is a basis of $\mathbb{Q}[\sqrt{5}]$ over $\mathbb{Q}$. Thus $[\mathbb{Q}(\sqrt{5}) : \mathbb{Q}] = 2 = \deg m_{\sqrt{5}}$.

2. $\mathbb{C} = \mathbb{R}[i] = \mathbb{R}(i)$, i is algebraic over $\mathbb{R}$. We have $m_i = x^2 + 1$ and $[\mathbb{C} : \mathbb{R}] = 2 = \deg(m_i)$. Thus $\{1, i\}$ is a basis of $\mathbb{C}/\mathbb{R}$. $\mathbb{C} = \mathbb{R}[x]/(x^2 + 1)$.

3. Let p be a prime integer. Let

$$\rho = \cos \frac{2\pi}{p} + i \sin \frac{2\pi}{p}$$

(i.e., ρ is a primitive p th root of 1). Then $\rho^p = 1$, i.e., ρ is a root of

$$x^p - 1 = (x - 1) \underbrace{(x^{p-1} + x^{p-2} + \cdots + 1)}_{\rho \text{ is a root of this polynomial}}.$$

By Eisenstein's criterion, $x^{p-1} + x^{p-2} + \cdots + 1$ is irreducible over $\mathbb{Q}$, so $m_\rho = x^{p-1} + \cdots + 1$. We conclude that $[\mathbb{Q}(\rho) : \mathbb{Q}] = p - 1$.

Let E/F be a field extension. We say that E/F is **algebraic** if every element of E is algebraic.

Corollary 2.6.

- (1) An element $\alpha \in E$ is algebraic over F if and only if $[F(\alpha) : F] < \infty$.
- (2) Every finite extension is algebraic.
- (3) Let $\alpha_1, \dots, \alpha_n \in E$ be algebraic over F . Then, $F(\alpha_1, \dots, \alpha_n) = F[\alpha_1, \dots, \alpha_n]$ is a finite extension over F .
- (4) Let α and β be algebraic element of E over F . Then, $\alpha\beta, \alpha \pm \beta, \alpha^{-1}$ are algebraic over F . Hence, the set of elements that are algebraic over F forms a subfield of E over F .

Proof. (1) $(\Rightarrow)$ By Theorem 2.5, $[F(\alpha) : F] = \deg(m_\alpha) < \infty$.

$(\Leftarrow)$ Let $n = [F(\alpha) : F]$. Then $\{1, \alpha, \dots, \alpha^n\}$ is linearly dependent over F . Thus

$$b_0 + b_1\alpha + b_2\alpha^2 + \cdots + b_n\alpha^n = 0, \quad b_i \in F \text{ not all zero.}$$

Set $g = b_0 + b_1x + \cdots + b_nx^n$. Then, $g(\alpha) = 0$, i.e., α is algebraic over F .

(2) Let $n = [E : F]$ and let $\alpha \in E$. Then, $[F(\alpha) : F] \leq [E : F] = n$. Hence, by (1), α is algebraic over n .

(3) We prove this by induction on n . If $n = 1$, then it follows from Theorem 2.5. Now observe that

$$\begin{aligned} F(\alpha_1, \dots, \alpha_n) &= F(\alpha_1, \dots, \alpha_{n-1})(\alpha_n) \\ &= F(\alpha_1, \dots, \alpha_{n-1})[\alpha_n] && \text{(by Theorem 2.5)} \\ &= F[\alpha_1, \dots, \alpha_{n-1}][\alpha_n] && \text{(by induction hypothesis)} \\ &= F[\alpha_1, \dots, \alpha_n]. \end{aligned}$$

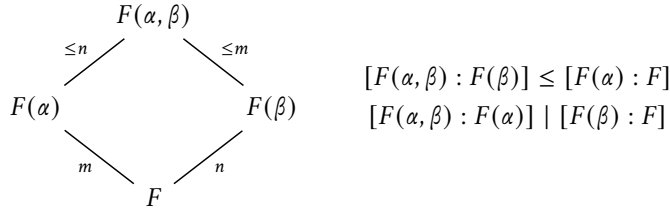
To consider the degree, $[F(\alpha_1, \dots, \alpha_n) : F(\alpha_1, \dots, \alpha_{n-1})]$ is finite by Theorem 2.5, and $[F(\alpha_1, \dots, \alpha_{n-1}) : F]$ is finite by the induction hypothesis. Therefore, $F(\alpha_1, \dots, \alpha_n)/F$ is finite.

(4) Let $\gamma = \alpha\beta, \alpha \pm \beta, \alpha^{-1}$. Then, $F \subset F(\gamma) \subset F(\alpha, \beta)$. By (3), we have $[F(\alpha, \beta) : F] < \infty$. By (1), γ is algebraic over F . ■

Remark. (1) Let $E = \{\alpha \in \mathbb{C} \mid \alpha \text{ is algebraic over } \mathbb{Q}\}$. Then, $E/\mathbb{Q}$ is a field extension. By Eisenstein's criterion, $x^n - p$ is irreducible over $\mathbb{Q}$ for any $n \geq 1$ and any prime p .

Let α be a root of $x^n - p$. Then, $m_\alpha = x^n - p$ and $[\mathbb{Q}(\alpha) : \mathbb{Q}] = n$. As n is arbitrary, $\infty = [E : \mathbb{Q}] \geq [\mathbb{Q}(\alpha) : \mathbb{Q}] = n$, i.e., not every algebraic extension is finite.

(2) Consider the following diagram:



Let $m_\alpha \in F[x]$ be the minimal polynomial of α over F . Then $m_\alpha \in F(\beta)[x]$. Hence, the minimal polynomial of α over $F(\beta)$ divides m_α .

Proposition 2.7. Let E/F be a field extension and let $\alpha \in E$. Then, the following statements are all equivalent:

- (1) α is transcendental over F .
- (2) $[F(\alpha) : F] = \infty$.
- (3) $F[\alpha] \simeq F[x]$.
- (4) $F[\alpha] \subsetneq F(\alpha)$.

Proof. (1) $\Leftrightarrow$ (2). It follows from Corollary 2.6(1).

(1) $\Rightarrow$ (3). Let $\varphi: F[x] \rightarrow F[\alpha]$ defined by $f \mapsto f(\alpha)$. As α is transcendental over F , $\ker \varphi = 0$. Hence, $F[x] \simeq F[\alpha]$.

(3) $\Rightarrow$ (4). As $F[x] \subsetneq F(\alpha)$ and $F[x] \simeq F[\alpha]$, we have $F[\alpha] \subsetneq F(\alpha)$.

(4) $\Rightarrow$ (1). If α is algebraic over F , then $F[\alpha] = F(\alpha)$ by Theorem 2.5(3). Hence, the statement follows by taking the contrapositive. ■

Proposition 2.8. Let L/E and E/F be field extensions. Then, L/F is algebraic if and only if both E/F and L/E are algebraic.

Proof. ($\Rightarrow$) L/E is algebraic as $f \in F[x]$ can be viewed as a polynomial in $E[x]$. E/F is algebraic as any element $\alpha \in E$ can be viewed as an element of L .

($\Leftarrow$) Let $\alpha \in L$. As L/E is algebraic, there exists a polynomial

$$f = b_n x^n + \cdots + b_0 \in E[x]$$

with $f(\alpha) = 0$. Let $K = F(b_0, b_1, \dots, b_n)$. As E/F is algebraic, b_i is algebraic over F for each i . Then, by the Corollary 2.6(1), we get that $[K : F] < \infty$. As α is algebraic over K , by the same corollary we have $[K(\alpha) : K] < \infty$. Hence, we have that

$$[K(\alpha) : F] = [K(\alpha) : K][K : F] < \infty,$$

which implies that α is algebraic over F . ■

3 Splitting Field

Definition. Let E/F be a field extension and let $f \in F[x]$. We say that f **splits over** E if $f = a(x - \alpha_1) \cdots (x - \alpha_n)$ for some $\alpha_i \in E$, $a \in F$ (i.e., f factors into a product of linear polynomials in $E[x]$). We say that E is a **splitting field of f** if f splits over E and $F = F(\alpha_1, \dots, \alpha_n)$, where $\alpha_1, \dots, \alpha_n$ are roots of f in E (i.e., the splitting field of f is the smallest field over which f splits).

Lemma 3.1 (Kronecker). Let $f \in F[x]$ be a non-constant polynomial, where F is a field. Then, there exists a field extension E/F such that f has a root in E and $[E : F] \leq \deg(f)$.

Proof. Let $p(x)$ be a monic irreducible factor of f . Let $E = F[x]/(p(x))$. As $p(x)$ is irreducible and $F[x]$ is a P.I.D., $(p(x))$ is maximal, thus E is a field. Consider

$$\begin{aligned} \varphi: F &\hookrightarrow F[x] \longrightarrow F[x]/(p(x)) = E \\ a &\longmapsto a \longmapsto \bar{a} = a + (p(x)). \end{aligned}$$

Then, $\ker \varphi = 0$ (because if $a = p(x)q(x)$ then $\bar{a} = 0$). Thus, φ is injective.

We can identify F with $\varphi(F)$ ($F \simeq \varphi(F)$). Then, E/F is a field extension. Let $\alpha = \bar{x} = x + (p(x)) \in E$. Write

$$p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0, \quad a_i \in F.$$

Let

$$\begin{aligned} \varphi(p(x)) &= x^n + \varphi(a_{n-1})x^{n-1} + \cdots + \varphi(a_0) \\ &= x^n + \overline{a_{n-1}}x^{n-1} + \cdots + \overline{a_0} \in \varphi(F)[x]. \end{aligned}$$

Then,

$$\begin{aligned} \varphi(p(x))(\alpha) &= \alpha^n + \overline{a_{n-1}}\alpha^{n-1} + \cdots + \overline{a_0} \\ &= \bar{x}^n + \overline{a_{n-1}}\bar{x}^{n-1} + \cdots + \overline{a_0} \\ &= (x^n + a_{n-1}x^{n-1} + \cdots + a_0) + (p(x)) = (p(x)), \end{aligned}$$

i.e., $\varphi(p)(\alpha) = 0$ in E , so α is a root of $\varphi(p)$. Under the identification $F \leftrightarrow \varphi(F)$, we have $E = F(\alpha)$ and

$$[E : F] = \deg \alpha = \deg(p) \leq \deg(f) = n. \quad \blacksquare$$

Remark. There exists some K such that

$$\begin{array}{ccc} K & \xrightarrow{\sim} & E \\ | & & | \\ F & \xrightarrow{\sim} & \varphi(F) \end{array}$$

Indeed, $K = F \cup (E - \varphi(F))$ is a disjoint union as sets. Then, K is bijective to E . We can give a natural field structure on K using the bijection.

Theorem 3.2. Let f be a non-constant polynomial in $F[x]$. Then there exists a splitting field E of f over F such that $[E : F] \leq (\deg f)!$.

Proof. Use induction on $n = \deg f$. If $n = 1$, then take $E = F$. Now consider $n > 1$. By Lemma 3.1, there exists L/F such that $[L : F] \leq n$ and $f(\alpha) = 0$ for some $\alpha \in L$. Then, $f(x) = (x - \alpha)g(x)$ for some $g \in L[x]$. As $\deg g = n - 1$, by induction hypothesis there exists a splitting field E of g over L such that $[E : L] \leq (\deg g)! = (n - 1)!$. Hence, E is a splitting field of f over F ($\alpha \in L \subset E$) and $[E : F] = [E : L][L : F] \leq (n - 1)!n = n!$. ■

Examples.

1. $f(x) = (x^2 - p)(x^2 - q) \in \mathbb{Q}[x]$, $p \neq q$ are primes. Then $\mathbb{Q}(\sqrt{p}, \sqrt{q})$ is a splitting field of f over $\mathbb{Q}$. We have $[\mathbb{Q}(\sqrt{p}, \sqrt{q}) : \mathbb{Q}] = 4$ as $\{1, \sqrt{p}, \sqrt{q}, \sqrt{pq}\}$ is a basis over $\mathbb{Q}$.

Note that $[\mathbb{Q}(\sqrt{p}, \sqrt{q}) : \mathbb{Q}] = 4 < 4!$. Hence, $n!$ in the theorem is not optimal.

2. $f(x) = (x^p - 1)/(x - 1) = x^{p-1} + \cdots + 1 \in \mathbb{Q}[x]$. Let $\eta = \cos(2\pi/p) + i \sin(2\pi/p)$. Then, $\mathbb{Q}(\eta)$ is a splitting field of f over $\mathbb{Q}$ as η^i ($1 \leq i \leq p - 1$) are roots of f and $[\mathbb{Q}(\eta) : \mathbb{Q}] = p - 1$.

Theorem 3.3. Let E/F be a splitting field of $f \in F[x]$ and let L/K be a splitting field of $\varphi(f) \in K[x]$, where $\varphi: F \rightarrow K$ is an isomorphism of fields.

$$f = a_n x^n + \cdots + a_0 \in F[x] \implies \varphi(f) = \varphi(a_n)x^n + \cdots + \varphi(a_0) \in K[x].$$

Then, there exists an isomorphism $\psi: E \rightarrow L$ which extends φ ($\psi|_F = \varphi$).

$$\begin{array}{ccc} E & \xrightarrow{\psi} & L \\ \downarrow & & \downarrow \\ F & \xrightarrow[\varphi]{\sim} & K \end{array}$$

Corollary 3.4. Let E and E' be splitting fields of f over F . Then, there exists an F -isomorphism $\psi: E \rightarrow E'$ (i.e., $\psi(a) = a$ for all $a \in F$). In particular, there exists a unique splitting field of f over F up to isomorphisms.

Proof. Take $\varphi = \text{id}$ with $K = F$ in the previous theorem. ■

Lemma 3.5. Let $E = F(\alpha)$ be a finite extension of F and let L/K be a field extension. Let $f = m_\alpha$ and let $\varphi: F \rightarrow K$ be a field homomorphism. Then,

- (1) For any root β of $\varphi(f)$ in L , there exists an extension $\psi: E \rightarrow L$ of φ with $\psi(\alpha) = \beta$ (i.e., $\psi|_F = \varphi$).
- (2) If $\psi: E \rightarrow L$ is a field extension of φ , then $\psi(\alpha)$ is a root of $\varphi(f)$.

Proof. (1) Let $\theta: F[x] \rightarrow L$ be defined as $g \mapsto \varphi(g)(\beta)$. As β is a root of $\varphi(f)$, we have $(f) \subseteq \ker \theta$, thus we have a homomorphism $\psi: F[x]/(f) \rightarrow L$ such that $\theta = \psi \circ \pi$ where $\pi: F[x] \rightarrow F[x]/(f)$.

$$\left. \begin{array}{l} \theta: R \rightarrow S \\ I \subset \ker \theta \end{array} \right\} \implies \begin{array}{ccc} R & \xrightarrow{\theta} & S \\ & \searrow \pi & \uparrow \tilde{\theta} \\ & & R/I \end{array}$$

As $F[x]/(f) \simeq F(\alpha) = E$, $x + (f) \leftrightarrow \alpha$. Thus

$$\psi(\alpha) = \psi(x + (f)) = \theta(x) = \beta.$$

(2) $f(\alpha) = 0 \implies \psi(f)(\psi(\alpha)) = \psi(f(\alpha)) = \psi(0) = 0$, i.e., $\psi(f)(\psi(\alpha)) = 0$. Then $\varphi(f)(\psi(\alpha)) = 0$ (as $\psi(f) = \varphi(f)$ and $f \in F[x]$). ■

Proof of Theorem 3.3. We prove this by induction on $n = \deg f$. If $n = 1$, then we have $E = F$ and $L = K$. If we let $\psi = \varphi$, we are done.

Now consider $n > 1$. Let α be a root of f in E . Then, $m_\alpha \mid f$ thus $\varphi(m_\alpha) \mid \varphi(f)$. As $\varphi(f)$ splits over L , $\varphi(m_\alpha)$ splits over L . Let β be a root of $\varphi(m_\alpha)$ in L . By Lemma 3.5, there exists a homomorphism $\theta: F(\alpha) \rightarrow K(\beta)$ with $\theta(\alpha) = \beta$. As φ is an isomorphism, so is θ . As $f(\alpha) = 0$, $f = (x - \alpha)g(x)$ for some $g \in F(\alpha)[x]$. Then, $\theta(f) = (x - \beta)\theta(g)$ with $\theta(g) \in K(\beta)[x]$. As $\deg g = n - 1$, E and L are splitting fields of g and $\theta(g)$, respectively. By the induction hypothesis, there exists an isomorphism $\psi: E \rightarrow L$ with $\psi|_{F(\alpha)} = \theta$. Thus, $\psi|_F = \varphi$ ($\theta|_F = \varphi$). ■

$$\begin{array}{ccc} E & \xrightarrow{\psi} & L \\ \downarrow & & \downarrow \\ F(\alpha) & \xrightarrow[\theta]{\sim} & K(\beta) \\ \downarrow & & \\ F & \xrightarrow{\sim} & K \end{array} \quad \begin{array}{l} (\psi \text{ exists by induction}) \\ (\theta \text{ exists by Lemma 3.5}) \end{array}$$

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Lemma 3.6. For any root $\beta \in L$ of $\varphi(m_\alpha)$, there exists $\psi: E \rightarrow L$, $\alpha \mapsto \beta$ such that $\psi|_F = \varphi$.

$$\begin{array}{ccc} E = F(\alpha) & \xrightarrow{\psi} & L \\ \downarrow & & \downarrow \\ F & \xrightarrow{\varphi} & K \end{array}$$

Lemma 3.7. Let E/F be a finite extension and let $\varphi: F \rightarrow K$ be a field homomorphism. Then, there exists a field extension L/K and there exists a field homomorphism $\psi: E \rightarrow L$ with $\psi|_F = \varphi$ (i.e., ψ is an extension of φ).

Proof. As E/F is finite, E is finitely generated over F , so $E = F(\alpha_1, \dots, \alpha_n)$ for some $\alpha_i \in E$. As every finite extension is algebraic, every α_i is algebraic over F .

We prove by induction on n . If $n = 1$, then let L be a splitting field of $\varphi(m_{\alpha_1})$. Then, by the previous lemma, there exists $\psi: F(\alpha_1) \rightarrow L$ with $\psi|_F = \varphi$.

If $n > 1$, as $E = F(\alpha_1)(\alpha_2, \dots, \alpha_n)$, by induction hypothesis, there exists L'/L and there exists $\psi': E \rightarrow L'$ with $\psi'|_{F(\alpha_1)} = \psi$.

$$\begin{array}{ccc} E = F(\alpha_1)(\alpha_2, \dots, \alpha_n) & \xrightarrow{\psi'} & L' \\ | & & | \\ F(\alpha_1) & \xrightarrow{\psi} & L \\ | & & | \\ F & \xrightarrow{\varphi} & K \end{array}$$

Proposition 3.8. Let E/F be a finite extension. Then, the following statements are all equivalent.

- (1) E is a splitting field of some polynomial over F .
- (2) For any extension L/E and any homomorphism $\varphi: E \rightarrow L$ with $\varphi|_F = \text{id}_F$, we have $\varphi(E) = E$.
- (3) Every irreducible polynomial $f \in F[x]$ which has a root in E splits over E .

Definition. A finite extension E/F is called **normal** if it satisfies (1) $\sim$ (2) $\sim$ (3) in the proposition.

Proof. (1) $\Rightarrow$ (2). Let E be a splitting field of $f \in F[x]$. Then, $E = F(\alpha_1, \dots, \alpha_n)$, where $\alpha_1, \dots, \alpha_n$ are roots of f . As $f(\alpha_i) = 0$ and $\varphi(f) = f$, we have

$$0 = \varphi(f)(\varphi(\alpha_i)) = f(\varphi(\alpha_i)),$$

thus $\varphi(\alpha_i) = \alpha_j$ for some j . Hence, $\varphi(E) \subset E$. As φ is injective, $[\varphi(E) : F] = [E : F]$, so $\varphi(E) = E$.

Let $\{\beta_1, \dots, \beta_m\}$ be a basis of E/F . Then $\{\varphi(\beta_1), \dots, \varphi(\beta_m)\}$ is a basis of $\varphi(E)/F$, i.e.,

$$\begin{aligned} 0 &= c_1\varphi(\beta_1) + \dots + c_m\varphi(\beta_m) \\ &= \varphi(c_1\beta_1 + \dots + c_m\beta_m) = \varphi(0). \end{aligned}$$

(2) $\Rightarrow$ (3). Let f be an irreducible polynomial with $f(\alpha) = 0$, $\alpha \in E$. Let L be a splitting field of f over E . Let β be a root of f in L . By Lemma 3.5, there exists an extension $\varphi: F(\alpha) \rightarrow L$ with $\varphi(\alpha) = \beta$. By Lemma 3.7, there exists a field extension K/L and an extension $\psi: E \rightarrow K$ with $\psi|_{F(\alpha)} = \varphi$. By the assumption (2), $\psi(E) = E$. Hence, $\psi(\alpha) = \varphi(\alpha) = \beta \in E$.

$$\begin{array}{ccc} E & \xrightarrow{\psi} & K \\ | & & | \\ F(\alpha) & \xrightarrow{\varphi} & L \\ | & & | \\ F & \xrightarrow{\text{id}} & F \end{array} \quad \begin{array}{l} \text{(by Lemma 3.7)} \\ \\ \text{(by Lemma 3.5)} \end{array}$$

(3) $\Rightarrow$ (1). As E/F is finite, $E = F(\beta_1, \dots, \beta_n)$ for some $\beta_i \in E$. Let $f_i = m_{\beta_i}$. As f_i is irreducible and $f_i(\beta_i) = 0$, by assumption (3), every f_i splits over E . Hence, E is a splitting field of $f := f_1 f_2 \cdots f_n$. ■

Corollary 3.9. *Let $E = F(\alpha_1, \dots, \alpha_n)/F$ be a finite extension. Then, E/F is normal if and only if every m_{α_i} , $1 \leq i \leq n$, splits over E .*

Proof. ($\Rightarrow$) It follows from Proposition 3.8(3).

($\Leftarrow$) Let $f = \prod_{i=1}^n m_{\alpha_i}$. Then, f splits over E . As E is generated by the roots of f , E is a splitting field of f , thus by Proposition 3.8(1) E/F is normal. ■

Examples.

1. Let E/F be a field extension with $[E : F] = 2$. Then E/F is normal.

Let $\alpha \in E \setminus F$. Then, $E = F(\alpha)$ and $\deg \alpha = 2$. Hence, $m_\alpha = (x - \alpha)(x - \beta)$ splits over E , so E/F is normal.

2. Let $E/L/F$ be finite extensions. If E/F is normal, then so is E/L .

E is a splitting field of $f \in F[x]$. Then, E is a splitting field of $f \in L[x]$, i.e., E/L is normal.

3. $E = \mathbb{Q}(\sqrt[3]{5})/\mathbb{Q}$ is not normal.

Let $\alpha = \sqrt[3]{5}$. Then,

$$\begin{aligned} m_\alpha &= x^3 - 5 = (x - \alpha) \underbrace{(x^2 + \alpha x + \alpha^2)}_{\text{has no real roots}} \\ &= (x - \alpha)(x - \alpha\eta)(x - \alpha\eta^2), \end{aligned}$$

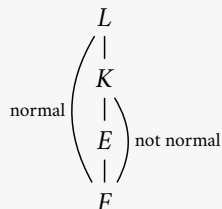
where η is the primitive 3rd root of 1, and $\mathbb{Q}(\alpha) \subset \mathbb{R}$, thus E/F is not normal.

Let $L = \mathbb{Q}(\sqrt[3]{5}, \eta)$. Then, L/F is normal.

$$\begin{array}{c} \text{normal} \left(\begin{array}{c} L \\ | \\ E \\ | \\ F \end{array} \right) \begin{array}{l} \text{normal} \\ \text{not normal} \end{array} \end{array}$$

4. Let $L = \mathbb{Q}(\sqrt[4]{5})/E = \mathbb{Q}(\sqrt[2]{5})/F = \mathbb{Q}$. Then, $[L : E] = [E : F] = 2$ by example (1). Both L/E and E/F are normal. But $m_{\sqrt[4]{5}} = x^4 - 5 = (x^2 - \sqrt{5})(x^2 + \sqrt{5})$ (here $x^2 + 5$ has no real roots). So L/F is not normal.

Definition. Let E/F be a finite extension. A field extension L/E is called a **normal closure** of E/F if L/F is normal and $[L : E]$ is minimal with respect to this property.



Examples.

1. If E/F is normal, then E/E is a normal closure of E/F .
2. Let $L = \mathbb{Q}(\sqrt[3]{5}, \eta)/E = \mathbb{Q}(\sqrt[3]{5})/F = \mathbb{Q}$. Then L/F is normal. As $[L : E] = 2$, L/E is a normal closure of E/F .

Theorem 3.10. Let E/F be a finite extension. Then, there exists a unique normal closure L/E up to E -isomorphisms.

Proof. Let $E = F(\alpha_1, \dots, \alpha_n)$, where $\alpha_i \in E$ and $m_{\alpha_i} \in F[x]$. Let L be a splitting field of $f := \prod_{i=1}^n m_{\alpha_i}$ over E . Then, L is a splitting field of f over F , i.e., L/E is normal. If there exists L'/E such that L'/F is normal, then m_{α_i} splits over L' , thus f splits over L' . Hence $L = L'$.

Let L'/E be another normal closure of E/F . As L'/F is normal and $L' \supset E$, f splits over L' . As L' is a normal closure, L' is a splitting field of f . By the uniqueness of splitting field, $L \simeq L'$ over E . ■

3.1 Finite Fields

27-09-23

Let F be a field. Consider a ring homomorphism $\varphi: \mathbb{Z} \rightarrow F$ given by

$$m \mapsto m \cdot 1_F = \underbrace{1 + \dots + 1}_{m \text{ times}}.$$

As $\ker \varphi$ is an ideal of $\mathbb{Z}$, we have that $\ker \varphi = n\mathbb{Z}$ for some integer n . Hence, $\bar{\varphi}: \mathbb{Z}/n\mathbb{Z} \hookrightarrow F$. As F is a domain, we have $n = 0$ or $n = p$ for some prime integer p .

Definition. The integer n with $\ker \varphi = n\mathbb{Z}$ is called the **characteristic of F** and is denoted by $\text{char}(F)$.

Remark. The integer n with $\ker \varphi = n\mathbb{Z}$ is the smallest positive integer n with $n \cdot 1 = 0$. (If there exists no such n , set $n = 0$.)

Examples.

1. If $F = \mathbb{Q}, \mathbb{R}, \mathbb{C}$, then $\text{char}(F) = 0$.
2. If $F = \mathbb{Z}/p\mathbb{Z}, (\mathbb{Z}/p\mathbb{Z})(x)$, then $\text{char}(F) = p$.

Indeed, if $\text{char}(F) = 0$, then the following holds:

$$\begin{array}{ccc} \mathbb{Z} & \xhookrightarrow{\quad} & F \\ & \searrow & \uparrow \\ & & (\mathbb{Z} \setminus \{0\})^{-1}\mathbb{Z} = \mathbb{Q} \end{array}$$

Thus, we have that $\mathbb{Q} \subseteq F$. If $\text{char}(F) = p$, then $\mathbb{Z}/p\mathbb{Z} \hookrightarrow F$.

Definition. The fields $\mathbb{Q}$ and $\mathbb{Z}/p\mathbb{Z}$ are called the **prime subfields** of F .

Definition. Let $f \in F[x]$ be a nonconstant polynomial and let α be a root of f in a splitting field K . We say that α is a **multiple root of f with multiplicity m** if $f = (x - \alpha)^m g$ for some $g \in K[x]$ such that $g(\alpha) \neq 0$. We say that α is **simple** if $m = 1$.

Lemma 3.11. Let $f \in F[x]$ be a nonconstant polynomial with a root α . Then, α is simple if and only if $f'(\alpha) \neq 0$. In particular, if $\gcd(f, f') = 1$, then every root of f is simple.

Proof. If $f(x) = (x - \alpha)g(x)$, then

$$f'(x) = g(x) + (x - \alpha)g'(x) \quad \text{or} \quad f'(\alpha) = g(\alpha).$$

If f and f' are relatively prime, then there exists g, h with $fg + f'h = 1$. Then,

$$\underbrace{f(\alpha)}_{=0} g(\alpha) + f'(\alpha)h(\alpha) = 1.$$

It implies that $f'(\alpha)h(\alpha) = 1$, or $f'(\alpha) \neq 0$. ■

Let E be a finite field (i.e., $|E| < \infty$). Then

$$\mathbb{Z}/p\mathbb{Z} \subset E \quad \text{and} \quad [E : \mathbb{Z}/p\mathbb{Z}] = n$$

for some integer $n \geq 1$. Then $|E| = p^n$ as for each $\alpha \in E$, we have that

$$\alpha = a_1\alpha_1 + \cdots + a_n\alpha_n$$

with $a_i \in \mathbb{Z}/p\mathbb{Z}$ for each i .

Theorem 3.12. For any prime p and any integer $n \geq 1$, there exists a unique finite field E of p^n elements up to isomorphism.

Proof. Proof of existence. Let $F = \mathbb{Z}/p\mathbb{Z}$ and let $f(x) = x^{p^n} - x \in F[x]$. Let E be a splitting field of f over F and let S be the set of all roots of f . If $\alpha, \beta \in S$, then $\alpha \pm \beta, \alpha\beta, \alpha^{-1} \in S$. (We have that $(\alpha \pm \beta)^{p^n} = \alpha^{p^n} \pm \beta^{p^n} = \alpha \pm \beta$ as $p \mid \binom{p^n}{i}$ for all $i \neq 0$ and for all p . Similarly, we have $(\alpha\beta)^{p^n} = \alpha^{p^n}\beta^{p^n} = \alpha\beta$ and $(\alpha^{-1})^{p^n} = \alpha^{-1}$.)

As E is a splitting field and $F \subset S \subset E$, $S = E$. As

$$f' = \underbrace{(x^{p^n})'}_{=p^n x^{p^n-1}=0 \text{ in } F} - x' = -1 \neq 0,$$

by Lemma 3.11, every root of f is simple, thus $|S| = |E| = p^n$.

Proof of uniqueness. By the uniqueness of splitting fields, it suffices to show that $|E| = p^n$, then E is a splitting field of F . Let E be a field with $|E| = p^n$. As $|E^\times| = p^n - 1$ and $E^\times$ is a multiplicative group, we have $\alpha^{p^n-1} = 1$ for any $\alpha \in E^\times$. Hence, $\alpha^{p^n} = \alpha$, i.e., every element of $E^\times$ is a root of f . Obviously, 0 is a root of f . Therefore, every element of E is a root of f . As $|E| = p^n$, E is a splitting field of f (i.e., $F(\alpha_1, \dots, \alpha_{p^n}) = E$). ■

Definition. $\mathbb{F}_q$ denotes the field with q elements, where $q = p^n$ for some $n \geq 1$ and some prime integer p .

We have that $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$.

Lemma 3.13. Let F be a field. Then, every finite subgroup of $F^\times$ is cyclic.

Proof. Let G be a finite subgroup of $F^\times$. As G is abelian, G is nilpotent, thus G is a product of Sylow p -subgroups (we can also use the fundamental theorem of finite abelian group), i.e.,

$$G = \mathbb{Z}/p_1^{n_1}\mathbb{Z} \times \cdots \times \mathbb{Z}/p_m^{n_m}\mathbb{Z}$$

where $p_1, \dots, p_m$ are not necessarily distinct prime integers. Let

$$r = \text{lcm}(p_1^{n_1}, p_2^{n_2}, \dots, p_m^{n_m}).$$

Then, $\alpha^r = 1$ for all $\alpha \in G$. As $x^r - 1$ has at most r roots, $|G| \leq r$. Hence, we get that $|G| = r$ (obviously, $|G| = p_1^{n_1} \cdots p_m^{n_m} \geq \text{lcm}(p_1^{n_1}, \dots, p_m^{n_m})$). Therefore, $p_1, \dots, p_m$ are distinct, thus

$$G = \mathbb{Z}/p_1^{n_1}\mathbb{Z} \times \cdots \times \mathbb{Z}/p_m^{n_m}\mathbb{Z},$$

i.e., G is cyclic. ■

Corollary 3.14.

- (1) $\mathbb{F}_q^\times$ is a finite cyclic group.
- (2) Every finite extension of finite fields is generated by a single element (i.e., if E/F , then $E = F(\alpha)$, where α is a generator of $E^\times$).
- (3) For every prime integer p and any $n \geq 1$, there exists an irreducible polynomial of degree n over $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$.
- (4) $\mathbb{F}_{p^m} \subset \mathbb{F}_{p^n}$ if and only if $m \mid n$.

Proof. (3) $\mathbb{F}_p \subset \mathbb{F}_q$, $[\mathbb{F}_q : \mathbb{F}_p] = n$ (i.e., $q = p^n$). Then, by (2) we have $\mathbb{F}_q = \mathbb{F}_p(\alpha)$, where $\alpha \in \mathbb{F}_q$. Hence, m_α is an irreducible polynomial of degree n over $\mathbb{F}_p$.

(4) $[\mathbb{F}_{p^n} : \mathbb{F}_{p^m}] = k$ if and only if $p^n = (p^m)^k$ if and only if $n = mk$. ■

Examples.

1. Consider $\mathbb{F}_4/\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$. Let $\mathbb{F}_4 = \mathbb{F}_2(\alpha)$ for $\alpha \in \mathbb{F}_4$. Let $m_\alpha = x^2 + ax + b$ with $a, b \in \mathbb{F}_2$.

- (i) If $a = b = 0$, then $m_\alpha = x^2$ is reducible.
- (ii) If $a = 1$ and $b = 0$, then $m_\alpha = x^2 + x = x(x + 1)$ is reducible.
- (iii) If $a = b = 1$, then $m_\alpha = x^2 + x + 1$ is irreducible.
- (iv) If $a = 0$ and $b = 1$, then $m_\alpha = x^2 + 1 = (x + 1)^2$ is reducible.

It implies that $\mathbb{F}_4 = \mathbb{F}_2[x]/(x^2 + x + 1) = \{a + b\alpha \mid a, b \in \mathbb{F}_2, \alpha^2 + \alpha + 1 = 0\}$.

2. Consider $\mathbb{F}_8/\mathbb{F}_2$. Let $\mathbb{F}_8 = \mathbb{F}_2(\alpha)$ for $\alpha \in \mathbb{F}_8$. Let $m_\alpha = x^3 + ax^2 + bx + c$. If $c = 0$, then $m_\alpha = x(x^2 + ax + b)$ is reducible. We assume that $c = 1$, i.e., $m_\alpha = x^3 + ax^2 + bx + 1$.

- (i) If $a = b = 0$, then $m_\alpha = x^3 + 1 = (x + 1)(x^2 + x + 1)$ is irreducible.
- (ii) If $a = b = 1$, then $m_\alpha = x^3 + x^2 + x + 1 = (x^2 + 1)(x + 1)$ is reducible.
- (iii) If $a = 1$ and $b = 0$, then $m_\alpha = x^3 + x^2 + 1$ is irreducible.
- (iv) If $a = 0$ and $b = 1$, then $m_\alpha = x^3 + x + 1$ is irreducible.

It implies that $\mathbb{F}_8 = \mathbb{F}_2[x]/(x^3 + x^2 + 1) \simeq \mathbb{F}_2[x]/(x^3 + x + 1)$. The two rings are isomorphic by the isomorphism $x \mapsto x + 1$.

Remark. $\mathbb{F}_4$ is a splitting field of $x^4 - x$. Observe that $x^4 - x = x(x - 1)(x^2 + x + 1)$ where $x, x - 1 = x + 1, x^2 + x + 1$ are all irreducible polynomials of degree dividing 2. Similarly,

$$x^{2^3} - x = x^8 - x = x(x + 1)(x^3 + x^2 + 1)(x^3 + x + 1)$$

where each factor is monic irreducible polynomial of degree dividing 3.

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3. $\mathbb{F}_{p^n}$ is a splitting field of $x^{p^n} - x$ over $\mathbb{F}_p$, and it is the set of roots of $x^{p^n} - x$.

We claim that $x^{p^n} - x$ is a product of all monic irreducible polynomials of degree d over $\mathbb{F}_p$ such that $d \mid n$.

Let f be a monic irreducible polynomial of degree d in $\mathbb{F}_p[x]$ such that $d \mid n$. Let α be a root of f . Then we have that $[\mathbb{F}_p(\alpha) : \mathbb{F}_p] = d$ and $\mathbb{F}_p(\alpha) = \mathbb{F}_{p^d}$. As $d \mid n$, we have that

$$\alpha \in \mathbb{F}_p(\alpha) = \mathbb{F}_{p^d} \subset \mathbb{F}_{p^n},$$

which is a splitting field of $x^{p^n} - x$. Thus α is a root of $x^{p^n} - x$, so f divides $x^{p^n} - x$.

Conversely, let $f \in \mathbb{F}_p[x]$ be a monic irreducible polynomial which divides $x^{p^n} - x$. Let α be a root of f . Then we have $[\mathbb{F}_p(\alpha) : \mathbb{F}_p] = \deg(f)$. As $\mathbb{F}_p(\alpha) \subset \mathbb{F}_{p^n}$ and $[\mathbb{F}_{p^n} : \mathbb{F}_p] = n$, we get $\deg(f) \mid n$.

Write $x^{p^n} - x = g_1(x)^{k_1} g_2(x)^{k_2} \cdots g_m(x)^{k_m}$, where $g_1, \dots, g_m$ are monic irreducible polynomials over $\mathbb{F}_p$. Then we have that $\deg(g_i) \mid n$ and $k_i = 1$ (as $x^{p^n} - x$ has no multiple roots) for each i . If f is a monic irreducible polynomial of degree d with $d \mid n$, then $f = g_i$ for some i . ■

For example, let $p = 3$ and $n = 2$. Then

$$x^{3^2} - x = x^9 - x = x(x - 1)(x + 1)(x^2 + 1)(x^2 + x - 1)(x^2 - x - 1).$$

4 Separable Field Extension

Lemma 4.1. Let $f \in F[x]$, where F is a field. Then the following statements are equivalent:

- (1) f and f' are relatively prime (i.e., $fF[x] + f'F[x] = F[x]$).
- (2) f has no multiple roots in any field extension L of F .
- (3) There exists a field extension E/F such that f splits over E and f has no multiple roots in E .

Proof. (1) $\Rightarrow$ (2). By (1), we have that $fL[x] + f'L[x] = L[x]$ for all L/F . If $f(\alpha) = 0$ for some $\alpha \in L$, then $f(\alpha)g(\alpha) + f'(\alpha)h(\alpha) = 1$ for some $g, h \in L[x]$. Thus, $f'(\alpha) \neq 0$. Hence, f has no multiple roots.

(2) $\Rightarrow$ (3). Take E be the splitting field of f over F .

(3) $\Rightarrow$ (1). Assume that f and f' are not relatively prime. Then, f and f' have a common root α in E . Hence, α is a multiple root of f in E , which is a contradiction. ■

Definition. A non-constant polynomial $f \in F[x]$ is called **separable over F** if it satisfies (1) $\sim$ (2) $\sim$ (3) in Lemma 4.1.

Corollary 4.2. If f is separable over F , then f is separable over any field extension L/F .

Proof. It follows from the Lemma 4.1(1). ■

Corollary 4.3. If f is separable over F , so is every non-constant divisor of f .

Proof. It follows from Lemma 4.1(3). ■

Corollary 4.4. Let $f \in F[x]$ be an irreducible polynomial. If $f' \neq 0$, then f is separable.

Proof. Assume that f is not separable. Then, f and f' are not relatively prime. As f is irreducible, $f \mid f'$. As $\deg(f') < \deg(f)$, $f' = 0$. ■

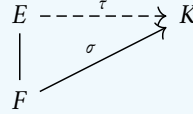
Remark. (1) Suppose that $\text{char } F = 0$. Then $f' \neq 0$, so every irreducible polynomial is separable.

- (2) Let $F = \mathbb{F}_p(t)$ be a quotient field of $\mathbb{F}_p[t]$. Let $f = x^p - t \in F[x]$. Let α be a root of f . Then, $f = x^p - t = x^p - \alpha^p = (x - \alpha)^p$ is not separable. By Eisenstein's criterion with t , which is a prime, f is irreducible over $\mathbb{F}_p[t]$. As the quotient field of $\mathbb{F}_p[t]$ is F , f is irreducible over F .

Definition. Let E/F be a finite field extension. An algebraic element α of E is called **separable over F** if the minimal polynomial m_α is separable over F .

Lemma 4.5.

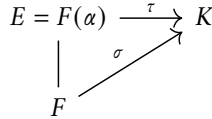
- (1) Let $K/E/F$ be field extensions. Let $\alpha \in K$ be algebraic over F . If α is separable over F , then α is separable over E .
- (2) Let E/F be a finite field extension and let $\sigma: F \rightarrow K$ be a field homomorphism. Then, there exists at most $[E : F]$ extensions $\tau: E \rightarrow K$ of σ (i.e., $\tau|_F = \sigma$).



Definition. A finite extension E/F is **separable** if there exists a field homomorphism $\sigma: F \rightarrow K$ that has exactly $[E : F]$ extensions of σ .

Proof of Lemma. (1) Let $m_{\alpha,F}$ be the minimal polynomial of α over F and $m_{\alpha,E}$ be the minimal polynomial of α over E . Then, $m_{\alpha,F} \mid m_{\alpha,E}$ as $m_{\alpha,F}(\alpha) = 0$ and $m_{\alpha,F} \in E[x]$. As $m_{\alpha,F}$ is separable, by Corollary 2, $m_{\alpha,E}$ is separable.

(2) Let $E = F(\alpha_1, \dots, \alpha_n)$ for some $\alpha_i \in E$. We prove this by induction on n . If $n = 1$, then let $\alpha = \alpha_1$. We have the following:

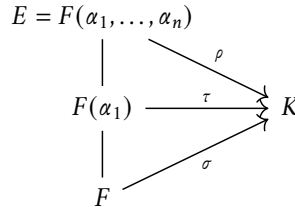


As $m_\alpha(\alpha) = 0$, we have

$$\sigma(m_\alpha)(\tau(\alpha)) = \tau(m_\alpha)(\tau(\alpha)) = \tau(0) = 0.$$

Thus, $\tau(\alpha)$ is a root of $\sigma(m_\alpha)$. Hence, the number of extensions τ of σ equals the number of distinct roots of $\sigma(m_\alpha)$ (as τ is determined by $\tau(\alpha)$), which is at most $\deg(\sigma(m_\alpha)) = \deg(m_\alpha) = [E : F]$.

Now suppose that $n > 1$. We have:



By the case of $n = 1$, the number of extensions τ of σ is at most $[F(\alpha_1) : F]$. By the induction hypothesis, the number of extensions ρ of τ is at most $[E : F(\alpha_1)]$. Therefore, the number of extensions ρ of σ is at most $[E : F(\alpha_1)][F(\alpha_1) : F] = [E : F]$. ■

The notion of ‘separability’ is defined for

polynomials, algebraic elements, and finite field extensions.

Proposition 4.6. *A finite extension $F(\alpha)/F$ is separable if and only if α is separable over F .*

Proof. By the definition, there exists some field homomorphism $\sigma: F \rightarrow K$ which has $[F(\alpha) : F]$ extensions of σ . Since $\deg(\sigma(m_\alpha)) = \deg(m_\alpha) = [F(\alpha) : F]$ equals the number of extensions of σ , and it equals the number of distinct roots of $\sigma(m_\alpha)$ in K , every root of $\sigma(m_\alpha)$ is simple and $\sigma(m_\alpha)$ splits over K . It is equivalent to the fact that $\sigma(m_\alpha)$ is separable over $\sigma(F)$, and it is equivalent to that m_α is separable over F . Hence, α is separable.

To prove the converse, let E be a splitting field of m_α over F . Let $\sigma: F \hookrightarrow E$. As m_α is separable, m_α has no multiple roots in E , thus m_α has exactly $[F(\alpha) : F]$ roots in E . Hence, σ has exactly $[F(\alpha) : F]$ extensions $\tau: F(\alpha) \rightarrow E$. ■

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Theorem 4.7 (Primitive Element Theorem). *Let E/F be a finite separable extension. Then $E = F(\alpha)$ for some $\alpha \in E$.*

For example, we have $\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \mathbb{Q}(\sqrt{2} + \sqrt{3})$.

Lemma 4.8. *Let F be an infinite field and let E/F be a field extension. Let $f \in E[x_1, \dots, x_n]$ be a nonzero polynomial. Then, there exist $a_1, \dots, a_n \in F$ such that $f(a_1, \dots, a_n) \neq 0$.*

Proof. If $n = 1$, f has at most $\deg(f)$ roots in F . As $|F| = \infty$, there exists an element $a \in F$ with $f(a) \neq 0$.

Now suppose that $n > 1$. Write

$$f = f_0 + f_1 x_n + \dots + f_m x_n^m$$

for some m , where $f_i \in E[x_1, \dots, x_{n-1}]$ for all $0 \leq i \leq m$ and $f_m \neq 0$. By induction hypothesis, there exist $a_1, \dots, a_{n-1} \in F$ such that $f_m(a_1, \dots, a_{n-1}) \neq 0$. Then, we get $f(a_1, \dots, a_{n-1}, x_n) \neq 0$ in $E[x_n]$. Hence, by the base case, there exists some $a_n \in F$ such that $f(a_1, \dots, a_{n-1}, a_n) \neq 0$. ■

Corollary 4.9. *Let F be an infinite field and let E/F be a field extension. If $f_1, \dots, f_m$ are distinct polynomials in $E[x_1, \dots, x_n]$, then there exists $a_1, \dots, a_n \in F$ such that*

$$f_1(a_1, \dots, a_n), \quad f_2(a_1, \dots, a_n), \quad \dots, \quad f_m(a_1, \dots, a_n)$$

are distinct.

Proof. Let

$$f = \prod_{1 \leq i \leq j \leq m} (f_i - f_j).$$

Then we have $f \neq 0$. By Lemma 4.8, there exist $a_1, \dots, a_n \in F$ such that $f(a_1, \dots, a_n) \neq 0$. Thus, $f_1(a_1, \dots, a_n), \dots, f_m(a_1, \dots, a_n)$ are distinct. ■

Proof of Theorem. First, suppose that $|F| < \infty$. Then $|E| < \infty$, so $E^\times$ is a finite cyclic group. Hence, $E = \mathbb{F}_p(\alpha)$ for some $\alpha \in E$. As $|F| < \infty$, we have $\mathbb{F}_p \subset F$, so $E = F(\alpha)$.

Now, suppose that $|F| = \infty$. Let $E = F(\alpha_1, \dots, \alpha_n)$ for some $\alpha_i \in E$. Set

$$f = \alpha_1 x_1 + \dots + \alpha_n x_n \in E[x_1, \dots, x_n].$$

As E/F is separable, there exists a field homomorphism $\sigma: F \rightarrow K$ that has exactly $m = [E : F]$ extensions. Let $\tau_1, \dots, \tau_m$ be such extensions of σ , i.e., $\tau_i: E \rightarrow K$ with $\tau_i|_F = \sigma$. Then

$$\tau_i(f) = \tau_i(\alpha_1)x_1 + \dots + \tau_i(\alpha_n)x_n \in K[x_1, \dots, x_n].$$

As $\alpha_1, \dots, \alpha_n$ generate E , $\tau_1(f), \dots, \tau_m(f)$ are distinct.

If $\tau_i(f) = \tau_j(f)$, then

$$\tau_i(\alpha_1)x_1 + \dots + \tau_i(\alpha_n)x_n = \tau_j(\alpha_1)x_1 + \dots + \tau_j(\alpha_n)x_n.$$

Thus, $\tau_i(\alpha_k) = \tau_j(\alpha_k)$ for all $1 \leq k \leq n$, i.e., $\tau_i = \tau_j$.

By Corollary 4.9, there exists $a_1, \dots, a_n \in F$ such that

$$\tau_i(f)(\sigma(a_1), \dots, \sigma(a_n)) = \tau_i(\alpha_1)\sigma(a_1) + \tau_i(\alpha_2)\sigma(a_2) + \dots + \tau_i(\alpha_n)\sigma(a_n)$$

are distinct for all $1 \leq i \leq m$.

Let $\beta = \alpha_1 a_1 + \alpha_2 a_2 + \dots + \alpha_n a_n \in E$. Then

$$\tau_i(\beta) = \tau_i(\alpha_1)\sigma(a_1) + \tau_i(\alpha_2)\sigma(a_2) + \dots + \tau_i(\alpha_n)\sigma(a_n)$$

and $\tau_1(\beta), \dots, \tau_m(\beta)$ are distinct. Now we show that $E = F(\beta)$. It is obvious that $F(\beta) \subset E$. As the restriction of τ_i on $F(\beta)$ are distinct for all $1 \leq i \leq m$, the number of extensions of σ on $F(\beta)$ is at least $m = [E : F]$. Therefore,

$$m = [E : F] \geq [F(\beta) : F] \geq \text{the number of extensions of } \sigma \text{ on } F(\beta) \geq m.$$

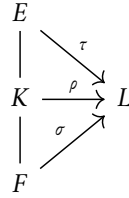
Thus $E = F(\beta)$. ■

$$\begin{array}{c} E \\ \text{separable} \left(\begin{array}{c} | \\ | \end{array} \right) \\ K \\ \text{separable} \left(\begin{array}{c} | \\ | \end{array} \right) \\ F \end{array} \text{separable}$$

Proposition 4.10. *Let $E/K/F$ be finite extensions. Then E/F is separable if and only if both E/K and K/F are separable.*

Proof. ($\Rightarrow$) There exists $\sigma: F \rightarrow L$ that has exactly $[E : F]$ extensions $\tau: E \rightarrow L$. Observe that

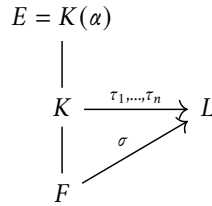
$$\text{the number of } \tau = [E : F] = [E : K][K : F].$$



Let $\rho: K \rightarrow L$ be an extension of σ . As

$$\begin{aligned} \text{the number of } \tau \text{ extending } \rho &\leq [E : K] & \text{and} & \text{the number of } \rho \text{ extending } \sigma \leq [K : F], \\ \text{the number of } \tau \text{ extending } \rho &= [E : K] & \text{and} & \text{the number of } \rho \text{ extending } \sigma = [K : F]. \end{aligned}$$

($\Leftarrow$) As E/K is separable, by Theorem 4.7, we have $E = K(\alpha)$ for some $\alpha \in E$. As K/F is separable, there exists $\sigma: F \rightarrow L$ that has exactly $n = [K : F]$ extensions, namely, $\tau_1, \dots, \tau_n$.



Let $f = m_\alpha \in K[x]$. As $E = K(\alpha)/K$ is separable, by Proposition 4.6, f is separable.

Let $f_i = \tau_i(f)$ for $1 \leq i \leq n$. Let M be a splitting field of $f_1 \cdots f_n$ over L . As f is separable, f_i has no multiple roots in m and f_i splits over M . Hence, each $[E : K]$ roots in M . Therefore, every f_i has exactly $[E : K]$ extensions $E \rightarrow M$, thus the composition $F \xrightarrow{\sigma} L \hookrightarrow M$ has exactly $n \cdot [E : K] = [E : F]$ extensions. Hence, E/F is separable. ■

Corollary 4.11. *Let E/F be a finite extension. Then, the following statements are equivalent:*

- (1) E/F is separable.
- (2) Every element of E is separable over F .
- (3) If $E = F(\alpha_1, \dots, \alpha_n)$, then each α_i is separable.

Proof. (1) $\Rightarrow$ (2). Let $\alpha \in E$. Then we have $E/F(\alpha)/F$. By Proposition, $F(\alpha)/F$ is separable. By Proposition 4.6, α is separable.

(2) $\Rightarrow$ (3). Obvious.

(3) $\Rightarrow$ (1). Let $E = F(\beta_1, \dots, \beta_n)$ for some $\beta_i \in E$. We proceed by induction on n . If $n = 1$, i.e., $E = F(\beta_1)$, then by the assumption of (3), β_1 is separable. By the same proposition, $F(\beta_1)/F$ is separable.

Now suppose that $n > 1$.

$$\begin{array}{ccc}
 E = F(\beta_1, \dots, \beta_n) = K(\beta_n) & & \\
 \downarrow & \left. \vphantom{\begin{array}{c} E \\ K \\ F \end{array}} \right\} \text{separable by case } n = 1 & \\
 K = F(\beta_1, \dots, \beta_{n-1}) & & \\
 \downarrow & \left. \vphantom{\begin{array}{c} E \\ K \\ F \end{array}} \right\} \text{separable by induction hypothesis} & \\
 F & &
 \end{array}$$

By Proposition 4.10, we conclude that E/F is separable. ■

Definition. A field is called **perfect** if $\text{char}(F) = 0$ or $\text{char}(F) = p$ with $F^p = F$, where $F^p = \{a^p \mid a \in F\}$ (i.e., for all $a \in F$, there exists $b \in F$ such that $b^p = a$).

Examples.

1. $\mathbb{Q}, \mathbb{R}$, and $\mathbb{C}$ are perfect fields.
2. Every finite field is perfect. Let F be a finite field and $\text{char}(F) = p$. Consider $\varphi: F \rightarrow F, a \mapsto a^p$. It is a field homomorphism. Clearly, φ is injective. As $|F| < \infty$, φ is surjective as well. Hence $F = F^p$.
3. Let $\text{char}(F) = p$. $F(x)$ is not perfect as $x \notin F(x)^p$.

$$\begin{aligned}
 x = f(x)^p / g(x)^p &\implies xg(x)^p = f(x)^p \\
 &\implies 1 + np = mp, \quad n = \deg(g), \quad m = \deg(f) \\
 &\implies 1 = (m - n)p,
 \end{aligned}$$

which is impossible.

4. Let $\text{char}(F) = p$. Let $K = F(x, x^{1/p}, x^{1/p^2}, \dots)$. Then, $K^p = K$, i.e., K is perfect.

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Proposition 4.12. *The following statements are equivalent:*

- (1) F is perfect.
- (2) Every irreducible polynomial in $F[x]$ is separable.
- (3) Every finite extension of F is separable.

Proof. (1) $\implies$ (2). Let $f \in F[x]$ be an irreducible polynomial. By Corollary 4.4, it suffices to show that $f' \neq 0$. If $\text{char}(F) = 0$, then $f' \neq 0$ (as $\deg(f') = \deg(f) - 1$ and $(x - a)' = 1 \neq 0$). If $\text{char}(F) = p$ and $f' = 0$, then

$$f = a_1 x^p + a_2 x^{2p} + \dots + a_n x^{np}$$

for some n and some $a_i \in F$.

As $F^p = F$, we have $a_i = b_i^p$ for some $b_i \in F$. Hence, we have

$$f = b_1^p x^p + b_2^p x^{2p} + \dots + b_n^p x^{np} = (b_1 x + b_2 x^2 + \dots + b_n x^n)^p$$

(as $\text{char}(F) = p$ and $\binom{p}{i_1 \dots i_k} \equiv p \pmod{p}$, $i_1 + \dots + i_k = p$). Therefore, f is not irreducible.

(2) $\Rightarrow$ (1). Assume that $\text{char}(F) = p$ and $F \neq F^p$. Let $a \in F \setminus F^p$ and let $f = x^p - a$. Let b be a root of f in a splitting field of f , i.e., $b^p = a$. Then $f = x^p - b^p = (x - b)^p$, so f is not separable.

Assume that $f = gh$. Then, $g = (x - b)^m$ for some $1 < m < p$. As $g \in F[x]$, we have $b^m \in F$. As $\gcd(m, p) = 1$, we have $am + cp = 1$ for some a, c . Then,

$$b^1 = b^{am+cp} = \underbrace{(b^m)^a}_{\in F} \cdot \underbrace{(b^p)^c}_{=a^c \in F} \in F,$$

so it is a contradiction. Therefore, f is not irreducible (since $a = b^p$ we have $a \in F^p$).

(2) $\Rightarrow$ (3). Let E/F be a finite extension. Let $\alpha \in E$. Then m_α is irreducible, so m_α is separable, or equivalently, α is separable. By Corollary 4.11, E/F is separable.

(3) $\Rightarrow$ (2). Let f be an irreducible polynomial in $F[x]$. Let $E = F(\alpha)$, where α is a root of f . Then E/F is finite, so it is separable, i.e., α is separable, or equivalently, m_α is separable. As $f = c \cdot m_\alpha$ for some $c \in F$, f is separable. ■

Part III

Galois Theory

5 Galois Extension

Recall. E/F is **normal** if all field homomorphism $\sigma: E \rightarrow K$, where K/E is a field extension and $\sigma|_F = \text{id}_F$, satisfies that $\sigma(E) = E$.

E/F is **separable** if there exists a field homomorphism $\tau: F \rightarrow K$ that has exactly $[E : F]$ extensions of τ .

Definition. Let E/F be a finite extension. The **Galois group** of E/F is defined by

$$\text{Gal}(E/F) := \text{Aut}_F E = \{\sigma: E \xrightarrow{\sim} E \mid \sigma|_F = \text{id}_F\}.$$

We say that E/F is **Galois** if $|\text{Gal}(E/F)| = [E : F]$.

Remark. Note that

$$|\text{Gal}(E/F)| \leq \text{the number of } F \xrightarrow{\text{id}} F \hookrightarrow E \leq [E : F].$$

$$\begin{array}{ccc} E & \xrightarrow{\sim} & E \\ | & & \uparrow \\ F & \xlongequal{\text{id}} & F \end{array}$$

Proposition 5.1. Let E/F be a finite extension. Then E/F is Galois if and only if E/F is normal and separable.

Proof. ($\Rightarrow$) Since $F \xrightarrow{\text{id}} F \hookrightarrow E$ has $|\text{Gal}(E/F)|$ extensions, E/F is separable by the definition of separable extension. Let $\tau: E \rightarrow K$ be a field homomorphism such that $\tau|_F = \text{id}_F$ and K/E is a field extension. Then, τ is an extension of $F \hookrightarrow K$.

Consider

$$A = \{E \xrightarrow[\sigma]{\sim} E \hookrightarrow K \mid \sigma \in \text{Gal}(E/F)\}.$$

Then, $|A| = |\text{Gal}(E/F)| = [E : F]$. Hence, A contains all extensions of $F \hookrightarrow K$ to E . So, we get $\tau \in A$, i.e., $\tau(E) = E$. Therefore, E/F is normal.

($\Leftarrow$) Since E/F is separable, there exists a field homomorphism $\sigma: F \rightarrow K$ that has $[E : F] = m$ extensions, say $\tau_1, \dots, \tau_m$.

$$\begin{array}{ccc} E & \xrightarrow{\tau_1, \tau_2, \dots, \tau_m} & K \\ | & \nearrow \sigma & \\ F & & \end{array}$$

Choose τ_1 . Then,

$$\begin{array}{ccc} E & \xrightarrow{\sim} & \tau_1(E) \hookrightarrow K \\ | & & | \nearrow \\ F & \xrightarrow{\sim} & \sigma(F) \end{array}$$

so

$$\begin{array}{c} K \\ | \\ \tau_1(E) \\ | \\ \sigma(F). \end{array}$$

Consider

$$\begin{array}{ccc} \tau_1(E) & \xrightarrow{\tau'_i = \tau_i \circ \tau_1^{-1}} & K \\ | & \nearrow \sigma & \\ \sigma(F) & & \end{array}$$

for $1 \leq i \leq m$. As $\tau_i|_F = \sigma$, we have $\tau'_i|_{\sigma(F)} = \text{id}_{\sigma(F)}$ for each i . As E/F is normal, $\tau_1(E)/\sigma(F)$ is normal, thus $\tau'_i(\tau_1(E)) = \tau_1(E)$ for each i . Hence, since $\tau'_i \in \text{Gal}(\tau_1(E)/\sigma(F))$ so we have

$$m = |\text{Gal}(\tau_1(E)/\sigma(F))| = |\text{Gal}(E/F)|,$$

i.e., E/F is Galois. ■

Examples.

1. $\mathbb{C}/\mathbb{R}$ is Galois. As $m_i = x^2 + 1$ is separable and $\mathbb{C} = \mathbb{R}(i) = \mathbb{R}[i]$, $\mathbb{C}/\mathbb{R}$ is separable. As $[\mathbb{C} : \mathbb{R}] = 2$, $\mathbb{C}/\mathbb{R}$ is normal (or $\mathbb{C}$ is a splitting field of m_i). Now observe that $\text{Gal}(\mathbb{C}/\mathbb{R}) = \{\text{id}, \sigma\} \simeq \mathbb{Z}/2\mathbb{Z}$, where $\sigma(a + bi) = a - bi$ and $a, b \in \mathbb{R}$.
2. Let F be a field with $\text{char } F \neq 2$. Assume that we have $a \in F^\times \setminus (F^\times)^2$. Then, $x^2 - a$ is irreducible. Let $E = F[x]/(x^2 - a)$. Then $[E : F] = 2$, so E/F is normal. As $(x^2 - a)' = 2x \neq 0$, $x^2 - a$ is separable, so E/F is separable. Therefore, E/F is Galois. Moreover, we have

$$E = \{x + y\sqrt{a} \mid x, y \in F\} \quad \text{and} \quad \text{Gal}(E/F) = \{\text{id}, \sigma\} \simeq \mathbb{Z}/2\mathbb{Z},$$

where $\sigma(x + y\sqrt{a}) = x - y\sqrt{a}$.

3. Let F be a field of $\text{char}(F) = 2$. Suppose that there exists $a \in F$ such that $x^2 + x + a$ has no root in F . Let $E = F[x]/(x^2 + x + a)$. Then, we have $[E : F] = 2$, so E/F is normal. As $(x^2 + x + a)' = 1 \neq 0$, E/F is separable, so E/F is Galois. Observe that

$$\text{Gal}(E/F) = \{\text{id}, \sigma\} \quad \text{where} \quad \sigma(\alpha) = \alpha + 1,$$

where α is a root of $x^2 + x + a$.

$$E = \{x + y\alpha \mid x, y \in F\}$$

$$\sigma(x + y\alpha) = x + y\sigma(\alpha) = x + y(\alpha + 1) = (x + y) + y\alpha.$$

4. Let $q = p^m$, where p is a prime and $m \geq 1$. Let $E = \mathbb{F}_{q^n}$ and $F = \mathbb{F}_q$. Then E/F is Galois. (As E/F is perfect, it is separable. As E is a splitting field of $x^{q^n} - x$, E/F is normal.) Let $\sigma: E \rightarrow E$ that maps a to a^q . For $b \in F$, we have $b^q = b$, thus $\sigma|_F = \text{id}_F$. It follows that $\sigma \in \text{Gal}(E/F)$. Now observe that

$$\sigma^n(a) = a^{q^n} = a$$

for all $a \in E$, so $\sigma^n = \text{id}_E$. As $E^\times$ is cyclic, we have $E^\times = \langle a \rangle$ for some $a \in E^\times$. For $0 < i < n$, $\sigma^i(a) = a^{q^i} \neq a$, thus $\sigma^i \neq \text{id}_E$. Hence, $\text{Gal}(E/F) = \langle \sigma \rangle \simeq \mathbb{Z}/n\mathbb{Z}$.

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5. Consider $E = \mathbb{Q}(\sqrt{p}, \sqrt{q})$, where p and q are distinct primes. Let $F = \mathbb{Q}$ and $f = (x^2 - p)(x^2 - q)$. Then, E is a splitting field of f , so E/F is normal. As $\text{char}(F) = 0$, E/F is separable (F is perfect). Therefore, E/F is Galois. Then, $|\text{Gal}(E/F)| = [E : F] = 4$ ($1, \sqrt{p}, \sqrt{q}, \sqrt{pq}$ is a basis over F).

Let $G = \text{Gal}(E/F)$ and $\sigma \in G$. Then we have that

$$\sigma(\sqrt{p})^2 = \sigma(\sqrt{p} \cdot \sqrt{p}) = \sigma(p) = p.$$

Thus $\sigma(\sqrt{p}) = \pm\sqrt{p}$ and $\sigma(\sqrt{q}) = \pm\sqrt{q}$. Then,

$$\begin{aligned} G &= \left\{ \text{id}, \quad \sigma: \begin{array}{l} \sqrt{p} \mapsto \sqrt{p} \\ \sqrt{q} \mapsto -\sqrt{q} \end{array}, \quad \tau: \begin{array}{l} \sqrt{p} \mapsto -\sqrt{p} \\ \sqrt{q} \mapsto \sqrt{q} \end{array}, \quad \sigma\tau = \tau\sigma \right\} \\ &\simeq \langle \sigma \rangle \times \langle \tau \rangle \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}. \end{aligned}$$

6. Let $E = \mathbb{Q}(\sqrt{2 + \sqrt{2}})$ and $F = \mathbb{Q}$. Let $\alpha = \sqrt{2 + \sqrt{2}}$. Then, we have $\alpha^2 = 2 + \sqrt{2}$ or $\alpha^4 - 4\alpha^2 + 2 = 0$. Observe that

$$\begin{aligned} f &= x^4 - 4x^2 + 2 = (x^2 - 2)^2 - (\sqrt{2})^2 \\ &= (x^2 - 2 - \sqrt{2})(x^2 - 2 + \sqrt{2}) \\ &= (x^2 - (2 + \sqrt{2}))(x^2 - (2 - \sqrt{2})) \\ &= (x - \alpha)(x + \alpha) \left(x - \sqrt{2 - \sqrt{2}} \right) \left(x + \sqrt{2 - \sqrt{2}} \right). \end{aligned}$$

Observe that $\alpha \cdot \sqrt{2 - \sqrt{2}} = \sqrt{2}$, so $\sqrt{2 - \sqrt{2}} = \sqrt{2}/\alpha \in E$. Also, $\alpha^2 - 2 = \sqrt{2} \in E$. Hence, E is a splitting field of f , so E/F is normal. As $\text{char}(F) = 0$, E/F is separable, thus E/F is Galois.

By Eisenstein's criterion with $p = 2$, f is irreducible over F . Let $G = \text{Gal}(E/F)$. Then $|G| = [E : F] = 4$. Let $\sigma \in G$ with $\sigma(\alpha) = \sqrt{2 - \sqrt{2}} = \sqrt{2}/\alpha$. Then,

$$\sigma^2(\alpha) = \sigma(\sqrt{2}/\alpha) = \frac{\sigma(\sqrt{2})}{\sigma(\alpha)} = \frac{-\sqrt{2}}{\sqrt{2}/\alpha} = -\alpha$$

(because $\sigma(\sqrt{2}) = \sigma(\alpha^2 - 2) = \sigma(\alpha)^2 - 2 = (2 - \sqrt{2}) - 2 = -\sqrt{2}$). Thus $\sigma^2 \neq \text{id}$. Hence $G = \langle \sigma \rangle \simeq \mathbb{Z}/4\mathbb{Z}$.

7. Let $E = \mathbb{Q}(\sqrt[3]{5}, \eta)$ and $F = \mathbb{Q}$, where η is a primitive third root of 1 (that is, $\eta^2 + \eta + 1 = 0$). Let

$$f = x^3 - 5 = (x - \sqrt[3]{5})(x - \eta\sqrt[3]{5})(x - \eta^2\sqrt[3]{5}),$$

i.e., $\sqrt[3]{5}$, $\eta\sqrt[3]{5}$, and $\eta^2\sqrt[3]{5}$ are roots of f . Hence, E/F is Galois.

$$\begin{array}{c} E = \mathbb{Q}(\sqrt[3]{5}, \eta) \\ \downarrow 2 \\ \mathbb{Q}(\sqrt[3]{5}) \\ \downarrow 3 \\ F = \mathbb{Q} \end{array}$$

Let $G = \text{Gal}(E/F)$. Then $|G| = |\text{Gal}(E/F)| = 6$. Let $\sigma \in G$. Then $\sigma(\sqrt[3]{5})^3 = \sigma(5) = 5$. Thus $\sigma(\sqrt[3]{5}) = \sqrt[3]{5}, \eta\sqrt[3]{5}, \eta^2\sqrt[3]{5}$ and $\sigma(\eta) = \eta, \eta^2$. Let

$$\sigma: \begin{array}{l} \sqrt[3]{5} \mapsto \eta\sqrt[3]{5} \\ \eta \mapsto \eta \end{array} \quad \text{and} \quad \tau: \begin{array}{l} \sqrt[3]{5} \mapsto \sqrt[3]{5} \\ \eta \mapsto \eta^2 \end{array}.$$

Then $\sigma^3 = \text{id} = \tau^2$. Observe that

$$\sigma \circ \tau(\sqrt[3]{5}) = \eta\sqrt[3]{5} \neq \eta^2\sqrt[3]{5} = \tau \circ \sigma(\sqrt[3]{5}),$$

i.e., $\sigma \circ \tau \neq \tau \circ \sigma$. Hence, $G = \{\text{id}, \sigma, \tau, \sigma^2, \sigma\tau, \sigma^2\tau\}$ is a non-abelian group of order six, which is isomorphic to S_3 .

5.1 Galois Correspondence

Lemma 5.2 (Artin). *Let G be a finite subgroup of the automorphism group of a field E (i.e., $G < \text{Aut}(E)$). Let $F = E^G := \{a \in E \mid \sigma(a) = a \text{ for all } \sigma \in G\}$. Then, E/F is Galois and $\text{Gal}(E/F) = G$.*

Proof. Since $[E : F] \geq |\text{Gal}(E/F)| \geq |G|$, it suffices to show that $|G| \geq [E : F]$. Let $\alpha \in E$. Then, we claim the following:

$$F(\alpha)/F \text{ is separable and } [F(\alpha) : F] \leq |G|. \quad (*)$$

To check $(*)$, let $A = \{\sigma(\alpha) \mid \sigma \in G\}$ and let $f = \prod_{\beta \in A} (x - \beta) \in E[x]$. Then, α is a root of f (i.e., $f(\alpha) = 0$) and f has no multiple roots in E . Let $\tau \in G$. As $A = \{\tau\sigma(\alpha) \mid \sigma \in G\}$, $\tau(f) = f$. Since $F = E^G$, $f \in F[x]$. As f is separable over F , m_α is separable. Hence, $F(\alpha)/F$ is separable. Therefore, we conclude that

$$[F(\alpha) : F] = \deg(m_\alpha) \leq \deg(f) \leq |G|$$

(since $|A| \leq |G|$), as desired.

Now, assume that $[E : F] > |G|$. Then, there exists $\alpha_1, \dots, \alpha_n \in E$ such that $[F(\alpha_1, \dots, \alpha_n) : F] > |G|$. By $(*)$, $F(\alpha_1, \dots, \alpha_n)/F$ is separable. Hence, by the primitive element theorem, there exists some $\beta \in E$ such that $F(\alpha_1, \dots, \alpha_n) = F(\beta)$. Then, by $(*)$, we have $[F(\beta) : F] \leq |G|$, which is a contradiction. Therefore, we have $|G| \geq [E : F]$, so $[E : F] = |\text{Gal}(E/F)|$, i.e., E/F is Galois. ■

Let E/F be a Galois extension with $G = \text{Gal}(E/F)$. Then, there is a one-to-one correspondence:

$$\begin{array}{ccc}
 \boxed{\text{the set of subfields of } E \text{ containing } F} & \xleftrightarrow{1-1} & \boxed{\text{the set of subgroups of } G} \\
 K & \xrightarrow{\varphi} & \text{Gal}(E/K) \\
 E^H & \xleftarrow{\psi} & H
 \end{array}$$

Here, $E^H = \{a \in E \mid \sigma(a) = a \text{ for all } \sigma \in H\}$, which is a fixed subfield by H .

Theorem 5.3. $\varphi \circ \psi = \text{id}$ and $\psi \circ \varphi = \text{id}$.

Proof. Note that

$$\begin{aligned}
 \varphi \circ \psi &= \text{id} \iff \text{Gal}(E/E^H) = H, \\
 \psi \circ \varphi &= \text{id} \iff E^{\text{Gal}(E/K)} = K.
 \end{aligned} \tag{*}$$

Here (*) follows from Artin's theorem.

Let $H = \text{Gal}(E/K)$. Then $K \subset E^H$. By Artin's theorem, we have $|H| = [E : E^H]$. As E/K is Galois (As E/F is normal, so is E/K . As E/F is separable, so is E/K), we get $[E : K] = |\text{Gal}(E/K)|$. Then

$$[E : E^H] = |H| = |\text{Gal}(E/K)| = [E : K],$$

which implies that $K = E^H$. ■

Properties. Let E/F be a Galois extension with $G = \text{Gal}(E/F)$. Then,

$$(1) \quad E/K/L/F \iff \{\text{id}\} = \text{Gal}(E/E) < \text{Gal}(E/K) < \text{Gal}(E/L) < \text{Gal}(E/F) = G, \\ E = E^{\{1\}}/E^H/E^T/E^G = F \iff \{\text{id}\} = 1 < H < T < G.$$

(2)

$$\begin{array}{ccc}
 E & & 1 \\
 \left(\begin{array}{c} | \\ | \end{array} \right) & & \left(\begin{array}{c} | \\ | \end{array} \right) \\
 K & \longleftrightarrow & \text{Gal}(E/K) \\
 \left(\begin{array}{c} / \\ | \\ \backslash \end{array} \right) & & \left(\begin{array}{c} | \\ / \\ \backslash \end{array} \right) \\
 F & & \text{Gal}(E/F) = G
 \end{array}$$

$$|\text{Gal}(E/K)| = [E : K]$$

$$[K : F] = \frac{[E : F]}{[E : K]} = \frac{|G|}{|\text{Gal}(E/K)|} = [G : \text{Gal}(E/K)].$$

(3) Let $K = E^H$ for some $H < G$ (i.e., $H = \text{Gal}(E/K)$). Then, for all $\sigma \in G$, we have $\sigma(K) \leftrightarrow \sigma H \sigma^{-1}$.

Proof. We show $\text{Gal}(E/\sigma(K)) = \sigma H \sigma^{-1}$. Let $a \in K$ and let $\tau \in H$. Then

$$\sigma \tau \sigma^{-1}(\sigma(a)) = \sigma \tau(a) = \sigma(a),$$

thus $\sigma H \sigma^{-1} < \text{Gal}(E/\sigma(K))$. Then, we have that

$$\begin{aligned} |\sigma H \sigma^{-1}| &= |H| = |\text{Gal}(E/K)| \\ &= [E : K] \\ &= [E : \sigma(K)] \\ &= |\text{Gal}(E/\sigma(K))|, \end{aligned}$$

so $\sigma H \sigma^{-1} = \text{Gal}(E/\sigma(K))$.

- (4) Let $K = E^H$. Then K/F is normal if and only if $H \triangleleft G$ (i.e., H is a normal subgroup of G). In this case, $G/H \simeq \text{Gal}(K/F)$.

Proof. Observe that

$$\begin{aligned} H \triangleleft G &\iff \sigma H \sigma^{-1} = H \text{ for all } \sigma \in G && \text{(by definition)} \\ &\iff \sigma(K) = K \text{ for all } \sigma \in G && \text{(by property (3))} \\ &\iff K/F \text{ is normal.} && (*) \end{aligned}$$

Here, the proof of $\xleftarrow{(*)}$ follows from the definition of normal extension K/F . The proof of $\xrightarrow{(*)}$ is given as follows:

$$\begin{array}{ccc} E = F(\beta) & \xrightarrow{\rho} & E \\ \downarrow & & \downarrow \\ K = F(\alpha) & \xrightarrow{\tau} & L \text{ splitting field of } m_\alpha \\ \text{separable} \downarrow & & \downarrow \\ F & \xrightarrow{id} & F \end{array}$$

As K/F is separable, by the primitive element theorem, we get $K = F(\alpha)$ for some α . As K/F is normal, or equivalently, K is a splitting field of m_α , it suffices to check that $\alpha' \in K$, where α' is a root of m_α . Then, there exists an extension $\tau: F(\alpha) \rightarrow L$ with $\alpha \mapsto \alpha'$ of id_F . As E/F is normal and $E = F(\beta)$ (by primitive element theorem), E is a splitting field of m_β and $L \subset E$. We have an extension $\rho: E \rightarrow E$ of σ such that $\rho(E) = E$, i.e., $\rho \in \text{Gal}(E/F) = G$. Hence, $\sigma(K) = \rho(K) = K$ (by assumption that $\sigma(K) = K$ for all $\sigma \in G$) and $\alpha' = \tau(\alpha) = \tau(K)$.

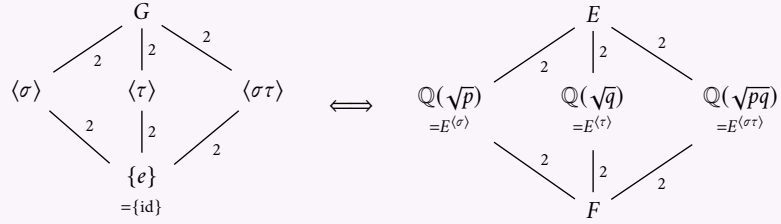
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Examples.

1. Consider E/F , where $E = \mathbb{Q}(\sqrt{p}, \sqrt{q})$ with p and q are distinct primes and $F = \mathbb{Q}$. Then we have

$$\begin{aligned} G = \text{Gal}(E/F) &= \left\{ \text{id}, \quad \sigma: \begin{array}{l} \sqrt{p} \mapsto \sqrt{p} \\ \sqrt{q} \mapsto -\sqrt{q} \end{array}, \quad \tau: \begin{array}{l} \sqrt{p} \mapsto -\sqrt{p} \\ \sqrt{q} \mapsto \sqrt{q} \end{array}, \quad \sigma\tau = \tau\sigma \right\} \\ &\simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}. \end{aligned}$$

three nontrivial proper subgroups $\iff$ three proper nontrivial subfields ($\neq F$)



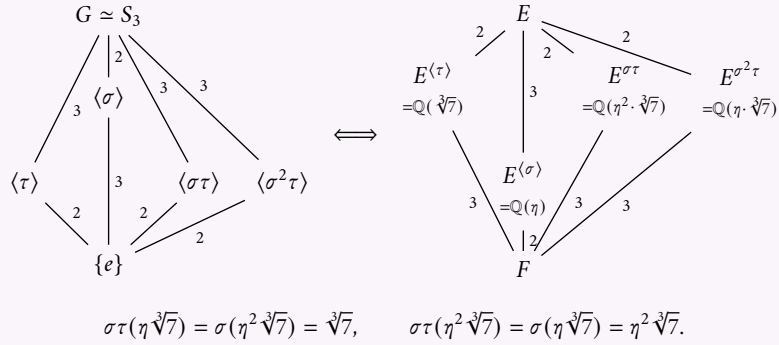
2. Consider E/F , where $E = \mathbb{Q}(\sqrt[3]{7}, \eta)$ and $F = \mathbb{Q}$. Here, η is the primitive third root of 1. Then, we have that

$$G = \text{Gal}(E/F) = \{\text{id}, \sigma, \tau, \sigma^2, \sigma\tau, \sigma^2\tau\} \simeq S_3,$$

where

$$\begin{aligned} \sigma: \sqrt[3]{7} &\mapsto \eta \cdot \sqrt[3]{7} & \text{and} & & \tau: \sqrt[3]{7} &\mapsto \sqrt[3]{7} \\ \eta &\mapsto \eta & & & \eta &\mapsto \eta^2. \end{aligned}$$

Note that $\eta^3 = \text{id} = \tau^3$.



6 Cyclotomic Field Extension

Let $n \geq 1$ be an integer and let F be a field, where $\text{char}(F) = 0$ or $\gcd(\text{char}(F), n) = 1$. Let $f = x^n - 1$. As $-f + (x/n) \cdot f' = 1$, f is separable over F . Let E be a splitting field of f over F . Then, E/F is Galois. What is $\text{Gal}(E/F)$?

Let μ_n be the group of n th root of 1. Then, $\mu_n < E^\times$ is cyclic and $\mu_n \simeq \mathbb{Z}/n\mathbb{Z}$. Moreover, we have a ring isomorphism $\mathbb{Z}/n\mathbb{Z} \rightarrow \text{End}(\mu_n)$ such that

$$[m] \mapsto \theta_m, \quad \theta_m(x) = x^m \quad \text{for all } x \in \mu_n.$$

($\mathbb{Z}/n\mathbb{Z} \simeq \text{End}(\mathbb{Z}/n\mathbb{Z})$.) Let $\sigma \in G = \text{Gal}(E/F)$. Then, $\sigma(x)^n - 1 = 0$ for all $x \in \mu_n$, so $\sigma|_{\mu_n}$ is a map from μ_n to μ_n , i.e., $\sigma|_{\mu_n} \in \text{Aut}(\mu_n)$. Therefore, we have

$$\psi: G \longrightarrow (\mathbb{Z}/n\mathbb{Z})^\times = \text{Aut}(\mu_n) (= \text{End}(\mu_n)^\times) \quad \sigma \longmapsto \sigma|_{\mu_n}.$$

Theorem 6.1. ψ is injective. In particular, $G \hookrightarrow (\mathbb{Z}/n\mathbb{Z})^\times$ is abelian.

Proof. Let

$$\begin{aligned}\psi(\sigma) = [m] &\longleftrightarrow \sigma(x) = x^m \\ \psi(\tau) = [k] &\longleftrightarrow \tau(x) = x^k \\ \psi(\sigma\tau) = [l] &\longleftrightarrow \sigma\tau(x) = x^l\end{aligned}\quad \text{for all } x \in \mu_n.$$

Then, $\sigma\tau(x) = \sigma(x^k) = x^{km} = x^l$. Thus $km = l$, i.e., $\psi(\sigma \circ \tau) = \psi(\sigma)\psi(\tau)$.

Let $\psi(\sigma) = [1] \leftrightarrow \sigma(x) = x$ for all $x \in \mu_n$. As $E = F(\mu_n)$, we obtain that $\sigma = \text{id}$. ■

From now on, let $F = \mathbb{Q}$ and let η be a primitive n th root of 1, i.e., $\mu_n = \langle \eta \rangle$ and $E = \mathbb{Q}(\mu_n) = \mathbb{Q}(\eta)$. Let Φ_n be the minimal polynomial of η .

Lemma 6.2.

- (1) $\Phi_n \in \mathbb{Z}[x]$.
- (2) If p is a prime integer with $\gcd(p, n) = 1$, then η^p is a root of Φ_n .

Proof. (1) As $\mathbb{Z}$ is a U.F.D., there exists some $\alpha \in \mathbb{Q}$ such that $\alpha\Phi_n \in \mathbb{Z}[x]$ is primitive (by Lemma 1.11(1)). Write $x^n - 1 = \alpha\Phi_n \cdot g$ for some $g \in \mathbb{Q}[x]$. Similarly, there exists some $\beta \in \mathbb{Q}$ such that $\beta \cdot g \in \mathbb{Z}[x]$ is primitive. Then,

$$x^n - 1 = \underbrace{\alpha\Phi_n}_{\substack{\in \mathbb{Z}[x] \\ \text{primitive}}} \cdot \underbrace{(\beta \cdot g)}_{\substack{\in \mathbb{Z}[x] \\ \text{primitive}}} \cdot \frac{1}{\beta}$$

so $\alpha\Phi_n \cdot (\beta \cdot g)$ is a primitive polynomial over $\mathbb{Z}$ (by Corollary 1.10). Therefore, we have $\beta = \pm 1$, which implies that $x^n - 1 = \alpha\Phi_n \cdot g$ with $\alpha\Phi_n, g \in \mathbb{Z}[x]$. Comparing the leading coefficients, we have that $1 = (\pm 1) \cdot (\pm 1)$. As Φ_n is monic, we have $\alpha = \pm 1$, thus $\Phi_n \in \mathbb{Z}[x]$.

(2) By (1), we can write $x^n - 1 = \Phi_n \cdot g$ where $g \in \mathbb{Z}[x]$. Assume $\Phi_n(\eta^p) \neq 0$. Then, $g(\eta^p) = 0$. Let $h(x) = g(x^p)$. Then, $h(\eta) = 0$ and $\Phi_n \mid h$. In $\mathbb{Z}/p\mathbb{Z}[x]$, we get $\bar{h} = \bar{g}^p$.

Here, the bar notation is the map $\mathbb{Z}[x] \xrightarrow{\bar{\cdot}} \mathbb{Z}/p\mathbb{Z}[x]$ defined as $a \in \mathbb{Z} \mapsto \bar{a} = [a] \in \mathbb{Z}/p\mathbb{Z}$, that is, if $l(x) = a_k x^k + \dots + a_0$, then we get $\bar{l} = \bar{a}_k x^k + \dots + \bar{a}_0$.

Write $g(x) = b_0 + b_1 x + \dots + b_k x^k$. Then we have

$$h(x) = g(x^p) = b_0 + b_1 x^p + b_2 x^{2p} + \dots + b_k x^{pk}$$

or

$$\begin{aligned}\bar{h} &= \bar{b}_0 + \bar{b}_1 x^p + \bar{b}_2 x^{2p} + \dots + \bar{b}_k x^{pk} \\ &= (\bar{b}_0 + \bar{b}_1 x + \bar{b}_2 x^2 + \dots + \bar{b}_k x^k)^p = \bar{g}^p\end{aligned}$$

because $p \mid \binom{p}{i}$ and $p \mid \binom{p}{i_1 \dots i_k}$. Let $\bar{r}$ be an irreducible divisor of $\bar{\Phi}_n$. Then $\bar{r} \mid \bar{\Phi}_n \mid \bar{h} = \bar{g}^p$, where $\bar{r}$ is prime. Thus, we have $\bar{r} \mid \bar{g}$, so $\bar{r}^2 \mid x^n - \bar{1}$, which contradicts the separability of $x^n - \bar{1}$ (because $\gcd(p, n) = 1$). ■

Theorem 6.3. $\text{Gal}(\mathbb{Q}(\eta)/\mathbb{Q}) = (\mathbb{Z}/n\mathbb{Z})^\times$ and $[\mathbb{Q}(\eta) : \mathbb{Q}] = \varphi(n)$, where $\varphi(n)$ denotes the Euler-phi function and η is a primitive n th root of 1.

Proof. As $\text{Gal}(\mathbb{Q}(\eta)/\mathbb{Q}) \hookrightarrow (\mathbb{Z}/n\mathbb{Z})^\times$, it suffices to show that $\deg \Phi_n \geq \varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$.

Let m be an integer with $\gcd(m, n) = 1$. Then, $m = p_1^{k_1} \cdots p_l^{k_l}$ are primes with $\gcd(p_i, n) = 1$ for each $i = 1, \dots, l$. By applying Lemma 6.2(2) several times, we get that η^m is a root of Φ_n . Hence, every primitive n th root of 1 is a root of Φ_n . ■

Definition. The field extension $\mathbb{Q}(\eta)/\mathbb{Q}$ is called the ***nth cyclotomic field extension***. The minimal polynomial Φ_n is called the ***nth cyclotomic polynomial***.

Note that

$$x^n - 1 = \prod_{\eta \in \mu_n} (x - \eta) = \prod_{d|n} \left[\prod_{\substack{\eta \in \mu_d \\ \mu \text{ primitive}}} (x - \eta) \right] = \prod_{d|n} \Phi_d = \Phi_d.$$

Corollary 6.4.

$$\Phi_n = \frac{x^n - 1}{\prod_{\substack{d|n \\ d \neq n}} \Phi_d}.$$

Examples.

n	1	2	3	4	5	6
Φ_n	$x - 1$	$\frac{x^2 - 1}{x - 1}$ $= x + 1$	$\frac{x^3 - 1}{x - 1}$ $= x^2 + x + 1$	$\frac{x^4 - 1}{(x - 1)(x + 1)}$ $= x^2 + 1$	$\frac{x^5 - 1}{x - 1}$ $= x^4 + x^3 + x^2 + x + 1$	$\frac{x^6 - 1}{(x - 1)(x + 1)(x^2 + x + 1)}$ $= x^2 - x + 1$

7 Algebraically Closed Field

Proposition 7.1. Let F be a field. Then, the following statements are equivalent:

- (1) Every finite field extension of F is F itself.
- (2) Every irreducible polynomial over F is linear.
- (3) Every non-constant polynomial has a root in F .
- (4) Every non-constant polynomial splits over F .

Definition. A field F is ***algebraically closed*** if it satisfies (1) $\sim$ (2) $\sim$ (3) $\sim$ (4) in Proposition 7.1.

Proof. (1) $\Rightarrow$ (2). Let f be an irreducible polynomial over F . Let $E = F[x]/(f)$. Then, we have $[E : F] = \deg(f)$. By (1), we have $E = F$, i.e., $[E : F] = 1$, so $\deg(f) = 1$, i.e., f is linear.

(2) $\Rightarrow$ (3). Let f be a non-constant polynomial. Let g be an irreducible divisor of f . By (2), g is linear, so g has a root in F , and so does f .

(3) $\Rightarrow$ (4). Let f be a non-constant polynomial. We use induction on $\deg(f)$. By (3), we can write $f = (x - \alpha)g$ for some g and some α . As $\deg(g) < \deg(f)$, induction hypothesis implies that g splits over F , and so does f .

(4) $\Rightarrow$ (1). Let E/F be a finite extension and let $\alpha \in E$ and let m_α be the minimal polynomial of α . They, by (4), m_α splits over F . As m_α is irreducible, m_α is linear. Thus, we have $\alpha \in F$, i.e., $E = F$. ■

Observations.

1. If $E/\mathbb{R}$ is a field extension with odd degree, then $E = \mathbb{R}$.

Proof. Let $E = \mathbb{R}(\alpha)$ and let $f = m_\alpha$. We assume that $\deg(f) = [E : \mathbb{R}]$ is odd and not 1. Since $f(-\infty)f(\infty) < 0$, the intermediate value theorem says that f has a root in $\mathbb{R}$. As f is irreducible, however, it gives a contradiction. ■

2. Every complex number is a square.

Proof. For any $\alpha \in \mathbb{C}$, observe that

$$\alpha = r(\cos \theta + i \sin \theta) = \left[\pm \sqrt{r} \left(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right) \right]^2$$

and $\pm \sqrt{r}(\cos \theta/2 + i \sin \theta/2) \in \mathbb{C}$. ■

Theorem 7.2. $\mathbb{C}$ is algebraically closed.

Proof. Let $L/\mathbb{C}$ be a finite extension, and let E/L be a normal closure of $L/\mathbb{R}$. Then, $E/\mathbb{R}$ is Galois. Let $G = \text{Gal}(E/\mathbb{R})$. Let H be a Sylow 2-subgroup of G . Then, by the Galois correspondence, we have

$$[E^H : \mathbb{R}] = [G : H] = \text{odd}.$$

By the first observation above, we have $E^H = \mathbb{R}$, i.e., $G = H$ (G is a 2-group).

$$\begin{array}{ccc} E & & G \\ | & & | \\ E^H & \Longleftrightarrow & H \\ | & & | \\ \mathbb{R} & & \{e\} \end{array}$$

Now, let $N = \text{Gal}(E/\mathbb{C})$. Assume that $N \neq \{e\}$. Then, by Sylow theorem, there exists a subgroup M of N with $[N : M] = 2$ (as G is a 2-group, N is a 2-group). Then, by Galois correspondence, we have that $[E^M : \mathbb{C}] = [N : M] = 2$. Hence, $E^M = \mathbb{C}(\sqrt{z})$ for some $z \in \mathbb{C}$. By the second observation above, we have $\mathbb{C}(\sqrt{z}) = \mathbb{C}$, which is a contradiction.

$$\begin{array}{c} E \\ \left(\begin{array}{c} | \\ L \\ | \\ \mathbb{C} \\ | \\ \mathbb{R} \end{array} \right) \\ \begin{array}{l} \text{normal} \\ \Rightarrow \text{Galois} \end{array} \quad \begin{array}{l} \text{normal} \\ \Rightarrow \text{Galois} \end{array} \end{array}$$

Therefore, we conclude that $E = L = \mathbb{C}$. ■

8 Galois Group of a Polynomial

Definition. Let $f \in F[x]$ be a non-constant separable polynomial. Let E be a splitting field of f . The **Galois group of f** is defined by $\text{Gal}(E/F)$ and is denoted by $\text{Gal}(f)$.

Remark. 1. Let $E = F(\alpha_1, \dots, \alpha_n)$ be a splitting field of a separable polynomial f of degree n , where $\alpha_i, i = 1, \dots, n$, is a root of f . Then, for every σ in $\text{Gal}(f)$, we have

$$f(\sigma(\alpha_i)) = \sigma(f(\alpha_i)) = \sigma(0) = 0,$$

i.e., $\sigma(\alpha_i)$ is a root of f . Hence, $\sigma(\alpha_i) = \alpha_j$ for some j . Thus, $\text{Gal}(f)$ acts on the set of roots of f by permutation.

2. $\text{Gal}(f) \hookrightarrow S_n$: For all i , we have $\sigma(\alpha_i) = \alpha_i$. So $\sigma = \text{id}_E$.

Lemma 8.1. Let f be a non-constant separable polynomial in $F[x]$. Then,

- (1) If α is a root of f , then $\deg(\alpha) = |\{\sigma(\alpha) \mid \sigma \in \text{Gal}(f)\}|$.
- (2) If f is irreducible, then $\text{Gal}(f)$ acts transitively (i.e., it has a single orbit) on the set of roots of f .

This is just a restatement of Orbit-stabilizer theorem, which states that if a group G acts on a set X , then $G/G_x \simeq G \cdot x$ for an element x of X . Here, $G_x = \{g \in G \mid g \cdot x = x\}$ is a stabilizer of x , and $G \cdot x = \{gx \mid g \in G\}$ is an orbit of x . The bijective map between G/G_x and $G \cdot x$ maps gG_x with gx . Note also that this theorem implies that $[G : G_x] = |G \cdot x|$, or equivalently, $|G|/|G_x| = |G \cdot x|$.

Proof. (1) Let E be a splitting field of f , and let $G = \text{Gal}(f) = \text{Gal}(E/F)$. Then, by orbit-stabilizer theorem, we have

$$\deg(\alpha) = [F(\alpha) : F] = \frac{[E : F]}{[E : F(\alpha)]} = \frac{|G|}{|\text{stabilizer of } \alpha|} = |\text{orbit of } \alpha| = |\{\sigma(\alpha) \mid \sigma \in G\}|.$$

Here, the orbit of α is $G \cdot \alpha = \{\sigma(\alpha) \mid \sigma \in G\}$, and the stabilizer of α is

$$G_\alpha = \{\sigma \in G \mid \sigma(\alpha) = \alpha\} = \{\sigma \in G \mid \sigma|_{F(\alpha)} = \text{id}_{F(\alpha)}\} = \text{Gal}(E/F(\alpha)).$$

(2) Let X be the set of all roots of f . As f is irreducible, by (1), we have that

$$\deg(f) = \deg(\alpha) = |\{\sigma(\alpha) \mid \sigma \in G\}|.$$

Hence, $X = G \cdot \alpha$ and $G = \text{Gal}(f)$. ■

Corollary 8.2. Let f be an irreducible polynomial over $\mathbb{Q}$ of degree p , where p is prime. If f has exactly two non-real complex roots (that is, there exists two roots α and β of f in $\mathbb{C} \setminus \mathbb{R}$), then $\text{Gal}(f) = S_p$.

If $f(a + bi) = 0$, then $f(a - bi) = f(\overline{a + bi}) = \overline{f(a + bi)} = \overline{0} = 0$. Thus, the two roots forms a conjugate pair.

Proof. Let X be the set of roots of f . By Lemma 8.1(2), we have $|X| = |G \cdot \alpha|$ for all α , where α is a root of f . Hence, $p = |X| = |G \cdot \alpha|$ divides $|G|$. By Sylow theorem, G has a p -cycle (G has an element of order p). By the assumption, the complex conjugation yields a transposition. As S_p is generated by a p -cycle and a transposition, we conclude that $\text{Gal}(f) = S_p$. ■

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Definition. Let $f \in F[x]$ be a monic polynomial. Let $\alpha_1, \dots, \alpha_n$ be roots of f . The **discriminant of f** is defined by

$$\prod_{1 \leq i < j \leq n} (\alpha_i - \alpha_j)^2$$

and is denoted by $D(f)$.

Note that $D(f) \neq 0$ if and only if f is separable.

Examples.

1. Let $f = x^2 + bx + c = (x - \alpha_1)(x - \alpha_2)$. Then

$$\begin{aligned} D(f) &= (\alpha_1 - \alpha_2)^2 = (\alpha_1 + \alpha_2)^2 - 4\alpha_1\alpha_2 \\ &= (-b)^2 - 4c = b^2 - 4c. \end{aligned}$$

2. Let $f = x^3 + ax^2 + bx + c$ over a field F with $\text{char}(F) \neq 3$. Then

$$f\left(x - \frac{a}{3}\right) = \left(x - \frac{a}{3}\right)^3 + a\left(x - \frac{a}{3}\right)^2 + b\left(x - \frac{a}{3}\right) + c = x^3 + b'x + c'.$$

Since $D(f(x)) = D(f(x - a/3))$, it suffices to compute $f = x^3 + bx + c$. Write

$$f = x^3 + bx + c = (x - \alpha_1)(x - \alpha_2)(x - \alpha_3)$$

so that we get

$$\begin{aligned} f' &= 3x^2 + b \\ &= (x - \alpha_2)(x - \alpha_3) + (x - \alpha_1)(x - \alpha_2) + (x - \alpha_1)(x - \alpha_3). \end{aligned}$$

Then, we obtain

$$\begin{aligned} D(f) &= -f'(\alpha_1)f'(\alpha_2)f'(\alpha_3) \\ &= -(3\alpha_1^2 + b)(3\alpha_2^2 + b)(3\alpha_3^2 + b) = -4b^3 - 27c^2. \end{aligned}$$

Lemma 8.3. *Let f be a monic separable polynomial in $F[x]$, where $\text{char}(F) \neq 2$ and $\deg(f) = n$. Then,*

$$\text{Gal}(f) < A_n \quad \text{if and only if} \quad D(f) \in (F^\times)^2.$$

Proof. Let E be a splitting field of f . Let $G = \text{Gal}(f) = \text{Gal}(E/F) < S_n$. Let $\sigma \in G$. Then,

$$\sigma(D(f)) = \prod_{1 \leq i < j \leq n} [\sigma(\alpha_i) - \sigma(\alpha_j)]^2 = \prod_{1 \leq i < j \leq n} (\alpha_i - \alpha_j)^2 = D(f).$$

As $F = E^G$, we have that $D(f) \in F^\times$ (because f is separable). Let $D' = \prod (\alpha_i - \alpha_j)$, that is, a square root of $D(f)$. Then, for any $\sigma \in G$, we have that $\sigma(D') = \text{sgn}(\sigma)D'$.

Assume that $G < A_n$. Then, we have $\text{sgn}(\sigma) = 1$ for all $\sigma \in G$, thus $\sigma(D') = D'$. As $F = E^G$, $D' \in F^\times$. Hence, $D(f) \in (F^\times)^2$.

Assume that $G \not< A_n$. Then, there exists some $\sigma \in G$ such that $\text{sgn}(\sigma) = -1$. Hence, we have $\sigma(D') = -D'$, thus $D(f) \notin (F^\times)^2$. ■

9 Cyclic and Abelian Extensions (Kummer's Extensions)

Definition. Let E/F be a Galois Extension. We say that E/F is **cyclic** (resp. **abelian**) if $\text{Gal}(E/F)$ is cyclic (resp. abelian).

Examples.

1. Let E/F be a finite extension of a finite field F . Then, $\text{Gal}(E/F)$ is cyclic, i.e., E/F is cyclic.
2. We have $\text{Gal}(\mathbb{Q}(\eta)/\mathbb{Q}) = (\mathbb{Z}/n\mathbb{Z})^\times$, where η is the primitive n th root of 1. Thus, $\mathbb{Q}(\eta)/\mathbb{Q}$ is abelian.

Lemma 9.1. *Let $\sigma_1, \dots, \sigma_n$ be distinct group homomorphisms $G \rightarrow F^\times$, where G is a group and F is a field. Then, the set $\{\sigma_1, \dots, \sigma_n\}$ is linearly independent over F (i.e., $\sum_{i=1}^n a_i \sigma_i = 0$ implies that $a_i = 0$ for each i).*

Proof. We prove this lemma by induction on n . If $n = 1$, then $a_1 \sigma_1 = 0$ implies that

$$0 = a_1 \sigma_1(e) = a_1 \cdot 1 = a_1,$$

where e is the identity of G .

Now consider $n > 1$. As $\sigma_1, \dots, \sigma_n$ are distinct, there exists some element y in G such that $\sigma_1(y) \neq \sigma_n(y)$. Assume that $\sum_{i=1}^n a_i \sigma_i = 0$, i.e.,

$$\sum_{i=1}^n a_i \sigma_i(x) = 0 \quad \text{for all } x \in G.$$

Consider

$$\begin{aligned} 0 = 0 - 0 &= \sum_{i=1}^n a_i \sigma_i(yx) - \sigma_n(y) \sum_{i=1}^n a_i \sigma_i(x) \\ &= a_1 [\sigma_1(y) - \sigma_n(y)] \sigma_1(x) + \cdots + a_{n-1} [\sigma_{n-1}(y) - \sigma_n(y)] \sigma_{n-1}(x). \end{aligned}$$

By the induction hypothesis, we have that $a_1 [\sigma_1(y) - \sigma_n(y)] = 0$. By the choice of y , we have $\sigma_1(y) - \sigma_n(y) \neq 0$. So we get $a_1 = 0$. By induction, we get $a_i = 0$ for the rest of i 's. ■

Corollary 9.2. Let $\sigma_1, \dots, \sigma_n: E \rightarrow F$ be distinct field homomorphisms. Then, the set $\{\sigma_1, \dots, \sigma_n\}$ is linearly independent.

Proposition 9.3. Let F be a field containing a primitive n th root of 1, i.e., $\mu_n \subset F$. Here, μ_n denotes the group of n th roots of 1. Then the following holds:

- (1) If E is a splitting field of a polynomial $x^n - a \in F[x]$, then E/F is cyclic (of degree that divides n).
- (2) If E/F is cyclic of degree n , then E is a splitting field of some irreducible polynomial $x^n - a \in F[x]$.

Proof. (1) Let η denote the primitive n th root of 1. As the roots of $x^n - 1$ are $1, \eta, \dots, \eta^{n-1}$, this polynomial is separable. So, we have $\text{char}(F) = 0$ or $\gcd(\text{char}(F), n) = 1$ (as $(x^n - 1)' = nx^{n-1} \neq 0$). As $(x^n - a)' = n \cdot x^{n-1} \neq 0$, the field extension E/F is separable, hence Galois.

Let α be a root of $x^n - a$ in E . Let $\sigma \in \text{Gal}(E/F)$. Then, $\sigma(\alpha)^n = \sigma(\alpha^n) = \sigma(a) = a$. Hence, we have that

$$\left(\frac{\sigma(\alpha)}{\alpha} \right)^n = \frac{a}{a} = 1, \quad \text{i.e.,} \quad \frac{\sigma(\alpha)}{\alpha} \in \mu_n.$$

Let $\varphi: \text{Gal}(E/F) \rightarrow \mu_n$ such that $\sigma \mapsto \sigma(\alpha)/\alpha$. Then φ is an injective homomorphism. If $\varphi(\sigma) = 1$, then $\sigma(\alpha) = \alpha$. As $E = F(\alpha)$ (because

$$x^n - a = (x - \alpha)(x - \alpha\eta) \cdots (x - \alpha\eta^{n-1})$$

and $\mu_n \subset F$), we get $\sigma = \text{id}_E$, i.e., φ is injective. Then,

$$\begin{aligned} \varphi(\sigma\tau) &= \frac{\sigma\tau(\alpha)}{\alpha} = \frac{\sigma\tau(\alpha)}{\sigma(\alpha)} \cdot \frac{\sigma(\alpha)}{\alpha} = \varphi(\sigma) \\ &= \sigma \left(\frac{\tau(\alpha)}{\alpha} \right) = \frac{\tau(\alpha)}{\alpha} = \varphi(\tau) \end{aligned}$$

(Here, $\sigma(\tau(\alpha)/\alpha) = \tau(\alpha)/\alpha$ because $\tau(\alpha)/\alpha \in \mu_n \subset F$ and $\sigma \in \text{Gal}(E/F)$.)

(2) Let $\text{Gal}(E/F) = \langle \sigma \rangle = \{\text{id}, \sigma, \dots, \sigma^{n-1}\}$. By Corollary 9.2, we have that

$$\sum_{i=0}^{n-1} \eta^i \sigma^i \neq 0.$$

Hence, there exists some $\alpha \in E$ such that

$$\sum_{i=0}^{n-1} \eta^i \sigma^i(\alpha) \neq 0.$$

Let $\beta = \sum_{i=0}^{n-1} \eta^i \sigma^i(\alpha)$, i.e.,

$$\beta = \alpha + \eta \sigma(\alpha) + \cdots + \eta^{n-1} \sigma^{n-1}(\alpha).$$

Then,

$$\sigma(\beta) = \sigma(\alpha) + \eta \sigma^2(\alpha) + \cdots + \eta^{n-1} \alpha.$$

Hence, we get $\eta \sigma(\beta) = \beta$ (as $\eta^n = 1$) or $\sigma(\beta) = \eta^{-1} \beta$, and $\sigma^i(\beta) = \eta^{-i} \beta$.

$$(\sigma^i(\beta) = \sigma^{i-1}(\sigma(\beta)) = \sigma^{i-1}(\eta^{-1} \beta) = \eta^{-i} \beta.)$$

In particular, we have $\sigma(\beta^n) = \sigma(\beta)^n = (\eta^{-1} \beta)^n = \beta^n$, i.e., $\beta^n \in F$.

Let $\alpha = \beta^n$. As

$$x^n - \alpha = (x - \beta)(x - \beta\eta) \cdots (x - \beta\eta^{n-1}),$$

$F(\beta)$ is a splitting field of $x^n - \alpha$. As $\text{Gal}(E/F(\beta)) = \{\text{id}\}$ (as $\sigma^i(\beta) = \eta^{-i} \beta$), Galois correspondence gives that $E = F(\beta)$. As $x^n - \alpha$ is the minimal polynomial, it is irreducible. ■

Remark. Indeed, if E/F is cyclic of degree (or order) d , where $d \mid n$, then E is a splitting field of $x^d - a \in F[x]$. (The proof is the same as above, with replacing η by $\eta^{n/d}$.) In conclusion, if $\mu_n \subset F$, then E/F is cyclic of degree dividing n if and only if $E = F(\sqrt[n]{a})$ for some $a \in F$.

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Proposition 9.4. Let F be a field containing μ_n . Let E/F and K/F be two cyclic extensions of degree n , i.e.,

$$E = F(\alpha) \quad \text{with} \quad m_\alpha = x^n - a \in F[x],$$

and

$$K = F(\beta) \quad \text{with} \quad m_\beta = x^n - b \in F[x].$$

Then, $E = K$ if and only if $\langle a \rangle = \langle b \rangle$ in $F^\times / (F^\times)^n$.

Proof. ($\Rightarrow$) Assume that $F(\alpha) = F(\beta)$. Let $\sigma \in \text{Gal}(F(\alpha)/F) = \text{Gal}(F(\beta)/F)$ with $\sigma(\alpha) = \eta \alpha$, where η denotes the primitive n th root of 1. Then $\sigma(\beta) = \eta^m \beta$ for some m with $\gcd(m, n) = 1$. For, if $\gcd(m, n) = d$, then observe that

$$\begin{aligned} \sigma(\beta^{n/d}) &= \sigma(\beta)^{n/d} \\ &= (\eta^m \beta)^{n/d} \\ &= \left(\eta^{m/d} \right)^n \cdot \beta^{n/d} = \beta^{n/d}. \end{aligned}$$

Thus, β is a root of $x^{n/d} - \beta^{n/d} \in F[x]$. It has β as its root, which is a contradiction because $m_\beta = x^n - b$.

Write $\beta = \sum_{i=0}^{n-1} c_i \alpha^i$ for some $c_i \in F$. Then, we have that

$$\sigma(\beta) = \eta^m \beta = \sum_{i=0}^{n-1} c_i \eta^m \alpha^i \quad \text{and} \quad \sum_{i=0}^{n-1} c_i \sigma(\alpha)^i = \sum_{i=0}^{n-1} c_i \eta^i \alpha^i$$

so that

$$\sum_{i=0}^{n-1} c_i \eta^m \alpha^i = \sum_{i=0}^{n-1} c_i \eta^i \alpha^i.$$

Comparing the coefficients, we have $c_i \eta^m = c_i \eta^i$ for each i . As η^m is a primitive n th root of 1, we have $\eta^m \neq \eta^i$ if $m \neq i$. Thus, we get $c_i = 0$ for all $i \neq m$, which gives that $\beta = c_m \alpha^m$. Hence, we get

$$b = \beta^n = c_m^n (\alpha^n)^m = c_m^n \alpha^m,$$

that is, $\langle a \rangle = \langle b \rangle$ in $F^\times / (F^\times)^n$.

($\Leftarrow$) Assume that $\langle a \rangle = \langle b \rangle$. As $a \in \langle b \rangle$, we can write $a = b^m c^m$ for some $c \in F^\times$. Then, we have that

$$a = (\beta^n)^m \cdot c^n = (\beta^m c)^n,$$

i.e., $\beta^m c = \eta^i \alpha$ for some i . This is because $\beta^m c$ is a root of $x^n - a$. Thus, we get $\alpha = \eta^{-i} c \cdot \beta^m$, or $\alpha \in F(\beta)$.

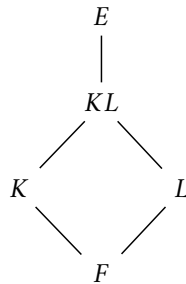
As $b \in \langle a \rangle$, we can write $b = a^m \cdot c^m$ for some $c \in F^\times$. Then, we have

$$b = (\alpha^n)^m \cdot c^n = (\alpha^m c)^n,$$

that is, $\alpha^m c = \eta^i \beta$ for some i . This is because $\alpha^m c$ is a root of $x^n - b$. Thus, we get $\beta = \eta^{-i} c \cdot \alpha^m$, or $\beta \in F(\alpha)$. Therefore, we conclude that $F(\alpha) = F(\beta)$. ■

Definition. Let E/F be a field extension and let K and L be intermediate fields. The **composite field** KL of K and L is the smallest subfield of E containing K and L .

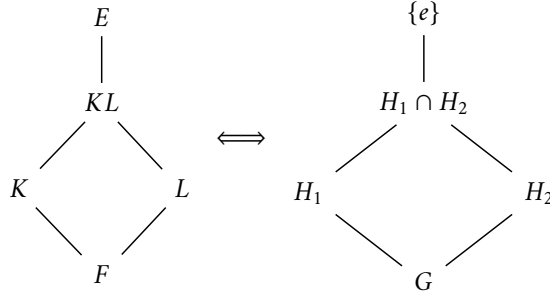
We have the following diagram:



Remark. 1. If $K = F(\alpha_1, \dots, \alpha_n)$, then $KL = L(\alpha_1, \dots, \alpha_n)$.

2. If E/F is Galois, then

$$\text{Gal}(E/KL) = \text{Gal}(E/K) \cap \text{Gal}(E/L).$$



Here, we let $H_1 = \text{Gal}(E/L)$ and $H_2 = \text{Gal}(E/K)$. Observe that $E^{H_1} = L$ and $E^{H_2} = K$.

By the fact that

$$H_1 \cap H_2 \subset H_1 \iff L = E^{H_1} \subset E^{H_1 \cap H_2}$$

and

$$H_1 \cap H_2 \subset H_2 \iff K = E^{H_2} \subset E^{H_1 \cap H_2},$$

we have that $LK \subset E^{H_1 \cap H_2}$. Conversely, $LK \subset E^{H_1 \cap H_2}$ if and only if $\text{Gal}(E/LK) \subset \text{Gal}(E/E^{H_1 \cap H_2})$. The latter statement holds because, if $\sigma \in \text{Gal}(E/LK)$, then $\sigma(a) = a$ for all $a \in LK$. It implies that $\sigma(a) = a$ for all $a \in E^{H_1 \cap H_2}$.

Definition. Let G be a group. The **exponent** of G , denoted by $\exp(G)$, is the least common multiple of orders of all elements of G , i.e., $\exp(G)$ is the smallest positive integer with $g^{\exp(G)} = 1$ for all $g \in G$.

Theorem 9.5. Let E/F be a finite extension such that $\mu_n \subset F$. Then, E/F is abelian with $\exp(\text{Gal}(E/F)) \mid n$ if and only if E is a splitting field of $(x^n - a_1) \cdots (x^n - a_m)$ for some $a_1, \dots, a_m \in F$.

Proof. ($\Rightarrow$) Let $G = \text{Gal}(E/F)$. Then, by the fundamental theorem of abelian groups, we have that $G = C_1 \times \cdots \times C_m$, where C_i is a cyclic group of order d_i . Hence, we get $d_i \mid \exp(G) \mid n$ and $\eta^{n/d_i} \in F$, i.e., the primitive d_i th root of 1 is contained in F .

Let $H_i = \prod_{j \neq i} C_j$. As G is abelian, we have that $\text{Gal}(E^{H_i}/F) = G/H_i$. For, if $H \triangleleft G$, then $\text{Gal}(E^H/F) = \text{Gal}(E/F)/\text{Gal}(E/E^H)$. By Proposition 9.3, $E^{H_i} = F(\alpha)$, where α is a root of $x^{d_i} - b_i$ for each i and $b_i \in F$. It implies that $\alpha^{d_i} = b_i$, or

$$\alpha^n = (\alpha^{d_i})^{n/d_i} = b_i^{n/d_i}.$$

Let $a_i = b_i^{n/d_i}$. Then, α is a root of $x^n - a_i$. Observe that

$$\begin{aligned} \{e\} &= \bigcap_{i=1}^m H_i && \text{(by the definition of } H_i) \\ &= \text{Gal}(E/E^{H_1} \cdots E^{H_m}). && \text{(by the second remark above)} \end{aligned}$$

Hence, we get $E = E^{H_1} \cdots E^{H_m} = E(\alpha_1, \dots, \alpha_n)$, which is the splitting field of $\prod_{i=1}^m (x^n - a_i)$ over F .

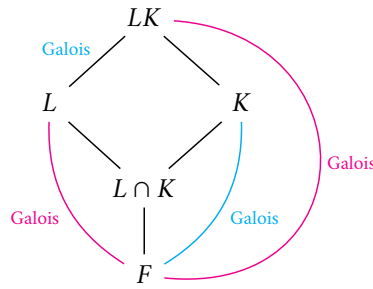
($\Leftarrow$) Let $E = F(\alpha_1, \dots, \alpha_m)$, where α_i is a root of $x^n - a_i$. Let $\sigma, \tau \in G = \text{Gal}(E/F)$. Then, $\sigma(\alpha_i) = \eta^{n_i} \cdot \alpha_i$ and $\tau(\alpha_i) = \eta^{k_i} \alpha_i$, for some integers n_i and k_i . Then, we get

$$\sigma\tau(\alpha_i) = \sigma(\eta^{k_i} \alpha_i) = \eta^{k_i} \cdot \eta^{n_i} \alpha_i = \eta^{k_i+n_i} \alpha_i = \eta^{n_i} \cdot \eta^{k_i} \alpha_i = \tau(\eta^{n_i} \alpha_i) = \tau\sigma(\alpha_i).$$

As $\sigma\tau = \tau\sigma$, G is abelian. As $F(\alpha_i)/F$ is a cyclic group of degree n , we have $\sigma^n(\alpha_i) = \alpha_i$ for all i . Thus, $\sigma^n = \text{id}$, i.e., $\exp(G) \mid n$. ■

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Consider the following diagram:



$$\text{Gal}(LK/L) \simeq \text{Gal}(K/L \cap K)$$

We have that $\text{Gal}(LK/F) \hookrightarrow \text{Gal}(L/F) \times \text{Gal}(K/F)$ if $L \cap K = F$.

Proposition 9.6. Let K/F and L/F be field extensions. If K/F is Galois, then LK/L is Galois. The restriction map from $\text{Gal}(LK/L)$ to $\text{Gal}(K/F)$ that maps σ to $\sigma|_K$ induces an isomorphism $\text{Gal}(LK/L) \simeq \text{Gal}(K/L \cap K)$.

Proof. Since K/F is normal, K is a splitting field of some $f \in F[x]$, thus L/K is a splitting field of $f \in L[x]$, i.e., LK/L is normal. Let $K = F(\alpha_1, \dots, \alpha_n)$. As K/F is separable, every α_i is separable over F , so α_i is separable over L . Hence, LK/L is separable.

As K/F is normal, we have $\sigma(K) = K$ for all σ in $\text{Gal}(LK/L)$. Let $\varphi: \text{Gal}(LK/L) \rightarrow \text{Gal}(K/F)$ such that $\sigma \mapsto \sigma|_K$. Let $\sigma \in \ker \varphi$. Then, $\sigma|_K = \text{id}$ and $\sigma|_L = \text{id}$, thus $\sigma = \text{id}_{LK}$. Hence, φ is injective. Obviously,

$$\text{Im } \varphi = \{\sigma|_K \mid \sigma \in \text{Aut}(LK), \sigma|_L = \text{id}_L\} \subset \text{Gal}(K/LK).$$

On the other hand, $\text{Im } \varphi \supset \text{Gal}(K/L \cap K)$ if and only if $K^{\text{Im } \varphi} \subset K^{\text{Gal}(K/L \cap K)} = L \cap K = LK^{\text{Gal}(LK/L)} \cap K$. As $K^{\text{Im } \varphi} \subset LK^{\text{Gal}(K/L)}$ (because if $a \in K$ and $\sigma \in \text{Gal}(LK/L)$, then $\sigma(a) = \sigma|_K(a) = a$), we get that $\text{Im } \varphi \supset \text{Gal}(K/L \cap K)$. ■

Proposition 9.7. If both K/F and L/F are Galois, then so is LK/F . The map $\varphi: \text{Gal}(LK/F) \rightarrow \text{Gal}(L/F) \times \text{Gal}(K/F)$ such that $\sigma \mapsto (\sigma|_L, \sigma|_K)$ is injective. Moreover, if $L \cap K = F$, then it is an isomorphism, i.e., $\text{Gal}(LK/F) \simeq \text{Gal}(L/F) \times \text{Gal}(K/F)$.

Proof. As K/F and L/F are normal, they are splitting fields of f and g over F , respectively. Hence, LK is a splitting field of fg . As $K = F(\alpha_1, \dots, \alpha_n)$, we have that $L = F(\beta_1, \dots, \beta_m)$ and that α_i, β_j are separable over F . Thus, we can say that $LK = F(\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_m)/F$ is separable. If $\sigma|_K = \text{id}$ and $\sigma|_L = \text{id}$, then $\sigma = \text{id}_L K$, thus φ is injective.

Assume $L \cap K = F$. Then, by Proposition 9.7, we have that

$$\text{Gal}(LK/L) \simeq \text{Gal}(K/F) \quad \text{and} \quad \text{Gal}(LK/K) \simeq \text{Gal}(L/F).$$

Let $(\tau, \rho) \in \text{Gal}(L/F) \times \text{Gal}(K/F)$. Then, there exist some $\tau' \in \text{Gal}(LK/K)$ and $\rho' \in \text{Gal}(LK/L)$ with $\tau'|_L = \tau$ and $\rho'|_K = \rho$. We then get

$$\varphi(\tau'\rho') = (\tau'\rho'|_L, \tau'\rho'|_K) = (\tau'|_L, \rho'|_K) = (\tau, \rho).$$

■

10 Radical Extension

Definition. A field extension E/F is a **radical extension** if there exists a chain of fields

$$F \subset F(\alpha_1) \subset F(\alpha_1, \alpha_2) \subset \dots \subset F(\alpha_1, \dots, \alpha_m) = E$$

such that $\alpha_1^{n_1} \in F$ and $\alpha_i^{n_i} \in F(\alpha_1, \dots, \alpha_{i-1})$, $i \geq 2$. If all the exponents $n_1 = \dots = n_m = n$ are the same, then we say that E/F is a **n -radical extension**.

Definition. Let $f \in F[x]$. We say f is **solvable by radicals** if there exists a radical extension E/F such that f splits over E .

Remarks. 1. The notion of radical extension is equivalent to the notion of n -radical extension (take $n = n_1 \dots n_m$).

2. Cyclotomic field extensions, cyclic extensions, and Kummer extensions are radical extensions.
3. The transitivity holds for radical extensions. That is, if E/F and K/E are radical extensions, then K/F is a radical extension.
4. $f \in F[x]$ is solvable by radicals if and only if the existence of all roots of f that can be expressed over F by using ordinary arithmetic operation $(+, -, \times, \div)$ and taking n th root (for variables in n).

For example, if $ax^2 + bx + c = 0$, then $x = (-b \pm \sqrt{b^2 - 4ac})/2a$.

Theorem 10.1 (Galois). Let F be a field of characteristic zero, and let $f \in F[x]$. Then, f is solvable by radicals if and only if $\text{Gal}(f)$ is solvable.

Recall. 1. A group G is **solvable** if there exists a chain

$$G = G_0 > G_1 > \cdots > G_n = \{e\}$$

such that $G_i \triangleright G_{i+1}$ and G_i/G_{i+1} is abelian for each i .

2. S_n for $n \leq 4$ is solvable, but S_n for $n \geq 5$ is not solvable.

3. Every subgroup of a solvable is solvable.

4. Let $N \triangleleft G$. Then, G is solvable if and only if both N and G/N are solvable.

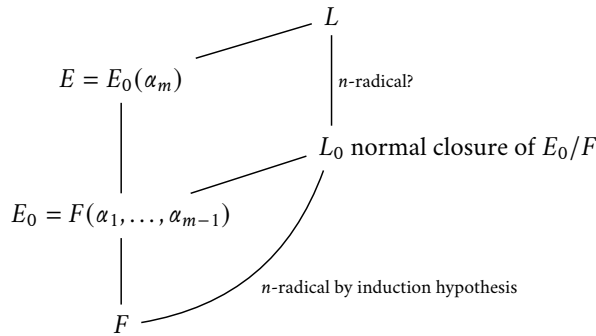
We postpone the proof to the next class.

Lemma 10.2. Let E/F be an n -radical extension and let L/E be a normal closure of E/F . Then, L/F is n -radical.

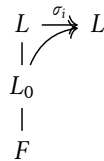
Proof. Let $E = F(\alpha_1, \dots, \alpha_m)$ with $\alpha_1^n \in F$ and $\alpha_i^n \in F(\alpha_1, \dots, \alpha_{i-1})$ for each $i \geq 2$. We show by induction on m .

If $m = 1$, then $E = F(\alpha)$ with $\alpha = \alpha_1$ and $L = F(\beta_1, \dots, \beta_n)$, where $\beta_1, \dots, \beta_n$ are roots of m_α (the minimal polynomial of α). Let $a = \alpha^n \in F$. Then $m_\alpha \mid x^n - a$ (α is a root of $x^n - a$). Thus, β_i are roots of $x^n - a$, i.e., $\beta_i^n = a \in F$ for each i . Hence, L/F is n -radical.

Now suppose that $m > 1$. Consider the following diagram:



Let L_0 be a radical closure of E_0 . Then, we have $L = L_0(\gamma_1, \dots, \gamma_s)$, where $\gamma_1, \dots, \gamma_s$ are roots of m_{α_m} (the minimal polynomial of α_m). Let $\sigma_i \in \text{Aut}(L/F)$ such that $\sigma_i(\alpha_m) = \gamma_i$ for each $i = 1, \dots, s$. Then, $\sigma_i(\alpha_m^n) = \sigma_i(\alpha_m)^n = \gamma_i^n$. As L_0/F is normal, we have $\sigma_i(\alpha_m^n) \in L_0$. Hence, L/L_0 is n -radical.



By induction hypothesis, we get that L_0/F is n -radical. By the transitivity of radical extension, L/F is n -radical. ■

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Proof of the theorem. Let E be the splitting field of f over F .

($\Rightarrow$) As f is solvable by radicals, there exists an n -radical extension L/F containing E . Let η be a primitive n th root of 1. Then, $L(\eta)/F$ is an n -radical extension (because $L(\eta)/L$ is an n -radical extension). Let $K/L(\eta)$ be a normal closure of $L(\eta)/F$. Then, by Lemma 10.2, K/F is an n -radical extension. Hence, $K/F(\eta)$ is an n -radical extension, i.e., there exists a chain

$$F = F_0 \subset F_1 = F(\eta) \subset F_2 = F_1(\alpha_1) \subset F_3 = F_2(\alpha_2) \subset \cdots \subset F_m = F_{m-1}(\alpha_{m-1}) = K$$

for some m , where $F_{i+1} = F_i(\alpha_i)$ and $\alpha_i^n = F_i$. Note that K/F is Galois.

Let $H_i = \text{Gal}(K/F_i)$. Then, by Galois correspondence, we have that

$$H_0 = \text{Gal}(K/F) \supset H_1 \supset \cdots \supset H_m = \{e\}.$$

As F_1/F_0 is a cyclotomic field extension, $\text{Gal}(F_1/F_0)$ is abelian. As F_{i+1}/F_i is cyclic (by Proposition 9.3(2)?), $\text{Gal}(F_{i+1}/F_i)$ is cyclic for each $i \geq 1$. As F_{i+1}/F_i is Galois, we have $H_i \triangleright H_{i+1}$.

Consider the following Galois correspondence:

$$\begin{array}{ccc} E & & \{e\} \\ | & & | \\ K & \leftrightarrow & \text{Gal}(E/K) \\ | & & | \\ F & & \text{Gal}(E/F) \end{array}$$

K/F is normal if and only if $\text{Gal}(E/F) \triangleright \text{Gal}(E/K)$, and

$$\frac{\text{Gal}(E/F)}{\text{Gal}(E/K)} = \text{Gal}(K/F).$$

As $\text{Gal}(F_{i+1}/F_i) = H_i/H_{i+1}$, H_i/H_{i+1} is abelian for each i . Hence, $\text{Gal}(K/F)$ is solvable.

Since

$$\text{Gal}(E/F) = \frac{\text{Gal}(K/F)}{\text{Gal}(K/F)},$$

$\text{Gal}(E/F)$ is solvable. (Recall that if G is a group and $N \triangleleft G$, G is solvable if and only if N and G/N are solvable.)

($\Leftarrow$) Let $G = \text{Gal}(E/F)$. Assume that G is solvable. Then, there exists a chain

$$G_0 = G \triangleright G_1 \triangleright G_2 \triangleright \cdots \triangleright G_m = \{e\},$$

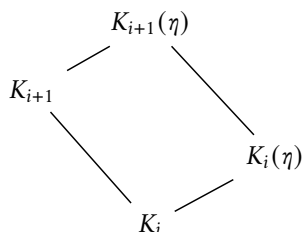
such that G_i/G_{i+1} is abelian. Let $n = \exp(G)$, where $\exp(G) = \text{lcm}_{g \in G} \text{ord}(g)$ denotes the exponent of G . Let $K_i = E^{G_i}$. Then

$$K_0 = F \subset K_1 \subset \cdots \subset K_m = E.$$

Consider

$$F \subset F(\eta) = K_0(\eta) \subset K_1(\eta) \subset \cdots \subset K_m(\eta) = E(\eta),$$

where η denotes a primitive n th root of 1. We show $K_{i+1}(\eta)/K_i(\eta)$ is an n -radical extension.



By Proposition 9.6, we get that $\text{Gal}(K_{i+1}(\eta)/K_i(\eta)) \subset \text{Gal}(K_{i+1}/K_i)$. As $\text{Gal}(K_{i+1}/K_i) = G_i/G_{i+1}$ and G_i/G_{i+1} is abelian, $\text{Gal}(K_{i+1}/K_i)$ is abelian such that $\exp(\text{Gal}(K_{i+1}/K_i)) \mid n$. Then, by Theorem 9.5, $K_{i+1}(\eta)$ is a splitting field of $f = (x^n - a_1) \cdots (x^n - a_l)$ for some $a_1, \dots, a_l \in K_i(\eta)$. Hence, $K_{i+1}(\eta)/K_i(\eta)$ is an n -radical extension for each i . As $F(\eta)/F$ is a radical extension, by the transitivity of radical extensions, $E(\eta)/F$ is a radical extension. Since $E(\eta) \supset E$, f is solvable by radicals. ■

Corollary 10.3. *Let F be a field of characteristic zero, and let $f \in F[x]$ be irreducible. If $\text{Gal}(f) = S_n$ and $n \geq 5$, then f is not solvable by radicals.*

Examples.

1. Let $f = x^5 - 6x + 3$ over $\mathbb{Q}$. By Eisenstein's criterion with $p = 3$, f is irreducible. $f' = 5x^4 - 6$ has exactly two roots. Thus, f has three real roots. By Corollary 8.2, we get $\text{Gal}(f) = S_5$. By Corollary 10.3, we conclude that f is not solvable by radicals.
2. It is known that $f = x^n - x + 1$ is irreducible over $\mathbb{Q}$ and $\text{Gal}(f) = S_n$ for $n \geq 2$. Hence, f is not solvable by radicals for $n \geq 5$.

10.1 Polynomials of Degrees Three and Four

(Cardano, 1545) Let $f = x^3 + a'x^2 + b'x + c'$, where F is a field with $\text{char}(F) \neq 2, 3$. By substituting x into $x - a'/3$, we get $f = x^3 + bx + c$. Let $x = u + v$. Then,

$$\begin{aligned} f(u + v) &= (u + v)^3 + b(u + v) + c \\ &= (u^3 + v^3 + c) + (u + v)(b + 3uv). \end{aligned}$$

We solve $f(u + v) = 0$. Set $u^3 + v^3 + c = 0$ and $b + 3uv = 0$. Then, we get $v = -b/3u$. Plugging it into the first equation, we get

$$u^3 + \left(-\frac{b}{3u}\right)^3 + c = 0 \quad \text{or} \quad u^6 + cu^3 - \frac{b^3}{27} = 0.$$

Then, u^3 is given as

$$u^3 = \frac{-c \pm \sqrt{c^2 + \frac{4b^3}{27}}}{2}.$$

Let $d = \sqrt{c^2 + 4b^2/27}$. (Indeed, $D(f) = -27d^2$.) Then, we get

$$u^3 = -\frac{c}{2} + \frac{d}{2} = \alpha \quad \text{and} \quad v^3 = -\frac{c}{2} - \frac{d}{2} = \beta.$$

It gives that

$$u = \alpha, \eta\alpha, \eta^2\alpha \quad \text{and} \quad v = \beta, \eta\beta, \eta^2\beta,$$

where η is a primitive third root of 1. Hence, the roots $\gamma_1, \gamma_2, \gamma_3$ of $f(x - a'/3)$ are given as

$$\gamma_1 = \alpha + \beta, \quad \gamma_2 = \eta\alpha + \eta^2\beta, \quad \gamma_3 = \eta^2\alpha + \eta\beta$$

by the equation $b + 3uv = 0$. The roots of f are given as $\gamma_i + a'/3, i = 1, 2, 3$.

Assume that a cubic polynomial f is irreducible. Then, by Lemma 8.1(2), $\text{Gal}(f)$ is a transitive subgroup of S_3 . That is, $\text{Gal}(f)$ acts transitively on the set of roots of f . For example, if $G = \langle \sigma = (1\ 2) \rangle$ and $\alpha_1, \alpha_2, \alpha_3$ are roots of f , then

$$\sigma(\alpha_1) = \alpha_2, \quad \sigma(\alpha_2) = \alpha_1, \quad \sigma(\alpha_3) = \alpha_3.$$

It gives that $G \cdot \alpha_1 = \{\alpha_1, \alpha_2\} \neq \{\alpha_1, \alpha_2, \alpha_3\}$, so G is not a transitive subgroup of S_3 . The transitive subgroups of S_3 are A_3 or S_3 . Hence,

$$\text{Gal}(f) \simeq \begin{cases} A_3 & \text{if } D(f) \in (F^\times)^2; \\ S_3 & \text{otherwise.} \end{cases}$$

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Let

$$\begin{aligned} f(x) &= x^4 + ax^3 + bx^2 + cx + d \\ &= (x - \alpha_1)(x - \alpha_2)(x - \alpha_3)(x - \alpha_4) \end{aligned}$$

be an irreducible separable polynomial over F . Set

$$\beta = \alpha_1\alpha_2 + \alpha_3\alpha_4, \quad \beta = \alpha_1\alpha_3 + \alpha_2\alpha_4, \quad \text{and} \quad \beta_3 = \alpha_1\alpha_4 + \alpha_2\alpha_3.$$

Let

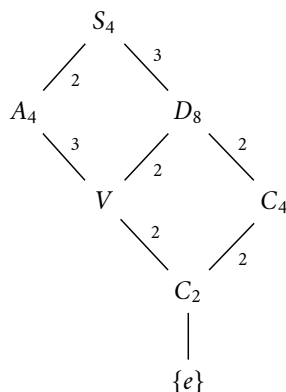
$$\begin{aligned} r(x) &= (x - \beta_1)(x - \beta_2)(x - \beta_3) \\ &= x^3 - bx^2 + (ac - 4d)x + (4bd - a^2d - c^2) \end{aligned}$$

be a polynomial over F (it is called the **resolvent cubic** of f).

Lemma 10.4. $D(f) = D(r)$.

Let $G = \text{Gal}(f)$. By Lemma 8.1(2), G is a transitive subgroup of S_4 , that is,

$$\begin{aligned} S_4 &= \langle (1\ 2\ 3\ 4), (1\ 2) \rangle, & A_4 &= \langle (1\ 2\ 3), (1\ 2)(3\ 4) \rangle, & D_8 &= \langle (1\ 2\ 3\ 4), (1\ 3) \rangle, \\ V &= \{e, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}, & \text{or} & & C_4 &= \langle (1\ 2\ 3\ 4) \rangle. \end{aligned}$$



(Here, $C_2 = \langle (1\ 3)(2\ 4) \rangle$. Note that D_8 and C_4 have three copies.)

Let $E = F(\alpha_1, \alpha_2, \alpha_3, \alpha_4)$ and $K = F(\beta_1, \beta_2, \beta_3)$ be the splitting fields of f and r , respectively. As $\alpha_1, \dots, \alpha_4$ are distinct and $\beta_1, \beta_2, \beta_3$ are distinct,

For example, we have

$$\begin{aligned}
 \beta_1 - \beta_2 &= (\alpha_1\alpha_2 + \alpha_3\alpha_4) - (\alpha_1\alpha_3 + \alpha_2\alpha_4) \\
 &= \alpha_1(\alpha_2 - \alpha_3) + \alpha_4(\alpha_2 - \alpha_3) \\
 &= (\alpha_2 - \alpha_3)(\alpha_1 - \alpha_4) \neq 0.
 \end{aligned}$$

we get that K/F is Galois.

Lemma 10.5. *If $G = \text{Gal}(f)$, we have that $E^{G \cap V} = K$.*

Proof. As every permutation in V fixes each β_i , we have that $K \subset E^{G \cap V}$.

If $\sigma \in G \setminus (G \cap V)$, then σ moves one of β_i , thus we get $K = E^{G \cap V}$. ■

$$\begin{array}{ccc}
 E & & \{e\} \\
 | & & | \\
 |G \cap V| & & \\
 K & \longleftrightarrow & \text{Gal}(E/K) = G \cap V \\
 | & & | \\
 [G:G \cap V] = |G|/|G \cap V| & & \\
 F & & G
 \end{array}$$

Theorem 10.6. *Let $m = |G|/|G \cap V| = [K : F]$, where $G = \text{Gal}(f)$. Then,*

- (1) $m = 6 \iff G = S_4$
- (2) $m = 3 \iff G = A_4$
- (3) $m = 1 \iff G = V$
- (4) $m = 2$ and f is irreducible over $K \iff G = D_8$
- (5) $m = 2$ and f is reducible over $K \iff G = C_4$.

Proof. ($\Leftarrow$) For (1), (2), and (3), we have that $G \cap V = V$, which follows from the lattice diagram.

For (4), we have $G \cap V = V$, so we get $m = 2$, i.e., $[E : K] = 4$. Let g_i be the minimal polynomial of α_i over K for each i . As $\text{Gal}(E/K) = \text{Gal}(E/E^V) = V$ and V is a transitive subgroup, there exists some $\sigma \in \text{Gal}(E/K)$ such that $\sigma(\alpha_i) = \alpha_j$. Hence, α_j is a root of g_i and thus $f \mid g_i$ over K . By the minimality of g_i , we get $f = g_i$, so f is irreducible over K .

For (5), we have $G \cap V = C_2$, so we get $m = [E : K] = 2$. Observe that $\text{Gal}(E/K) = C_2$ and C_2 is not a transitive subgroup of S_4 . If f is irreducible over K , then $\text{Gal}(f)$ acts transitively on the set of roots of f , which is impossible as C_2 is not a transitive subgroup.

($\Rightarrow$) Since $|G \cap V| \cdot m = |G|$, so we get $4 \mid |G|$ for $G = S_4, A_4, D_8, V$, and C_4 .

For (1), $m = 6$ implies that $12 \mid |G|$ so $G = S_4$ or $G = A_4$. As $|A_4|/|A_4 \cap V| = 3$, we get $G = S_4$.

For (2), $m = 3$ implies that $12 \mid |G|$ so $G = S_4$ or $G = A_4$. As $|S_4|/|S_4 \cap V| = 6$, we get $G = A_4$.

For (3), $m = 1$ implies that $G = G \cap V$ and hence $G = V$.

For (4) and (5), $m = 2$ implies that $|G \cap V| \cdot 2 = |G|$, so $|G \cap V|$ is either 2 or 4. If $|G \cap V| = 2$, then $|G| = 4$ so we get $G = C_4$. If $|G \cap V| = 4$, then $|G| = 8$ so we get $G = D_8$. If f is irreducible over K , then $\text{Gal}(E/K) = G \cap V$ acts transitively on $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$, thus $|G \cap V| = 4$. If f is irreducible over K , we get $|G \cap V| = 2$ (because if $|G \cap V| = 4$, then f is irreducible). ■

11 Construction with Ruler and Compass

Questions: Using only a ruler and a compass,

- (1) Given a cube with volume V , can we construct a cube with volume $2V$?
- (2) Can we construct a square whose area is π ?
- (3) Can we trisect a given angle?

Answer: No, in general.

Using a ruler and compass, the following constructions are allowed if two points P_1 and P_2 are given:

(A1) Draw a line through P_1 and P_2 .

(A2) Draw a circle centered at P_1 whose radius is $|\overline{P_1 P_2}|$, the length of the line $\overline{P_1 P_2}$.

(★) If n points $\{P_1, \dots, P_n\}$ are given, new points are obtained by intersecting pairs of lines, intersecting pairs of circles, or intersecting a line and a circle from (A1) and (A2).

Let $K(P_1, \dots, P_n)$ be the set of all points (containing $P_1, \dots, P_n$) obtained as in (★). We simply write K_n for $K(P_1, \dots, P_n)$. We say that a point p is **constructible** from $\{P_1, \dots, P_n\}$ if $p \in K_n$.

For example, consider Figure 11.1. The points P and Q are constructible from $\{P_1, \dots, P_n\}$, i.e., $P, Q \in P_7$.

Observations.

- (1) We can draw a line passing through $P \in K_n$ which is perpendicular to a line l through $Q \in K_n$. See Figure 11.2.
 - (a) Draw a circle centered at Q whose radius is $|\overline{PQ}|$.
 - (b) Let R be a point on l such that $|\overline{PQ}| = |\overline{PR}|$.
 - (c) Draw a circle centered at R whose radius is $|\overline{PR}|$.
 - (d) Let $S \neq P$ be an intersection of the two circles.
 - (e) Draw a line passing through P and S .

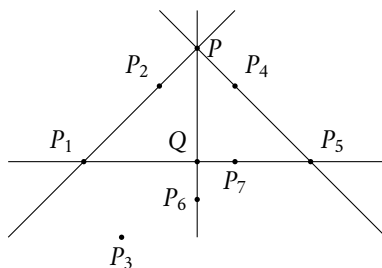
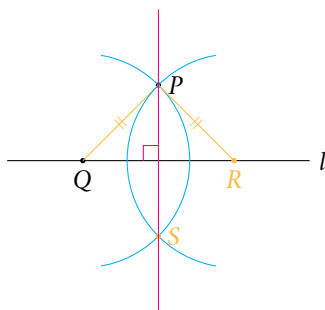


Figure 11.1: Example of constructible points

Figure 11.2: Drawing a line passing through P perpendicular to l

(2) We can draw a line parallel to l containing $P \in K_n$. See Figure 11.3.

- (a) Draw the line $\overline{PQ}$.
- (b) Choose a point R on $\overline{PQ}$.
- (c) Draw a circle C_1 centered at P whose radius is $|\overline{PR}|$.
- (d) Draw a circle C_2 centered at Q whose radius is $|\overline{PR}|$.
- (e) let T be an intersection of the circle C_2 and the line m .
- (f) let U be a point on C_2 such that $|\overline{TU}| = |\overline{RS}|$.
- (g) Draw a line l' which passes through Q and U .

(3) Given two rays of angles θ and ϕ , we can construct rays of angles $\theta \pm \phi$, $-\theta$, and $\theta/2$. See Figure 11.4.

For example, we can draw a ray of angle $\theta/2$ as follows:

- (a) Draw a circle centered at P .
- (b) Let R and S be the intersections of the rays with the circle.
- (c) Draw a line passing through R and S .
- (d) By (1), we can draw a line containing P perpendicular to $\overline{RS}$.

As $P_1, \dots, P_n$ are in the plane, we may assume that $P_j = z_j = x_j + y_j i \in \mathbb{C}$ and $z_1 = 0, z_2 = 1$. We set $K_n := K(z_1, \dots, z_n) \in \mathbb{C}$.

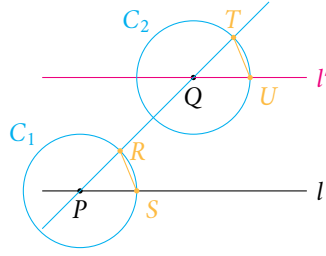


Figure 11.3: Drawing a line parallel to l

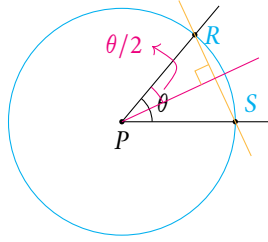


Figure 11.4: Drawing a ray of angle $\theta/2$

Lemma 11.1.

- (1) For any $n \in \mathbb{Z}$, n and ni are in K_n .
- (2) $z = x + yi \in K_n$ if and only if $x, yi \in K_n$.

Proof. (1) We need to show that $(n, 0)$ and $(0, n)$ are constructible. We use induction on n . To show that $(n, 0) \in K_n$, draw a circle centered at $(n - 1, 0)$ with radius 1. One of the intersections of x -axis with the circle is the point $(n, 0)$. See Figure 11.5a.

To show that $(0, n) \in K_n$, first draw a circle centered at $z_1 = (0, 0)$ passing through z_2 . One of the intersections of the circle and the y -axis is the point $(0, 1)$. Now, draw a circle centered at $(0, n - 1)$ with radius 1. One of the intersections of y -axis with the circle is the point $(0, n)$. see Figure 11.5b.

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(2) ($\Rightarrow$) Assume $x + yi \in K_n$. Draw a line passing (x, y) and perpendicular to each axis. Find the intersection of each line with the axis. Therefore, $x = (x, 0)$ and $y = (0, y)$ are in K_n . See Figure 11.6a.

($\Leftarrow$) Assume that $(x, 0), (0, y) \in K_n$. Draw a line passing $(x, 0)$ and perpendicular to x -axis. Draw a line passing $(0, y)$ and perpendicular to y -axis. Find the intersection of the two lines. Therefore, we get that $(x, y) \in K_n$. See Figure 11.6b. ■

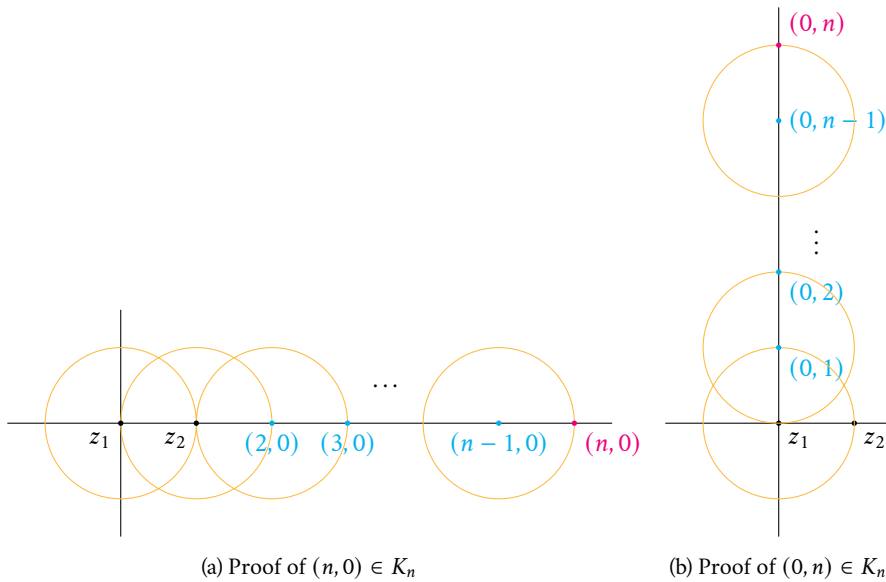


Figure 11.5: Proof of Lemma 11.1(1)

Proposition 11.2. Suppose $n \geq 2$ and let $K_n = K(z_1, \dots, z_n)$, where $z_1 = 0$ and $z_2 = 1$. Then, K_n is a subfield of $\mathbb{C}$ satisfying

$$\text{If } z \in K_n, \text{ then } \sqrt{z} \text{ and } \bar{z} \text{ are in } K_n. \quad (*)$$

If E is a subfield of $\mathbb{C}$ containing $z_1, \dots, z_n$ satisfying $(*)$, then $K_n \subset E$.

Proof. Step 1. We show that K_n is a subfield of $\mathbb{C}$.

Let $z = a + bi$ and $w = c + di$ be two points in K_n . Then,

$$\begin{aligned} z \pm w &= (a \pm c) + (b \pm d)i, \\ zw &= (ac - bd) + (ad + bc)i, \\ \frac{1}{z} &= \frac{1}{a^2 + b^2}(a - bi). \end{aligned}$$

By Lemma 11.1(2), it suffices to show that

- (i) if $(a, 0)$ and $(b, 0)$ are constructible, then $(a + b, 0)$ is constructible.
- (ii) if $(a, 0)$ and $(b, 0)$ are constructible, then $(ab, 0)$ is constructible.
- (iii) If $(a, 0)$ is constructible, then $(1/a, 0)$ is constructible.

For (i), draw a line l passing $(0, 1)$ and $(a, 0)$. Draw a line passing through $(b, 0)$ and perpendicular to the x -axis. Draw a line passing through $(0, 1)$ and perpendicular to the y -axis. Let X be the intersection of these two orthogonal lines. Draw a line l' passing through X and parallel to l . The intersection of l' and the x -axis is the point $(a + b, 0)$, because the distance from this point to $(b, 0)$ is a . See Figure 11.7.

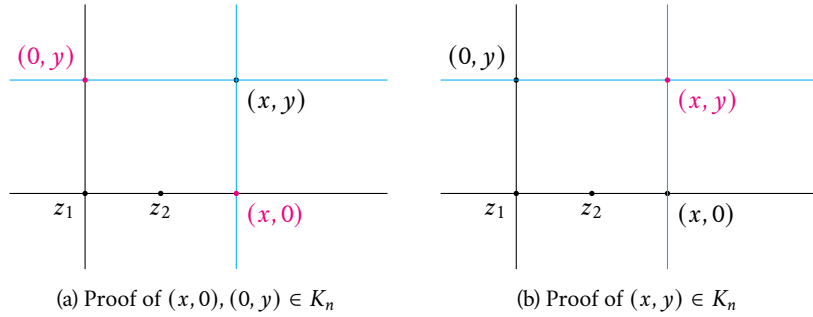
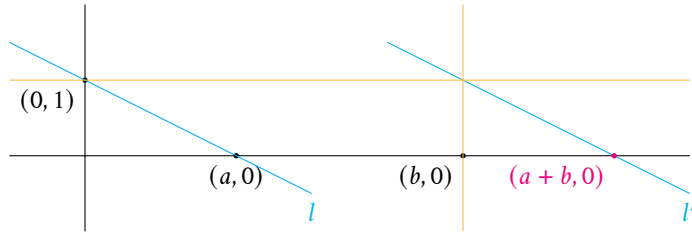


Figure 11.6: Proof of Lemma 11.1(2)


 Figure 11.7: Proof of $(a + b, 0) \in K_n$

For (ii), draw a line l connecting $(0, a)$ and $(1, 0)$. Since $(b, 0)$ is in K_n , $(1 + b, 0)$ is also in K_n by (i). Draw a line l' passing through $(1 + b, 0)$ and parallel to l . Let $(0, y)$ be the y -section of l' . Comparing the inclination, we have that

$$\frac{a}{1} = \frac{y}{1+b} \quad \text{or} \quad y = a + ab.$$

It gives that $(0, y) = (0, a + ab) \in K_n$. By (i), we have that $(0, ab) \in K_n$, or $(ab, 0) \in K_n$. See Figure 11.8.

For (iii), draw a line l connecting $(0, a)$ and $(1, 0)$. Since $(0, a)$ is in K_n , $(0, 1 + a)$ is also in K_n by (i). Draw a line l' passing through $(0, 1 + a)$ and parallel to l' . Let $(x, 0)$ be the x -section of l' . Comparing the inclination, we have that

$$\frac{a}{1} = \frac{1+a}{x} \quad \text{or} \quad x = \frac{1}{a} + 1.$$

It gives that $(x, 0) = (1/a + 1, 0) \in K_n$. By (i), we have that $(1/a, 0) \in K_n$. See Figure 11.9.

Step 2. We show that K_n satisfies (*).

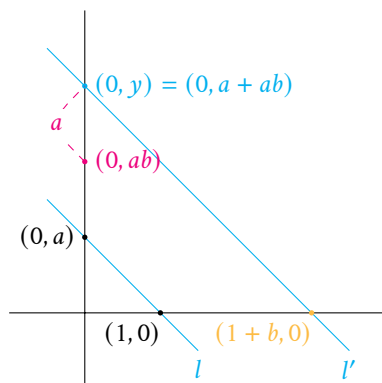
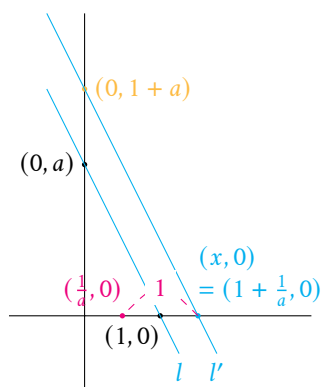
We show if $z \in K_n$, then $\sqrt{z} \in K_n$. Write $z = a + b\sqrt{-1} = re^{i\theta} = r(\cos \theta + i \sin \theta)$, where $r^2 = a^2 + b^2$. Then, we get

$$\sqrt{z} = \sqrt{r} e^{i\theta/2}.$$

We first show that if $(a, 0) \in K_n$, then $(\sqrt{a}, 0) \in K_n$. The point $((1+a)/2, 0)$ is in K_n , by (i), (ii), and (iii). See Figure 11.10.

Since

$$\left(\frac{1+a}{2}\right)^2 = x^2 + \left(\frac{a-1}{2}\right)^2,$$

Figure 11.8: Proof of $(ab, 0) \in K_n$ Figure 11.9: Proof of $(1/a) \in K_n$

we get that $x = \sqrt{a}$. It gives that $(1, \sqrt{a}) \in K_n$, so $(\sqrt{a}, 0) \in K_n$. If $r = \sqrt{a^2 + b^2} \in K_n$, then $r^{-1}z = e^{i\theta} \in K_n$. By Observations, we get $e^{i\theta/2} \in K_n$. Therefore, we conclude that $\sqrt{z} = \sqrt{r}e^{i\theta/2} \in K_n$.

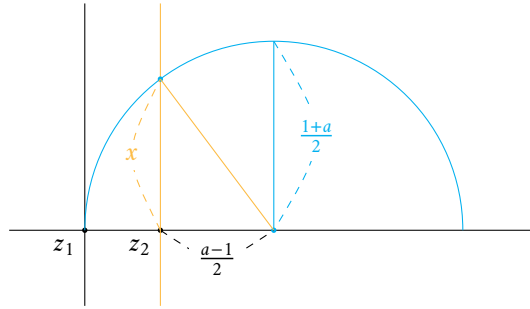
If $z = a + bi \in K_n$, then $\bar{z} = a - bi \in K_n$, as $(0, b) \in K_n$.

Step 3. We show that $K_n \subset E$ if E satisfies (*).

It suffices to show that the points obtained as in $\star$ are contained in E , i.e., for each of the following, the intersection points are in E .

- (i)
$$\begin{cases} ax + by + c = 0 \\ a'x + b'y + c' = 0 \end{cases}$$
- (ii)
$$\begin{cases} ax + by + c = 0 \\ (x - a')^2 + (y - b')^2 = c^2 \end{cases}$$
- (iii)
$$\begin{cases} (x - a)^2 + (y - b)^2 = c^2 \\ (x - a')^2 + (y - b')^2 = c'^2 \end{cases}$$

To show (i), it is enough to check $a, b, c \in E$ as E is a field. Now we show this.

Figure 11.10: Proof of $(\sqrt{a}, 0) \in K_n$

Let $z = x_1 + y_1\sqrt{i}$ and $w = x_2 + y_2\sqrt{i}$ are points in E such that the line through z and w is given by $ax + by + c = 0$. As

$$\bar{z} = x_1 - y_1i \in E \quad \text{and} \quad \frac{z + \bar{z}}{2} = x_1 \in E,$$

thus

$$\frac{\bar{z}i - zi}{2} = y_i \in E.$$

Hence, $x_1, y_1, x_2, y_2 \in E$. It gives that

$$\frac{y_2 - y_1}{x_2 - x_1}(x - x_1) = y - y_1,$$

which implies that $a, b, c \in E$ in $ax + by + c = 0$.

Similarly, one can easily check (ii) and (iii). ■

Recall. A field extension E/F is 2-radical if there exists a chain

$$F \subset F(\alpha_1) \subset F(\alpha_1, \alpha_2) \subset \cdots \subset F(\alpha_1, \dots, \alpha_n) = E$$

such that $\alpha_1^2 \in F$ and $\alpha_i^2 \in F(\alpha_1, \dots, \alpha_{i-1})$ for each $i \geq 2$.

Theorem 11.3. Let $F = \mathbb{Q}(z_1, \dots, z_n, \bar{z}_1, \dots, \bar{z}_n)$, where $z_i \in \mathbb{C}$, $z_1 = 0$, and $z_2 = 1$. Then, $z \in K_n$ if and only if there exists a 2-radical extension E/F with $z \in E$, where $K_n = K(z_1, \dots, z_n)$.

Proof. Let

$$L = \{z \in \mathbb{C} \mid \text{there exists a 2-radical extension } E/F \text{ with } z \in E\}.$$

We show $L = K_n$. By Proposition 11.2, $\bar{z}_n \in K_n$ for each i , so $F \subset K_n$. Let $z \in L$. Then, there exists a chain

$$F \subset F(\alpha_1) \subset \cdots \subset F(\alpha_1, \dots, \alpha_n) = E$$

with $\alpha_1^2 \in F$ and $\alpha_i^2 \in F(\alpha_1, \dots, \alpha_{i-1})$ for each $i \geq 2$. As $\alpha_1^2 \in F \subset K_n$, Proposition 11.2 gives that $\alpha_1 \in K_n$. For $i \geq 2$, as $\alpha_i^2 \in F(\alpha_1, \dots, \alpha_{i-1}) \subset K_n$, Proposition 11.2 gives that $\alpha_i \in K_n$. Hence, we get that $L \subset K_n$.

Now we show $K_n \subset L$. By Proposition 11.2, it suffices to show that L is a field satisfying (*). Let $z_1, z_2 \in L$. Then there exist 2-radical Extensions E_1/F and E_2/F with $z_i \in E_i$, $i = 1, 2$. Then, $E_1 E_2 / F$ is a 2-radical extension containing $z_1 z_2$, $z_1 \pm z_2$, and $1/z_1$. As $E_1 E_2 \subset L$, L is a field. Since $E_1(\sqrt{z_1})/E_1$ is a 2-radical extension, so is $E_1(\sqrt{z_1})/F$, thus $\sqrt{z_1} \in L$. As $F = \bar{F}$ (conjugate of F , i.e., $\bar{F} = \{\bar{a} \mid a \in F\}$), $\bar{E}_1/F = \bar{F}$ is a 2-radical extension, so $\bar{z}_1 \in L$. ■

Corollary 11.4. *Let $z \in K_n$. Then, z is algebraic over F , where $F = \mathbb{Q}(z_1, \dots, z_n, \bar{z}_1, \dots, \bar{z}_n)$ and $[F(z) : F] = 2^m$ for some $m \geq 0$.*

Proof. By Theorem 11.3, there exists a 2-radical extension E/F with $z \in E$. Hence, $[F(z) : F] \mid [E : F] = 2^k$ for some positive integer k . ■

Question 1. Given a cube with volume 1, can we construct a cube with volume 2? Equivalently, given $z_1 = 0$ and $z_2 = 1$, is $\sqrt[3]{2} \in K_2$ holds?

No, it is impossible, as $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$ by Corollary 11.4.

Question 2. Can we construct a square whose area is π ? Equivalently, given $z_1 = 0$ and $z_2 = 1$, is $\sqrt{\pi} \in K_2$ holds?

No, as π is transcendental over $\mathbb{Q}$, so by Corollary 11.4, we get that $\sqrt{\pi} \notin K_2$.

Question 3. Can we trisect an angle? Equivalently, given $z_1 = 0$, $z_2 = 1$, and $z_3 = \cos \theta$, is $\cos(\theta/3) \in K_3$ holds?

Let $F = \mathbb{Q}(\cos \theta)$. From

$$\cos \theta = 4 \cos^3 \left(\frac{\theta}{3} \right) - 3 \cos \left(\frac{\theta}{3} \right),$$

$\cos(\theta/3)$ is a root of $f = 4x^3 - 3x - \cos \theta \in F[x]$. Observe that

f is reducible over F

if and only if f has a root in F

if and only if for any root α of f , we have that $[F(\alpha) : F] \leq 2$

if and only if $\cos(\theta/3) \in K_3$.

In particular, if $\theta = \pi/3$, then we get $f = 4x^3 - 3x + 1/2$, which is irreducible over F . Thus, $\pi/3$ cannot be trisected. But, if $\theta = \pi/2$, then we get $f = 4x^3 - 3x$ is reducible, thus $\pi/2$ can be trisected.

Question 4. Can we construct a regular n -gon? Equivalently, given $z_1 = 0$ and $z_2 = 1$, is $\eta_n = \cos(2\pi/n) + i \sin(2\pi/n) \in K_2$ holds?

For example, a regular 3-gon is an equilateral triangle, and a regular 4-gon is a square. See Figure 11.11 for constructing a regular 3-gon.

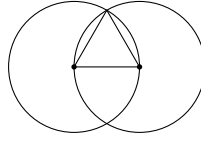


Figure 11.11: Constructing an equilateral triangle

Observations.

1. If $n = 2^m$, then $\eta_n \in K_2$, thus a regular 2^m -gon can be constructed.

If $m = 2$, then $\eta_4 \in K_2$. Use induction on m . Assume $\eta_{2^m} \in K_2$. Then by drawing a bisecting line, we get that $\eta_{2^{m+1}} \in K_2$.

2. $\eta_n \in K_2$ if and only if $\eta_{2n} \in K_2$, as angles can be doubled or bisected.

3. Let $\gcd(m, n) = 1$. Then, $\eta_{mn} \in K_2$ if and only if $\eta_n, \eta_m \in K_2$.

Since $2\pi/n = n \cdot (2\pi/mn)$ and $2\pi/n = m \cdot (2\pi/mn)$, the right direction holds. Since there exists some integer a and b such that $1 = am + bn$, we have $2\pi/mn = a \cdot (2\pi/n) + b \cdot (2\pi/m)$. Thus the left direction holds.

Lemma 11.5. *Let F be a field of characteristic zero, and let L/F be a normal extension. Then, L/F is 2-radical if and only if $[L : F] = 2^m$ for some $m \geq 0$.*

Proof. ($\Rightarrow$) It follows from the definition of 2-radical extension.

($\Leftarrow$) Let $[L : F] = 2^m$ and $G = \text{Gal}(L/F)$ with $|G| = 2^m$. By Sylow's theorem, there exists some $H < G$ with $|H| = 2^{m-1}$. Since $[G : H] = 2$, $H \triangleleft G$, thus $[L^H : F] = 2$. Hence, we get $L^H = F(\alpha)$ for some α . Let $m_\alpha = x^2 + ax + b$. As $F(\alpha) = F(2\alpha + a)$, we get $L^H = F(2\alpha + a)$. As

$$(2\alpha + a)^2 = 4\alpha^2 + 4\alpha a + a^2 = -4b + a^2 \in F,$$

L^H/F is a 2-radical extension. As $[L : L^H] = 2^{m-1}$, by induction on m , L/L^H is a 2-radical extension. Thus, L/F is a 2-radical extension. ■

Let $F_n = 2^{2^n} + 1$ with $n \geq 0$ (called a Fermat number). For example, $F_0 = 3$, $F_1 = 5$, $F_2 = 17$, and $F_3 = 257$. If F_n is a prime integer, F_n is called a Fermat prime. Indeed, if $n \leq 4$, then F_n is a prime integer, and

$$F_5 = 4294967297$$

is not a prime (Euler). It's an open question whether F_n is a prime integer for $n \geq 6$.

Theorem 11.6. *The regular n -gon is constructible from two points if and only if*

$$n = 2^m \quad (m \geq 2) \quad \text{or} \quad n = 2^l p_1 \cdots p_k \quad (l \geq 0)$$

for some distinct Fermat primes $p_1, \dots, p_k$.

Proof. Let p be an odd prime and let $\eta = \cos(2\pi/p^r) + i \sin(2\pi/p^r)$ be a primitive p^r th root of unity of 1, for some $r \geq 1$. By Theorem 6.3, we have

$$[\mathbb{Q}(\eta) : \mathbb{Q}] = [\mathbb{Q}(\mu_{p^r}) : \mathbb{Q}] = p^r(p-1).$$

Thus, $\mathbb{Q}(\eta)$ is 2-radical if and only if $(p-1)p^r = 2^s$ for some $s \geq 0$. It gives that $r = 1$ and $p = 2^s + 1$. If $s = ab$ with odd a , then $2^b + 1 \mid 2^s + 1$, so p cannot be prime. It gives that p must be a Fermat prime. By Theorem 11.3, $\eta \in K_2$ if and only if $\mathbb{Q}(\eta)/\mathbb{Q}$ is 2-radical. Hence, $\eta \in K_2$ if and only if $r = 1$ and p is a Fermat prime. Therefore, by observations (1)–(3), the statement follows. ■

Example.

Regular 9-gon and 25-gon cannot be constructed as

$$n = 3^2 \neq 2^m \cdot p_1 \cdots p_k \quad \text{and} \quad n = 5^2 \neq 2^m \cdot p_1 \cdots p_k.$$

A regular $(257 \cdot 17)$ -gon can be constructed as $17 = F_2$ and $257 = F_3$.

Part IV

Module Theory

Definition. Let R be a ring. A **left module over R** (a left R -module) is an abelian group $(M, +)$ together with a product map

$$R \times M \longrightarrow M, \quad (r, m) \longmapsto r \cdot m = rm$$

such that

- (i) $(r_1 + r_2)m = r_1m + r_2m, r(m_1 + m_2) = rm_1 + rm_2$
 - (ii) $(r_1r_2)m = r_1(r_2m)$
 - (iii) $1 \cdot m = m$
- for any $r_1, r_2 \in R$ and $m, m_1, m_2 \in M$.

Remarks. 1. One can define a right R -module structure on M :

$$M \times R \longrightarrow M, \quad (m, r) \longmapsto mr.$$

- 2. If R is commutative and M is a left R -module, then M becomes a right R -module. That is, $mr = rm$. It means that the left R -module equals the right R -module. In this case, we simply say R -module.
- 3. We have $0 \cdot m = 0, r \cdot 0 = 0$, and $-(rm) = (-r)m = r(-m)$.

Examples.

- 1. Let $R = F$, a field. Then an F -module equals a vector space over F .
- 2. Let $R = \mathbb{Z}$. Then a $\mathbb{Z}$ -module equals an abelian group.
- 3. Let I be a left ideal of R . Then, I is a left R -module.
- 4. Let $f: R \rightarrow S$ be a ring homomorphism. Let M be a left S -module. Then, M becomes a left R -module by defining $r \cdot m = f(r)m$.
- 5. Let A be an abelian group, and let $R = \text{End}(A) = \{f: A \rightarrow A \mid f \text{ is a group homomorphism}\}$. Define $(f + g)(a) = f(a) + g(a)$ and $(fg)(a) = f(g(a))$. Then, A is a left R -module with $f \cdot a = f(a)$.

Indeed, to give a structure of a left R -module on an abelian group M is equivalent to give a ring homomorphism $R \rightarrow \text{End}(M)$. Define $\varphi: R \rightarrow \text{End}(M)$ by $\varphi(r)(m) = rm$. Then, we get

$$\varphi(r_1 + r_2)m = (r_1 + r_2)m = r_1m + r_2m = \varphi(r_1)(m) + \varphi(r_2)(m) = (\varphi(r_1) + \varphi(r_2))(m)$$

and

$$\varphi(r)(m_1 + m_2) = r(m_1 + m_2) = rm_1 + rm_2 = \varphi(r)(m_1) + \varphi(r)(m_2).$$

Thus, we have that $\varphi \in \text{End}(M)$. Also, since

$$\varphi(r_1)\varphi(r_2) = \varphi(r_1r_2) \quad \text{and} \quad \varphi(1) = 1_M,$$

φ is a ring homomorphism.

Conversely, if $\varphi: R \rightarrow \text{End}(M)$ is a ring homomorphism, then define a left R -module structure on M as follows:

$$r \cdot m = \varphi(r)(m).$$